Rethinking the Southern European Productivity Declines

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Abstract

I study productivity in Spain and Italy over 1970–2019. Accounting for distortions and their propagation through production networks roughly halves Spain's measured decline and shifts its onset from 1988 to 1995. A cumulative decomposition separates within-sector productivity, between-sector misallocation, and structural change: how shifts in activity across sectors alter the contribution of subsequent sectoral productivity growth to aggregate TFP. In Spain, the shift toward services before 1995 made the subsequent productivity slowdown more costly, an effect missed by decompositions beginning in 1995. In Italy, the decline arose mainly within sectors and is less sensitive to the starting production structure.

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1 Introduction

Measured total factor productivity (TFP) in Southern European countries has declined over the past several decades according to widely used statistical sources such as EU KLEMS and the Penn World Table. As Figure 1 illustrates, conventional estimates based on the Solow residual suggest that aggregate productivity peaked in the late 1980s in Spain and in the mid-1990s in Italy, before declining substantially thereafter.

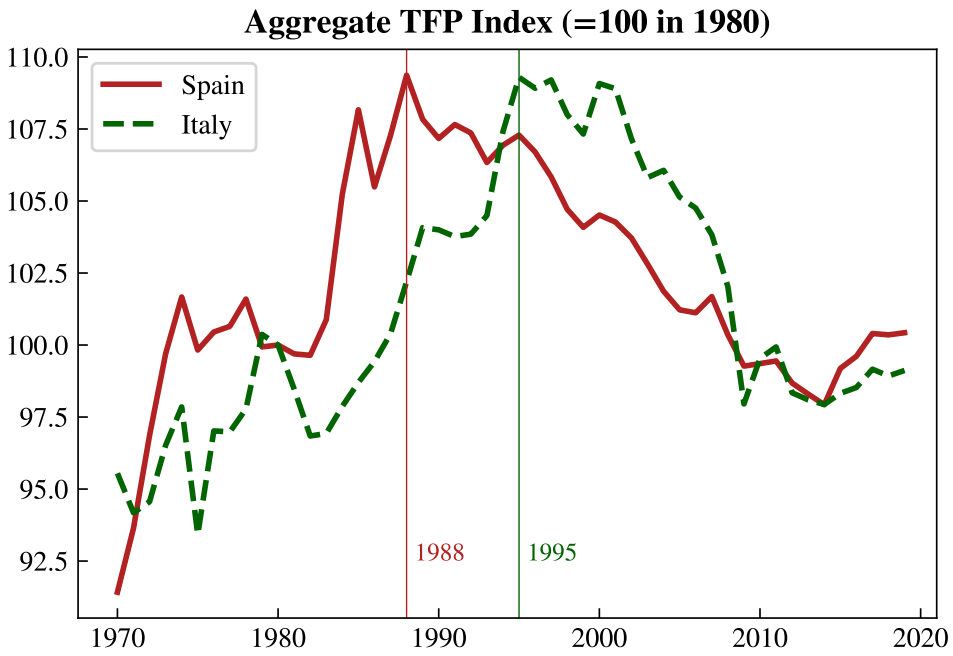


Figure 1: Evolution of Aggregate TFP in Spain and Italy.

Figure Notes. Each index is constructed for the total economy using data from *EU KLEMS Growth and Productivity Accounts* and follows a bottom-up approach that aggregates sectoral Solow residuals. The resulting series are normalized to 100 in 1980. Vertical lines indicate the years in which each TFP series peaks.

This apparent deterioration in productive efficiency has generated a large literature emphasizing structural transformation, capital misallocation, financial frictions, institutional decline, and technological stagnation. Yet no unified explanation has emerged. Existing studies often focus on specific sectors or mechanisms and frequently examine periods beginning in the mid-1990s, after much of the structural transformation of these economies had already occurred. This matters because structural change affects aggregate productivity not only when resources move. By changing the composition of production, it can alter how subsequent sectoral productivity growth translates into aggregate TFP. A study beginning after that transformation takes the resulting production structure as given and can therefore miss these delayed effects.

A second limitation is that the productivity declines these studies seek to explain need not coincide with declines in productive efficiency. When markups, taxes, and other distortions drive wedges between factor payments and marginal products, the

traditional Solow residual weights factor growth using revenue-based factor shares that no longer reflect the marginal contributions of inputs to output growth (Hall, 1988). The usual correction is to replace each producer's revenue-based factor shares with cost-based shares. With intermediate inputs, however, this producer-level correction is not enough. A factor's share of GDP is lowered not only by the wedge of the producer that employs it but also by the wedges of every producer that buys that producer's output, directly or indirectly, and a producer-by-producer correction removes only the first wedge.[1] The correct factor shares must therefore not only be cost-based but also account for the entire production network (Baqaee and Farhi, 2020). Standard measures, such as those reported by EU KLEMS and the Penn World Table, account for neither distortions nor production networks, while studies of Southern Europe that model distortions rarely propagate them through the production network. Part of the measured declines may thus be measurement artifacts.

These considerations leave two important and related questions largely unresolved. First, how much of the measured productivity declines reflects actual deterioration in productive efficiency? Second, once productivity is correctly measured, how much of its evolution reflects productivity changes within sectors, the delayed consequences of changes in the production structure, and the reallocation of inputs across distorted activities?

I answer these questions for Spain and Italy over 1970–2019.[2] The two countries experienced similar aggregate productivity declines but for substantially different reasons. In Spain, correcting the measurement of TFP makes the decline about half as large as conventional measures imply. Looking back before 1995 also changes its interpretation: the earlier shift toward services altered the production structure in a way that made the subsequent sectoral productivity slowdown substantially more costly for aggregate TFP. In Italy, by contrast, the decline arose mainly from productivity losses within sectors, and its interpretation is much less sensitive to the starting production structure. Misallocation across industries contributed little in either country.

To reach these conclusions, I construct aggregate TFP from the bottom up, combining industry accounts for Spain and Italy with a global input-output network that covers 23 countries and a rest-of-the-world region, each divided into the same 23 industries, and benchmarking real GDP to the national accounts. Within a general accounting framework that follows the production side of Baqaee and Farhi (2024), I measure factor contributions using cost-based shares that account for distortions and their propagation

[1] Take workers at a steel mill that sells its steel to a carmaker. Both firms' markups lower the workers' share of GDP, but the steel mill's cost share of labor corrects only for the mill's own markup.

[2] Greece and Portugal, the other Southern European economies, are part of the production network but are not studied, since their industry accounts before 1995 lack the factor data needed to measure aggregate productivity over such a long horizon: labor services start only in 1992 for both countries, and Greece has no capital services or capital stocks by asset.

through the production network. I then develop a novel cumulative decomposition of aggregate TFP growth into within-sector productivity, structural change, and the reallocation of factors and domestic and imported intermediate inputs across producers facing different wedges. The structural-change term measures how shifts in production across sectors alter the contribution of subsequent sectoral productivity growth to aggregate TFP. The decomposition therefore identifies the productivity consequences of observed changes in the production structure, not the forces that caused those changes.

The results substantially revise the conventional account of Spain's productivity decline. On measurement, the decline is about half as large as standard measures imply, and it begins later. Under the traditional Solow residual, aggregate TFP peaks in 1988 and falls 8.5 percentage points through 2019. Once distortions and their propagation through the production network are taken into account, TFP also stops growing in 1988, but it stagnates until 1995, and in 2019 is 4.7 points below its 1988 level. Because part of revenue accrues to rents generated by distortions, the traditional Solow residual places too much weight on capital, the fastest-growing input, and understates productivity growth while capital deepens rapidly. Network propagation is quantitatively important: ignoring it, by correcting each industry's factor payments only for its own wedge, would overstate TFP growth over 1970–2019 by 6.5 percentage points and show a decline of only 1.4 points after 1995, instead of 4.5.

On attribution, the central result is that Spain's productivity decline reflects two intertwined forces: a steady drag from structural change and the end of productivity growth within sectors. Crucially, much of the relevant structural transformation occurred before the decline began. With 1970 as the reference period, structural change subtracts similar amounts from aggregate TFP before and after 1995, 7.4 and 6.1 percentage points, mostly through the shrinking of tradables, whose productivity kept growing. What changes is productivity within sectors. It adds 37 points over 1970–1995, almost all in tradables, and nothing over 1995–2019, when slower productivity growth in tradables is offset by losses in non-tradables. The long horizon is what makes the role of structural change visible. At the production structure of 1995, the sectoral productivity changes of 1995–2019 subtract 5.1 points, so a study beginning in 1995 sees primarily a decline within sectors. The earlier shift toward services did not generate a large contemporaneous productivity loss, but it made the later slowdown more costly by changing which sectors contributed most to aggregate productivity. By contrast, worsening misallocation across industries plays little role: over 1995–2019, inefficient factor reallocation subtracts 0.7 percentage points from aggregate TFP, while the reallocation of intermediate inputs adds 2.4 percentage points.

Italy provides a revealing contrast. Correcting for distortions raises measured productivity growth but does not change the timing of the decline: aggregate TFP

stops growing in 1995 under both measures and by 2019 is 9 to 10 percentage points below its 1995 level. About 70 percent of Italy's decline reflects losses in within-sector productivity, while structural change accounts for a third. Moreover, unlike in Spain, the diagnosis changes little when the reference production structure is moved from 1970 to 1995. Productivity fell broadly across sectors, so the earlier shift toward services did not greatly alter the aggregate cost of the subsequent slowdown. Spain and Italy therefore experienced similar productivity declines for substantially different reasons.

The paper contributes to three closely related literatures. First, it contributes to the decomposition of productivity in multi-sector economies by developing a cumulative decomposition that separates productivity changes within sectors from the delayed consequences of earlier changes in sectoral weights. Standard first-order decompositions evaluate sectoral productivity growth using the prevailing production structure and therefore attribute these delayed effects to technical efficiency (Hulten, 1978; Basu and Fernald, 2002; Baqaee and Farhi, 2020, 2024). Locally, these effects are of second order—the product of the change in sectoral weights and productivity growth—and are closely related to the second-order term in Baqaee and Farhi (2019). Over long horizons, however, they can cumulate into a quantitatively important force behind aggregate TFP: in Spain, structural change subtracts 13.5 percentage points over 1970–2019, three fifths of cumulative growth, and more than the entire decline after 1995. The decomposition here extends to distorted network economies the cross term of shift-share analysis (Baily et al., 1992), the channel through which Baumol's cost disease lowers aggregate productivity growth (Baumol, 1967; Nordhaus, 2008).

The paper also contributes to the measurement of productivity in distorted economies with production networks (Hall, 1988; Basu and Fernald, 2002; Baqaee and Farhi, 2020). Empirically, the results show that accounting jointly for distortions and their propagation through production networks materially changes both the timing and the magnitude of the measured productivity declines. Closest in measurement is Comin et al. (2025), who in contemporaneous work construct markup- and utilization-adjusted TFP series for several European economies, including Spain and Italy, over 1995–2018, and show that standard measures overstate the volatility and cyclicality of TFP. This paper shares the insight that conventional TFP measures conflate technology with non-technological wedges, but differs in objective and horizon. The contribution here is not primarily to construct a new productivity series but to identify the sources of long-run productivity declines while accounting for the propagation of distortions through production networks.

Finally, the paper brings together competing explanations of the Southern European productivity decline. Previous studies have emphasized between-sector misallocation associated with falling interest rates and increased access to foreign capital (Reis, 2013;

Benigno and Fornaro, 2014), within-sector misallocation driven by financial frictions, institutional deterioration, or declining allocative efficiency (Gopinath et al., 2017; Calligaris et al., 2018; Fu and Moral Benito, 2018; García-Santana et al., 2020), and technological stagnation, managerial inefficiencies, and policy distortions (Díaz and Franjo, 2016; Pellegrino and Zingales, 2017; Schivardi and Schmitz, 2020). In Spain, the decline after 1995 reflects the end of within-sector productivity growth, which had offset a steady drag from structural change, rather than worsening misallocation between industries, including the reallocation of capital that followed the fall in interest rates. The long horizon shows that the production structure inherited in 1995 was itself part of the explanation for why this slowdown translated into an aggregate decline. Misallocation across firms within industries, which firm-level studies document, helps explain the within-sector productivity term and so complements this account.

More generally, the mechanisms uncovered here extend beyond Southern Europe. Conventional Solow residuals can exaggerate and misdate productivity slumps when factor payments differ materially from marginal products and distortions propagate through production networks. And structural transformation can shape aggregate productivity decades after it occurs by changing the economy's exposure to subsequent sectoral productivity growth.

Layout of the Paper. The rest of the paper is organized as follows. Section 2 shows why distortions, production networks, and structural change matter for measuring and interpreting aggregate TFP. Section 3 presents the accounting framework and Section 4 the decomposition. Section 5 describes the data and the estimation. Section 6 presents the empirical results, and Section 7 concludes. The Appendix contains proofs, robustness analysis, and further details on the data.

2 TFP Measurement and Structural Change

This section uses three simple economies to explain the two key ideas on which the empirical analysis rests: how distortions and production networks affect the measurement of aggregate TFP, and how structural change shapes its path over time.

2.1 Measuring TFP in Distorted Economies

TFP growth is commonly measured as the Solow residual: output growth minus a revenue-weighted average of input growth. In distorted economies, however, this traditional Solow residual does not generally provide a reliable measure of TFP growth.

To see this, consider an economy with a representative producer that generates output using a constant-returns-to-scale production technology combining multiple inputs. Output is given by

$$Y = AF(L_1, \ldots, L_N), \tag{1}$$

where A denotes Hicks-neutral productivity and L_n is the quantity of input $n \in \{1, \ldots, N\}$.

Suppose the producer has market power and sets price above marginal cost, so that the markup $\mu := P/\text{MC}$ exceeds one. In this environment, the traditional Solow residual can be written as:[3]

$$\underbrace{\Delta \log Y - \sum_{n=1}^{N} \Lambda_n \Delta \log L_n}_{\text{Traditional Solow residual}} = \underbrace{\Delta \log A}_{\text{TFP growth}} + \underbrace{(\mu - 1) \sum_{n=1}^{N} \Lambda_n \Delta \log L_n}_{\text{Distortion bias}}, \tag{2}$$

where $\Lambda_n \equiv w_n L_n/(PY)$ denotes the revenue share of input n.

This expression shows that the traditional Solow residual generally differs from true TFP growth in distorted economies. As a result, movements in the traditional Solow residual may conflate changes in technology with changes in distortions and the mismeasurement of input contributions. The larger the wedge, the larger the bias.

Equation (2) takes factor payments as correctly measured, so that revenue shares sum to $1/\mu$ and the bias is positive whenever inputs grow. Empirical implementations, however, typically compute the capital share residually, as $1 - WL/(PY)$, so that shares sum to one and the capital share includes the producer's profits. With labor and capital as the only inputs, the traditional Solow residual is then

$$\underbrace{\Delta \log Y - \Lambda_L \Delta \log L - (1 - \Lambda_L) \Delta \log K}_{\substack{\text{Traditional Solow residual} \\ \text{(empirical implementation)}}} = \underbrace{\Delta \log A}_{\text{TFP growth}} + \underbrace{(\mu - 1) \Lambda_L (\Delta \log L - \Delta \log K)}_{\text{Distortion bias}}, \tag{3}$$

which understates TFP growth whenever capital grows faster than labor. This is because the residual capital share overweights capital relative to its cost-based share.

Weighting input growth by *cost-based input shares* yields an unbiased measure of productivity growth (Hall, 1988). Defining cost shares as

$$\tilde{\Lambda}_n := \frac{w_n L_n}{\sum_k w_k L_k} = \mu \Lambda_n,$$

[3] See Appendix A.1 for a derivation.

aggregate productivity growth can be written as:[4]

$$\underbrace{\Delta \log Y - \sum_{n=1}^{N} \tilde{\Lambda}_n \Delta \log L_n}_{\text{Distorted Solow residual}} = \underbrace{\Delta \log A.}_{\text{TFP growth}} \tag{4}$$

2.2 Measuring TFP in Distorted Economies with IO Networks

Measuring aggregate TFP correctly requires taking into account not only distortions but also the underlying input-output network. With intermediate inputs, a factor's payment is lowered relative to its contribution to output not only by the markup of the producer that employs it but also by those of every producer downstream. Correcting a factor's payment only for the markup of the producer that pays it, as Section 2.1 does for a single producer, leaves part of the distortion in the factor shares.

To see this, consider an economy with two sectors linked in a vertical chain. Sector 2 produces an intermediate input from labor, and sector 1 produces the final good from capital and the intermediate input:

$$y_2 = A_2 L, \qquad y_1 = A_1 K^{\alpha} x^{1-\alpha},$$

where A_1 and A_2 are Hicks-neutral productivities and all of sector 2's output is sold to sector 1, $x = y_2$. Each sector i prices its output at a markup $\mu_i \geq 1$ over marginal cost.

Real GDP is the output of the final good, $Y = y_1$. Substituting the intermediate input gives the economy's aggregate production function,

$$Y = A K^{\alpha} L^{1-\alpha}, \qquad A := A_1 A_2^{1-\alpha}, \tag{5}$$

so that aggregate TFP growth is $\Delta \log A = \Delta \log A_1 + (1-\alpha)\Delta \log A_2$, and the elasticities of real GDP with respect to capital and labor are α and $1-\alpha$, whatever the markups. These elasticities are the factor shares with which a Solow residual recovers TFP growth.

The factor payments observed in the data are different. Sector 1, whose revenue is nominal GDP, spends a fraction $1/\mu_1$ of it on its inputs, a fraction α of that on capital and $1-\alpha$ on the intermediate input, which is sector 2's revenue. Sector 2 spends a fraction $1/\mu_2$ of its revenue on labor. The shares of capital and labor in nominal GDP, the counterparts of the revenue shares of Section 2.1, are therefore

$$\Lambda_K := \frac{w_K K}{PY} = \frac{\alpha}{\mu_1}, \qquad \Lambda_L := \frac{w_L L}{PY} = \frac{1-\alpha}{\mu_1 \mu_2}.$$

[4] See Appendix A.1, Proposition 1.

The labor share is reduced by both markups because of double marginalization.

Ignoring the input-output network, the correction of Section 2.1 would scale each factor's share by the markup of the sector that employs it, which gives each sector's cost share of its own primary input: $\hat{\Lambda}_K := \mu_1 \Lambda_K = \alpha$ for capital and $\hat{\Lambda}_L := \mu_2 \Lambda_L = (1-\alpha)/\mu_1$ for labor. Capital's share is right, but labor's is too small, because the correction misses the markup of sector 1. The resulting Solow residual can be written as:[5]

$$\underbrace{\Delta \log Y - \hat{\Lambda}_K \Delta \log K - \hat{\Lambda}_L \Delta \log L}_{\substack{\text{Distorted Solow residual} \\ \text{(ignoring the IO network)}}} = \underbrace{\Delta \log A}_{\text{TFP growth}} \underbrace{+ (1-\alpha)\left(1 - \frac{1}{\mu_1}\right) \Delta \log L.}_{\text{Network bias}} \tag{6}$$

Ignoring the network therefore overstates TFP growth whenever labor grows. The bias is larger the larger the markup downstream, μ_1, and it vanishes only when the downstream sector has no markup, so that labor's product reaches final demand without passing through a second distorted producer.

The solution is to scale each factor's share by the markup of the producer that employs it and by those of every producer downstream: μ_1 for capital and $\mu_1 \mu_2$ for labor.[6] Defining these *network-adjusted cost-based shares* as

$$\tilde{\Lambda}_K := \mu_1 \Lambda_K = \alpha, \qquad \tilde{\Lambda}_L := \mu_1 \mu_2 \Lambda_L = 1 - \alpha,$$

aggregate TFP growth can be written as:[7]

$$\underbrace{\Delta \log Y - \tilde{\Lambda}_K \Delta \log K - \tilde{\Lambda}_L \Delta \log L}_{\substack{\text{Distorted Solow residual} \\ \text{(accounting for the IO network)}}} = \underbrace{\Delta \log A.}_{\text{TFP growth}} \tag{7}$$

2.3 Structural Change and Aggregate TFP

Even when aggregate productivity is correctly measured and the economy is efficient, the reallocation of resources across sectors can change the path of aggregate TFP. The mechanism does not operate on impact, when resources move, but through the weights with which sectoral productivity contributes to aggregate TFP in later periods.

Figure 2 illustrates this distinction in a two-sector economy producing wine (W) and textiles (T). Each sector employs labor and a sector-specific fixed factor. Sectoral

[5] See Appendix A.2, Proposition 2, for a derivation, and Proposition 3 for a general network.

[6] Measuring each factor's share as its cost over total factor cost, so that the shares sum to one, does not solve the problem either. The normalization removes μ_1, which lowers both factor payments alike, but not μ_2, which lowers only labor's, so the labor share remains too small relative to capital's and the residual understates TFP growth when capital grows faster than labor (Proposition 3(iii)).

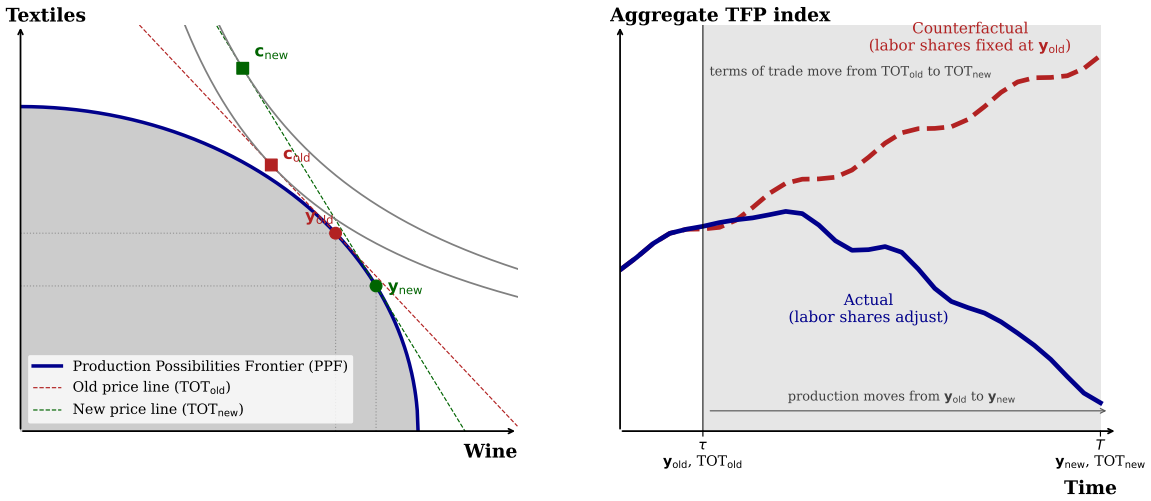
[7] See Appendix A.2, Proposition 2.

gross output in period t is given by

$$y_{i,t} = A_{i,t} L_{\ell i,t}^{\phi}, \qquad 0 < \phi < 1, \quad i \in \{W, T\},$$

where $A_{i,t}$ denotes sector-specific productivity and $L_{\ell i,t}$ is labor employed in sector i. Aggregate labor supply is fixed.

Figure 2: Trade–TFP Mechanism.



The left panel of Figure 2 illustrates how a change in the terms of trade affects production. The economy initially operates at a trade equilibrium with terms of trade TOT_{old}, producing at $\mathbf{y}_{\text{old}}$ on the production possibility frontier and consuming at $\mathbf{c}_{\text{old}}$ on the corresponding trade budget line. An increase in the relative price of wine shifts the terms of trade to TOT_{new}, inducing a reallocation of production along the frontier toward wine, to $\mathbf{y}_{\text{new}}$. Consumption moves to $\mathbf{c}_{\text{new}}$ on the new, steeper budget line.

The left panel holds technology fixed and isolates the movement along a given frontier. The right panel adds productivity growth, which shifts the frontier over time, and tracks aggregate TFP as labor moves from the shares at $\boldsymbol{y}_{\text{old}}$ to those at $\boldsymbol{y}_{\text{new}}$ over the shaded transition, during which the terms of trade gradually move from TOT_{old} to TOT_{new}. The panel shows two things. First, the reallocation does nothing to aggregate TFP on impact. Labor is paid the value of its marginal product in both sectors, so in every period the value of output lost in textiles equals the value of output gained in wine, and aggregate TFP growth—real output growth, a Divisia index of sectoral output, net of cost-share-weighted factor growth—equals the Domar-weighted sum of sectoral productivity growth,

$$\Delta \log \text{TFP}_t = \sum_{i \in \{T,W\}} \lambda_{i,t-1} \Delta \log A_{i,t}, \tag{8}$$

where $\lambda_{i,t-1}$ is sector i's share of nominal output. This is Hulten's theorem, which holds here as an exact identity when aggregate labor is measured, as in the data, by the Divisia index of its sectoral uses.[8]

The reallocation nevertheless matters for the path of aggregate TFP, because it changes the weights in (8). Let τ be the last period at TOT_{old}, and cumulate (8) from τ to any later period T of the transition. Adding and subtracting the weights of τ splits cumulative technical efficiency, the Domar-weighted sum of sectoral productivity growth, exactly into two parts

$$\sum_{t=\tau+1}^{T}\sum_{i}\lambda_{i,t-1}\Delta\log A_{i,t} = \underbrace{\sum_{t=\tau+1}^{T}\sum_{i}\lambda_{i,\tau}\Delta\log A_{i,t}}_{\text{within-sector productivity}} + \underbrace{\sum_{t=\tau+1}^{T}\sum_{i}(\lambda_{i,t-1}-\lambda_{i,\tau})\Delta\log A_{i,t}}_{\text{structural change}}. \tag{9}$$

The first term is cumulative aggregate TFP growth with the production structure held at its level in τ: the dashed counterfactual in the right panel of Figure 2. The second is structural change: the vertical distance between the counterfactual and the actual path at T. It is zero in the first period, when the weights are still those of τ, and from then on records the observed change in the production structure, whatever its cause.

In the example, the terms-of-trade improvement raises wine's share above its level in τ, but the same logic extends beyond trade-induced efficient reallocations: structural change driven by shifts in demand—as in Baumol (1967)'s cost disease—as well as changes in sectoral markups, regulation, or other wedges can reallocate resources across activities with heterogeneous productivity growth. Whenever the sector that gains weight has the slower productivity growth, every term of the second sum of (9) is negative and the gap widens with T: the reallocation lowers the *level of aggregate TFP* along the whole transition path.[9]

This is the effect that standard first-order decompositions of aggregate TFP growth do not identify (Hulten, 1978; Basu and Fernald, 2002; Baqaee and Farhi, 2020, 2024). In a competitive economy, the envelope theorem implies that a marginal reallocation of factors has no impact effect on aggregate TFP: because each factor's marginal product is the same in every use, the output gained in the sector it moves to equals the output lost in the sector it leaves. Applied period by period with updated weights, a first-order decomposition therefore attributes the whole of (8) to technical efficiency. In the example it would report, correctly, that every period's TFP growth is technical efficiency, but the reason the path ends lower—the movement of labor toward the slower-growing sector—would not appear anywhere. Identifying this effect requires comparing the

[8] Appendix B.1 derives equation (8).
[9] Appendix B.1 formalizes both results.

weights with those of a reference period, as in (9). How much of the declines in measured TFP in Southern Europe is of this kind, rather than productivity losses within sectors or the reallocation of resources across producers facing different wedges, is a central empirical question of the paper. Answering it requires a decomposition that identifies the second term of (9) in open economies with distortions and production networks, where inefficient reallocations also have an impact effect. Section 4 provides such a decomposition, within the framework introduced next.

3 General Accounting Framework

This section sets out the accounting framework on which the decomposition rests. It follows the production-side environment of Baqaee and Farhi (2024), with input-output networks, distortions, and international trade, but does not require solving for equilibrium. The decomposition in Section 4 is an accounting identity: given the definitions of real GDP, aggregate TFP, and sectoral productivity, it holds for whatever prices and quantities are observed in the data. Two assumptions give its terms their economic interpretation. Constant returns and cost minimization imply that cost-based input shares equal output elasticities, so sectoral productivity measures technical efficiency. Pricing at a wedge over marginal cost gives the reallocation terms their interpretation as movements of inputs across uses with different social returns.

Environment. The world economy consists of a set of countries $\mathcal{C}$, a set of producers $\mathcal{I}$, and a set of primary factors $\mathcal{F}$. Each producer and factor is located in a particular country. The sets located in country c are $\mathcal{I}_c$ and $\mathcal{F}_c$, and factors do not move across countries. Producer $i \in \mathcal{I}_c$ produces with a constant-returns technology

$$y_i = A_i F_i\Big(\{x_{ij}\}_{j\in\mathcal{I}}, \{L_{fi}\}_{f\in\mathcal{F}_c}\Big),$$

where A_i is Hicks-neutral productivity, x_{ij} is its use of good j, domestic or imported, and L_{fi} its use of factor f, so that $L_f = \sum_{i\in\mathcal{I}_c} L_{fi}$. The producer minimizes costs taking input prices as given and sets its price at a wedge over marginal cost,

$$p_i = \mu_i \, \mathrm{mc}_i,$$

where the wedge μ_i captures markups, taxes, and other frictions. With constant returns, marginal cost equals average cost, so the producer's total costs are $p_i y_i / \mu_i$.

Nominal and real output. Nominal gross domestic product of country c is the value of its final output,

$$\mathrm{GDP}_c := \sum_{i \in \mathcal{I}} p_i q_{ci} = \sum_{f \in \mathcal{F}_c} w_f L_f + \sum_{i \in \mathcal{I}_c} \left(1 - \frac{1}{\mu_i}\right) p_i y_i,$$

where

$$q_{ci} := y_i \mathbf{1}_{\{i \in \mathcal{I}_c\}} - \sum_{k \in \mathcal{I}_c} x_{ki}$$

is country c's net output of good i. For a domestic good, it consists of domestic final uses and exports; for a foreign good, it is minus the quantity imported for intermediate use. As in the System of National Accounts, GDP equals the income generated in production: payments to primary factors plus income generated with wedges. Real GDP is measured by the corresponding Divisia quantity index, with the GDP deflator given by its dual price index.

Input-output matrices and Domar weights. The revenue-based input-output matrix $\boldsymbol{\Omega}$ records each producer's expenditure on each input as a share of its revenue:

$$\Omega_{ij} = \frac{p_j x_{ij}}{p_i y_i}, \qquad \Omega_{if} = \frac{w_f L_{fi}}{p_i y_i},$$

for goods $j \in \mathcal{I}$ and factors $f \in \mathcal{F}_c$. The cost-based matrix records the same expenditures as shares of the producer's costs,

$$\tilde{\boldsymbol{\Omega}} = \mathrm{diag}(\boldsymbol{\mu})\, \boldsymbol{\Omega}.$$

The two coincide in an efficient economy, where $\boldsymbol{\mu} = \mathbf{1}$. For country c, let $\boldsymbol{\Omega}^c$ and $\tilde{\boldsymbol{\Omega}}^c$ be the blocks of these matrices that record purchases by domestic producers from domestic producers, $i, j \in \mathcal{I}_c$, and define the corresponding local Leontief inverses

$$\boldsymbol{\Psi}^c := (\boldsymbol{I} - \boldsymbol{\Omega}^c)^{-1}, \qquad \tilde{\boldsymbol{\Psi}}^c := (\boldsymbol{I} - \tilde{\boldsymbol{\Omega}}^c)^{-1}.$$

The revenue-based and cost-based Domar weights of domestic producer $i \in \mathcal{I}_c$ are

$$\lambda_i^{Y_c} = \sum_{k \in \mathcal{I}_c} \Omega_{Y_c,k} \Psi_{ki}^c = \frac{p_i y_i}{\mathrm{GDP}_c} \qquad \text{and} \qquad \tilde{\lambda}_i^{Y_c} = \sum_{k \in \mathcal{I}_c} \Omega_{Y_c,k} \tilde{\Psi}_{ki}^c,$$

where $\Omega_{Y_c,k} = p_k q_{ck} / \mathrm{GDP}_c$. The revenue-based weight is producer i's sales as a share of GDP, and the cost-based weight measures its exposure to final output through the

domestic cost network. Imported intermediates enter producers' costs through Ω_{ij}, for $j \notin \mathcal{I}_c$, but not the local Leontief inverses, so foreign wedges do not enter the domestic network weights of country c.

Factor shares. The revenue-based and cost-based shares of domestic factor $f \in \mathcal{F}_c$ in country c's GDP are

$$\Lambda_f^{Y_c} = \sum_{i \in \mathcal{I}_c} \lambda_i^{Y_c} \Omega_{if} = \frac{w_f L_f}{\text{GDP}_c} \qquad \text{and} \qquad \tilde{\Lambda}_f^{Y_c} = \sum_{i \in \mathcal{I}_c} \tilde{\lambda}_i^{Y_c} \tilde{\Omega}_{if}.$$

The first is the income of factor f as a share of GDP. The second is its network-adjusted cost-based share: factor payments are adjusted not only for the wedge of the producer employing the factor, but also for wedges transmitted through its domestic buyers. These are the factor shares used to measure aggregate TFP in the next section.

4 Decomposing Aggregate TFP

I now present the decomposition of aggregate TFP growth used in the empirical analysis. Throughout, all objects are expressed using log changes, $\Delta \log z_t := \log z_t - \log z_{t-1}$.

Definition 1 (Aggregate TFP Growth)**.** *Aggregate TFP growth of country c is defined as*

$$\Delta \log TFP_{c,t} := \Delta \log Y_{c,t} - \sum_{f \in \mathcal{F}_c} \tilde{\Lambda}_{f,t-1}^{Y_c} \Delta \log L_{f,t},$$

where $\Delta \log Y_{c,t}$ is real GDP growth, the Divisia index of final output with weights at $t-1$, $\tilde{\Lambda}_{f,t-1}^{Y_c}$ is the cost-based factor share of domestic factor $f \in \mathcal{F}_c$ in country c's GDP, and $\Delta \log L_{f,t}$ is the growth of that factor, the Divisia index of its sectoral uses.

This definition generalizes the Solow residual to economies with multiple sectors, distortions, international trade, and arbitrary input-output linkages. Its factor shares are cost-based rather than revenue-based, which removes the distortion bias of Section 2.1, and they are computed from the domestic cost-based Leontief inverse, which removes the network bias of Section 2.2. The wedges enter both of these factor shares and the reallocation terms of the TFP decomposition through a single statistic for each producer, which measures by how much the distortions along its supply chain raise the social returns to its inputs above their cost.

Definition 2 (Network-Adjusted Wedge). *For each domestic producer $i \in \mathcal{I}_c$ and domestic factor $f \in \mathcal{F}_c$, let*

$$\kappa_{i,t} := \mu_{i,t} \frac{\tilde{\lambda}_{i,t}^{Y_c}}{\lambda_{i,t}^{Y_c}}, \qquad \overline{\kappa}_{f,t} := \frac{\tilde{\Lambda}_{f,t}^{Y_c}}{\Lambda_{f,t}^{Y_c}} = \sum_{i \in \mathcal{I}_c} s_{fi,t} \kappa_{i,t},$$

where $s_{fi,t} := \lambda_{i,t}^{Y_c} \Omega_{if,t} / \Lambda_{f,t}^{Y_c}$ is producer i's share of total payments to factor f.

Raising any input k of producer i by one percent, holding everything else fixed, raises real GDP to first order by $\tilde{\lambda}_i^{Y_c} \tilde{\Omega}_{ik}$ percent, while the input's cost is $\lambda_i^{Y_c} \Omega_{ik}$ percent of GDP. Their ratio is κ_i: the social return per unit of cost of inputs used by producer i. In an efficient economy, $\kappa_i = 1$ for every producer. The following lemma shows how κ_i accumulates the wedges along the supply chain.

Lemma 1 (Cumulative Wedges). *Let $\phi_{ik} := x_{ki}/y_i$ denote the share of producer i's output sold to domestic producer $k \in \mathcal{I}_c$, and $\phi_i^Y := q_{ci}/y_i = 1 - \sum_{k \in \mathcal{I}_c} \phi_{ik}$ the share sold to final uses, where q_{ci} is producer i's output net of sales to domestic producers, that is, domestic final demand and exports. Then*

$$\kappa_{i,t} = \mu_{i,t} \left[\phi_{i,t}^Y + \sum_{k \in \mathcal{I}_c} \phi_{ik,t} \kappa_{k,t} \right].$$

Consequently, $\kappa_{i,t} = 1$ for all i when $\mu_{i,t} = 1$ for all i; $\kappa_{i,t} \geq \mu_{i,t} \geq 1$ when all wedges are at least one; and $\kappa_{i,t} = \mu_{i,t}$ when producer i sells only to final uses. See Proof in Appendix A.4.

In the vertical chain of Section 2.2, the lemma gives $\kappa_1 = \mu_1$ and $\kappa_2 = \mu_1 \mu_2$. The other building block of the decomposition is sectoral productivity, measured by each producer's Solow residual.

Definition 3 (Sectoral Productivity Growth). *For each domestic producer $i \in \mathcal{I}_c$, productivity growth is the gross-output Solow residual with cost-based input shares,*

$$\Delta \log A_{i,t} := \Delta \log y_{i,t} - \sum_{f \in \mathcal{F}_c} \mu_{i,t-1} \Omega_{if,t-1} \Delta \log L_{fi,t} - \sum_{j \in \mathcal{I}} \mu_{i,t-1} \Omega_{ij,t-1} \Delta \log x_{ij,t}.$$

Under cost minimization, the cost shares $\mu_i \Omega_{if}$ and $\mu_i \Omega_{ij}$ equal the output elasticities of the inputs, so that the residual measures Hicks-neutral technology growth to first order (Hall, 1988; Basu and Fernald, 2002). The shares are evaluated at $t-1$, as are all weights in the decompositions below.

Theorem 1 (Decomposition of Aggregate TFP Growth). *Over any window* $(\tau, T]$*, cumulative aggregate TFP growth in country* c *satisfies*

$$\begin{aligned}\sum_{t=\tau+1}^{T} \Delta \log TFP_{c,t} = & \underbrace{\sum_{t=\tau+1}^{T} \sum_{i \in \mathcal{I}_c} \tilde{\lambda}_{i,\tau}^{Y_c} \Delta \log A_{i,t}}_{\text{(I) within-sector technical efficiency}} + \underbrace{\sum_{t=\tau+1}^{T} \sum_{i \in \mathcal{I}_c} (\tilde{\lambda}_{i,t-1}^{Y_c} - \tilde{\lambda}_{i,\tau}^{Y_c}) \Delta \log A_{i,t}}_{\text{(II) structural change}} \\ & + \underbrace{\sum_{t=\tau+1}^{T} \sum_{f \in \mathcal{F}_c} \sum_{i \in \mathcal{I}_c} \lambda_{i,t-1}^{Y_c} \Omega_{if,t-1} (\kappa_{i,t-1} - \overline{\kappa}_{f,t-1}) \left(\Delta \log L_{fi,t} - \Delta \log L_{f,t} \right)}_{\text{(III) factor reallocation}} \\ & + \underbrace{\sum_{t=\tau+1}^{T} \sum_{i \in \mathcal{I}_c} \sum_{j \in \mathcal{I}_c} \lambda_{i,t-1}^{Y_c} (\kappa_{i,t-1} - 1) \Omega_{ij,t-1} (\Delta \log x_{ij,t} - \Delta \log y_{j,t})}_{\text{(IV) reallocation in domestic intermediates}} \\ & + \underbrace{\sum_{t=\tau+1}^{T} \sum_{i \in \mathcal{I}_c} \sum_{j \notin \mathcal{I}_c} \lambda_{i,t-1}^{Y_c} (\kappa_{i,t-1} - 1) \Omega_{ij,t-1} \Delta \log x_{ij,t}}_{\text{(V) reallocation in foreign intermediates}}, \end{aligned} \tag{10}$$

where sectoral productivity growth is measured as in Definition 3. For a single period, $T = \tau + 1$*, term (II) is zero. See Proof in Appendix A.5.*

Interpretation. Term (I) measures within-sector technical efficiency. It aggregates sectoral productivity growth using the cost-based Domar weights of the reference period, thus maintaining the production structure of that period and isolating changes in productivity within sectors. Economically, it answers the counterfactual question: *How would aggregate TFP have evolved if the economy had experienced the observed sectoral productivity changes while keeping its production structure fixed at the reference-period level?* Under the assumptions of the model, within-sector productivity growth is technology growth. In the data, measured with industry accounts, it also contains whatever else changes productivity inside a sector, including the allocation of resources across firms.

Term (II) measures structural change. It weights the same productivity growth by the change in cost-based Domar weights since the reference period, and so records how the shift in the production structure has altered the contribution of sectoral productivity growth to aggregate TFP. Economically, it answers the question: *How much does aggregate TFP growth differ from what it would have been with the reference-period structure, because Domar weights have shifted toward sectors with different productivity growth?* This term is negative when the economy has shifted toward sectors whose productivity grows more slowly than average, and it accumulates: once the weights have moved,

the shift alters the contribution of every later period's productivity growth.[10] It requires no distortions: it is the channel through which efficient reallocation, such as the trade-induced reallocation of Section 2.3, affects aggregate TFP. The term is an accounting object: it measures the productivity consequences of observed changes in the production structure, whatever their cause. Differential productivity growth across sectors, as in Baumol's cost disease, shifts in demand, trade, or changes in wedges can all change Domar weights. Term (II) records how those changes alter the contribution of subsequent sectoral productivity growth, not why the production structure changed or why productivity evolved differently across sectors.

Term (III) measures factor reallocation. Economically, it answers the question: *How much does aggregate TFP change because labor and capital move between sectors whose inputs have different social returns?* It is a covariance between network-adjusted wedges and factor flows. It generalizes the reallocation term of Basu and Fernald (2002), which weights factor flows by each producer's own markup, to a production network, where the weight is the network-adjusted wedge κ_i, which also includes the wedges of the producer's buyers (Lemma 1). It is positive when factors move toward producers whose network-adjusted wedge is above the factor's average—producers whose inputs earn a social return further above their cost, because they or their buyers are more distorted. It is zero when all sectors expand factor use proportionally, and zero whenever κ_i is the same for all users of the factor.

Term (IV) measures reallocation in domestic intermediates. Economically, it answers the question: *How much of aggregate TFP growth is accounted for by changes in domestic input-output linkages across sectors?* It records how the output of each domestic good j is redistributed among its buyers. When buyer i's purchases of j grow faster than j's output, i receives a larger share of the good, and the term rises in proportion to $\kappa_i - 1$, the amount by which the social return to i's inputs exceeds their cost. In a competitive economy ($\kappa_i = 1$), an additional unit of intermediate input costs exactly what it contributes at the margin, so the effect is zero. With distortions, redirecting a good toward buyers whose inputs earn a higher social return raises aggregate TFP, because the distortions lead those buyers to use too little of it; a productivity gain upstream that raises all purchases in proportion is technical efficiency, not reallocation.

Term (V) measures reallocation in foreign intermediates, the contribution of international trade through imported intermediates. Economically, it answers the question: *How much does aggregate TFP change because sectors alter their use of foreign inputs?* Expanding imported inputs in a sector whose inputs earn a social return above their cost, $\kappa_i > 1$, raises real GDP by more than the imports cost, for the same reason given above.

[10] In the vocabulary of shift-share analysis, term (II) is the cross term of Baily et al. (1992) and the Baumol effect of Nordhaus (2008), computed with cost-based Domar weights.

Trade can also affect aggregate TFP indirectly by reshaping the allocation of primary factors across domestic sectors, but those effects are conceptually different and are captured by terms (II) and (III), not by term (V).

Terms (II)–(V) all measure effects of reallocation. Terms (III)–(V) are its *impact effect*, one term per kind of input that can move: they record the current period's flows of factors, domestic intermediates, and imported intermediates, each weighted by how much more the input contributes to output where it goes than where it comes from. Term (II) is its *delayed effect*: a reallocation changes the Domar weights, and with them the contribution of every later period's sectoral productivity growth. In an efficient economy an input contributes the same everywhere at the margin, so $\kappa_i = \overline{\kappa}_f = 1$ and terms (III)–(V) vanish: the whole effect of reallocation arrives with a delay, through term (II), and Theorem 1 reduces to Hulten (1978)'s theorem. In a distorted economy, both effects are present. The empirical analysis reports each term separately and distinguishes three channels: within-sector productivity, term (I); structural change, the delayed effect of reallocation, term (II); and allocative efficiency, its impact effect, terms (III)–(V). Appendix B.2 presents stylized examples that isolate each term.

Relation to Baqaee–Farhi. The first-order decomposition of Baqaee and Farhi (2024) splits aggregate TFP growth into technical efficiency and allocative efficiency. Cumulated over a window, technical efficiency equals terms (I) and (II) together: it weights sectoral productivity growth by the current production structure, so it cannot separately identify productivity changes within sectors from changes in the production structure relative to the reference period. For ease of comparability to Baqaee and Farhi, and to illustrate the importance of distinguishing terms (I) and (II) in long-run productivity analyses, I sometimes group terms (I) and (II) together as technical efficiency. Their allocative efficiency equals terms (III)–(V) to first order but is expressed in terms of prices, through changes in wedges and factor income shares. Theorem 1 expresses it in quantities, which has the advantage of showing which resources moved where: term (III) separates the reallocation of labor from that of capital, and terms (IV) and (V) separate domestic from imported intermediates. This makes it possible to ask, for example, whether the capital that flowed into Southern Europe after the fall in interest rates lowered aggregate TFP by moving toward producers whose inputs earn a lower social return.

Weighting convention. The decompositions evaluate all weights at the beginning of each period. This convention has a convenient economic interpretation: each term measures the contribution of a given margin to aggregate TFP or welfare growth relative to the production structure inherited from the previous period. In this sense,

the decompositions answer a sequence of counterfactual questions of the form: starting from the structure of the previous period, how much would aggregate TFP or welfare have changed if only this margin had moved as observed?

In applied growth accounting, midpoint (Törnqvist) weights are often preferred because they provide a more accurate second-order approximation to the underlying continuous-time Divisia representation. I instead use base-period weights because midpoint weights absorb interaction terms—arising from simultaneous changes in primitives and endogenous equilibrium weights—into the measured contributions of each component. For the purposes of this paper, separating those interaction effects is useful both for interpretation and for highlighting the role of sectoral reallocation. Appendix E nonetheless reports robustness to these alternative weighting choices.

5 Data and Estimation

This section describes the data sources and estimation procedures used to measure aggregate productivity and welfare.

Data sources. I rely on four groups of data sources. World input-output tables from the Long-Run WIOD, the 2013 release of WIOD, and Eurostat's FIGARO tables provide nominal output, intermediate purchases, and final demand. I use these data to construct input-output matrices and expenditure shares, and to estimate sector-level distortions. These data are complemented by the EU KLEMS Growth and Productivity Accounts and the national accounts, which provide industry-level measures of output, capital, labor, and intermediate inputs. Asset-specific depreciation rates come from the BEA Fixed Asset Tables. The cost of capital combines long-term government bond yields and equity and bond returns from the Jordà–Schularik–Taylor Macrohistory Database, the US corporate bond spread from FRED, the liabilities of non-financial corporations from Eurostat's financial balance sheets, and corporate tax rates and depreciation allowances from the Oxford CBT Tax Database. See Appendix C for more details.

The analysis spans 1970–2019, ending before the pandemic, and uses annual data. The units of observation are 24 countries or regions (23 individual countries plus a rest-of-world aggregate), each comprising 23 production sectors and two primary factors (capital and labor). The resulting world production network therefore contains roughly 550 producers in each year, summarized in a Leontief matrix with about 300,000 entries.

Estimation. To compute and decompose aggregate TFP and welfare, I estimate three key inputs for each sector-country-year observation:

(i) *Distortions*. A central challenge in measuring aggregate TFP in distorted economies is to quantify distortions. In the baseline specification, I estimate distortions non-parametrically for each sector i as

$$\mu_i = 1 + \frac{\text{wedge margin}_i}{1 - \text{wedge margin}_i},$$
$$\text{wedge margin}_i := \frac{p_i y_i - \sum_{j \in \mathcal{I}} p_j x_{ij} - w_i \ell_i - R_i k_i}{p_i y_i},$$

where $p_i y_i$ denotes gross revenue, $\sum_j p_j x_{ij}$ is expenditure on intermediate inputs, $w_i \ell_i$ is labor compensation, and $R_i k_i$ is the user cost of capital. Thus, the wedge margin is the share of revenue that remains after subtracting the cost of inputs. As such, it can reflect monopolistic markups, taxes, adjustment frictions, or other frictions that drive a wedge between price and marginal cost. By construction, $\mu_i = 1$ under marginal-cost pricing, while $\mu_i > 1$ indicates positive distortions. This approach allows distortions to vary flexibly across sectors, countries, and over time. Negative margins are set to zero, so that the estimated wedges are never below one.

(ii) *Cost of capital*. While revenue, expenditure on intermediate inputs, and labor compensation are observed, the cost of capital must be imputed. In my baseline specification, I use the tax-adjusted rental rate of Hall and Jorgenson (1967) for Spain and Italy,

$$R_i = \sum_j \omega_{ij} \Big(\max\{\rho, 0\} + \delta_j \Big) \frac{1 - \tau z_j}{1 - \tau},$$
$$\rho = \frac{D}{D+E}(b+s)(1-\tau) + \frac{E}{D+E}(b+e) - \pi,$$

where j indexes assets, ω_{ij} is asset j's share in the sector's capital stock, δ_j its depreciation rate, τ the corporate tax rate, and z_j the present value of the tax depreciation allowances on a unit of the asset. The real after-tax cost of capital ρ weights the cost of debt and equity by their shares in the liabilities of non-financial corporations, $D/(D+E)$ and $E/(D+E)$. Debt costs the long-term government bond yield b plus the corporate credit spread s, net of the deduction of interest from taxable profits, and equity costs the bond yield plus the country's historical equity premium e. Expected capital-goods inflation π is the trailing three-year mean of the realized change in the price of the market economy's capital stock.

(iii) *Depreciation rates*. Sector-specific depreciation rates are weighted averages of the BEA asset-specific rates, with weights given by the composition of each sector's capital stock in each year. Appendix D.2 provides all implementation details.

6 Aggregate Productivity in Spain and Italy

This section applies the productivity decomposition derived above to Spain and Italy over 1970–2019. I first show that Spain's productivity decline was substantially smaller than conventional measures imply and decompose its sources, highlighting how earlier structural transformation shaped the aggregate consequences of the post-1995 slowdown in sectoral productivity. I then contrast the Spanish experience with Italy, where the decline arose mainly within sectors. Finally, I compare the two countries directly, showing that the inherited production structure played a much larger role in Spain.

6.1 Aggregate Productivity in Spain

Figure 3 plots three measures of aggregate TFP in Spain from 1970 to 2019. The solid line accounts for both distortions and their propagation through the production network, and thus correctly measures aggregate TFP. The dotted line corrects for distortions but ignores their propagation through the network, correcting each industry's factor payments only by its own wedge. The dashed line reports a TFP series based on the traditional Solow residual, which ignores distortions altogether. The two omissions work in opposite directions: ignoring distortions understates TFP growth, while correcting for them but ignoring the network overstates it. Measured correctly, Spain's productivity stops growing in 1988, as under the traditional Solow residual, but the subsequent decline is only about half as large: 4.7 rather than 8.5 percentage points.

6.2 The Sources of Spain's Productivity Decline

Figure 4 decomposes cumulative aggregate TFP growth based on Theorem 1, with 1970 as the reference period. The decomposition separates three forces. Within-sector productivity captures technological change and the allocation of resources across firms within sectors. Structural change captures how changes in the production structure alter the aggregate effect of sectoral productivity growth. Allocative efficiency captures the contemporaneous effect of reallocating inputs across producers whose inputs earn different social returns. Within-sector productivity was the dominant source of growth, accounting for more than the cumulative gain (+168%). Structural change worked persistently in the opposite direction (−62%): as production shifted away from sectors with high productivity growth and toward sectors with weak or negative productivity growth, sectoral productivity improvements translated into progressively smaller aggregate gains. Allocative efficiency was much less important (−6%).

Figure 3: The Evolution of Aggregate TFP in Spain, 1970–2019.

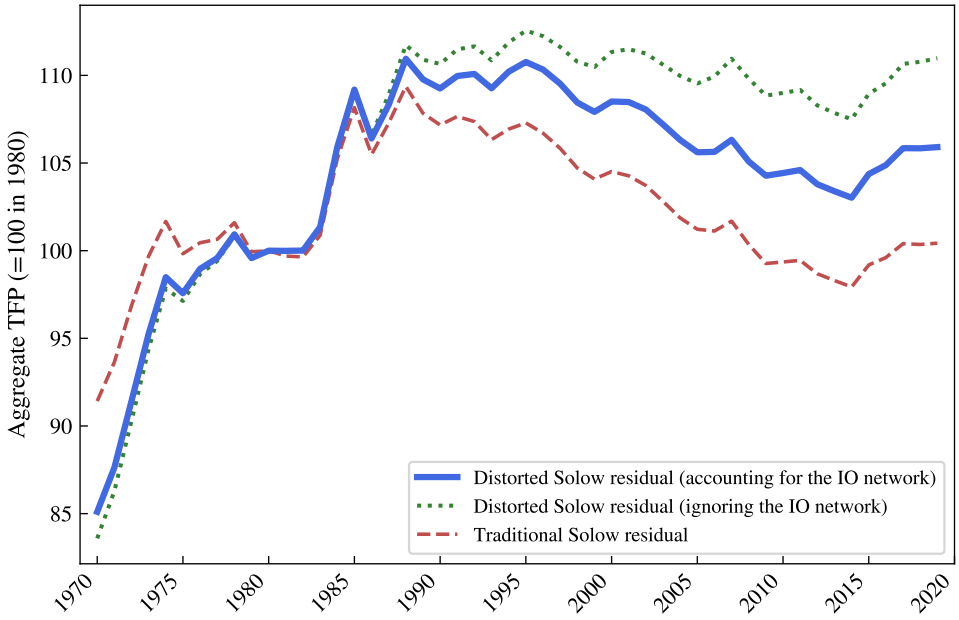


Figure Notes. All three indices are Divisia aggregates of the sectoral data, normalized to 100 in 1980. The distorted Solow residual accounting for the IO network weights factors by their cost-based shares, which scale factor payments by the network-adjusted wedges. The distorted Solow residual ignoring the IO network weights each factor by its payments scaled only by the wedge of the industry that employs it. The traditional Solow residual weights labor by its compensation over value added at basic prices and capital by the residual share.

This decomposition gives a simple account of Spain's productivity path. Until the late 1980s, rapid within-sector productivity growth more than offset the drag from structural change. Thereafter, within-sector productivity stalled: it fell during the housing boom and recovered after the crisis, ending in 2019 close to its level of the late 1980s. Structural change, however, kept subtracting from aggregate productivity. Spain's aggregate TFP decline therefore reflects the disappearance of the within-sector productivity gains that had offset this persistent drag from structural change.

Table 1 provides a more detailed breakdown over two subperiods, 1970–1995 and 1995–2019, separating within-sector productivity and structural change into tradables and non-tradables, and allocative efficiency into factor reallocation, domestic intermediates, and imported intermediate inputs. The first two columns use 1970 as the reference period, while the final column evaluates 1995–2019 using 1995 as the reference period. With 1970 as the reference period, structural change subtracts similar amounts in both subperiods: 7.4 percentage points before 1995 and 6.1 afterward. In both, the drag comes from the declining weight of tradables, whose productivity kept growing. The contribution of tradables to structural change is 6.2 percentage points before 1995 and 7.0 points afterward. What changes sharply is within-sector productivity. It adds 36.6 points over 1970–1995, almost all in tradables, when aggregate TFP rises 26.4 points. Over 1995–2019, by contrast, it makes essentially no contribution, and aggregate TFP

Figure 4: Cumulative Decomposition of Aggregate TFP in Spain, 1970–2019.

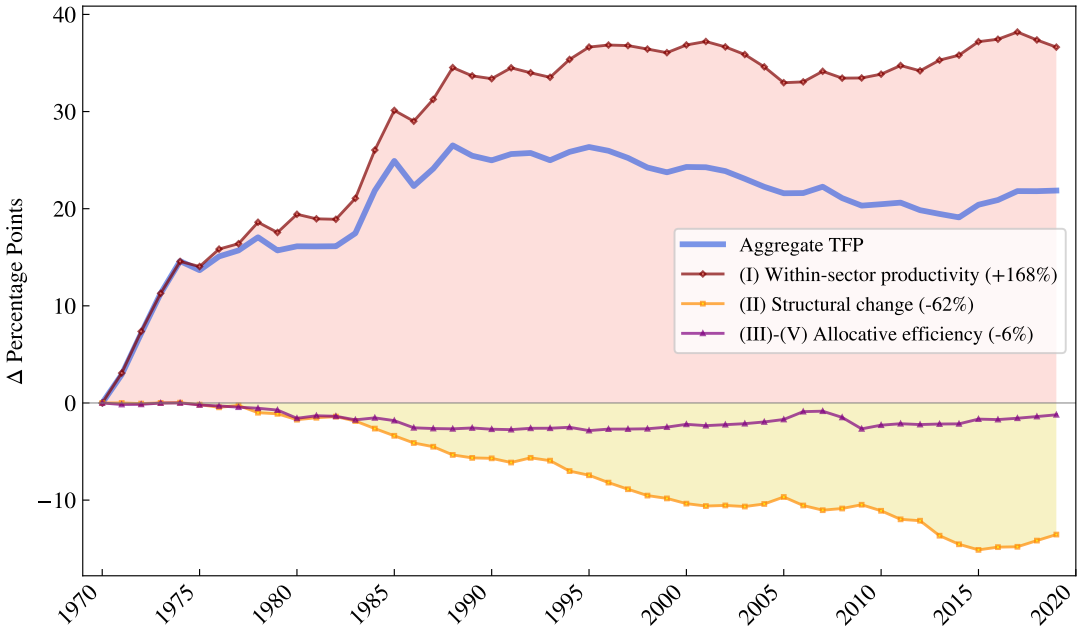


Figure Notes. The figure reports the cumulative decomposition of aggregate TFP given in Theorem 1, with base-period weights and 1970 as the reference period. Each channel's share of cumulative growth over 1970–2019 is rounded to integers in parentheses.

falls 4.5 points. During this period, productivity growth in tradables slowed and non-tradables registered productivity losses. These within-sector patterns are consistent with the firm-level evidence for Spain (Gopinath et al., 2017; García-Santana et al., 2020).

Table 1: Decomposition of Aggregate TFP Growth in Spain, 1970–2019.

Reference period	**1970–1995** $\tau = 1970$	**1995–2019** $\tau = 1970$	**1995–2019** $\tau = 1995$
Δ TFP	**+26.4**	**−4.5**	**−4.5**
Technical efficiency	**+29.2** (+111%)	**−6.1** (+136%)	**−6.1** (+136%)
(I) Within-sector productivity	+36.6 (+139%)	0.0 (0%)	−5.1 (+114%)
Tradables	+36.4 (+138%)	+10.2 (-228%)	+4.2 (-94%)
Non-tradables	+0.2 (+1%)	−10.2 (+228%)	−9.4 (+209%)
(II) Structural change	−7.4 (-28%)	−6.1 (+136%)	−1.0 (+22%)
Tradables	−6.2 (-24%)	−7.0 (+155%)	−1.0 (+21%)
Non-tradables	−1.2 (-5%)	+0.9 (-19%)	0.0 (0%)
Allocative efficiency	**−2.8** (-11%)	**+1.6** (-36%)	**+1.6** (-36%)
(III) Factor reallocation	−2.1 (-8%)	−0.7 (+16%)	−0.7 (+16%)
Labor	−1.5 (-6%)	−0.6 (+13%)	−0.6 (+13%)
Capital	−0.6 (-2%)	−0.2 (+3%)	−0.2 (+3%)
(IV) Domestic-intermediates reallocation	−0.8 (-3%)	+1.3 (-29%)	+1.3 (-29%)
(V) Foreign-intermediates reallocation	0.0 (0%)	+1.1 (-24%)	+1.1 (-24%)

Table Notes. Cumulative changes in percentage points, from Theorem 1 with base-period weights; in parentheses, the share of the change in aggregate TFP in the column, so that when aggregate TFP falls a positive share is a contribution to the decline. The first two columns evaluate terms (I) and (II) with 1970 as reference period; the last column evaluates the 1995–2019 window with 1995 as reference period, as a study that starts in 1995 would. The reference period changes only the split between terms (I) and (II): technical efficiency, their sum, and allocative efficiency, terms (III)–(V), do not depend on it. Tradables are agriculture, mining, and manufacturing; non-tradables are utilities, construction, and services. Each component of term (I) is the productivity growth of that group of industries, and each component of term (II) the change in their cost-based Domar weights since the reference period times their productivity growth; in every column, on average over the window, tradables lost weight and non-tradables gained it.

Allocative efficiency across industries matters much less. It subtracts 2.8 points before 1995, mostly through factor reallocation, and adds 1.6 points afterward, mostly through the reallocation of domestic and foreign intermediate inputs. Within factor reallocation across sectors, labor accounted for most of the losses in both periods.

What standard decompositions miss. A standard first-order decomposition cumulated over time combines within-sector productivity growth and structural change in a single technical-efficiency term, obscuring their separate roles. In Table 1, technical efficiency rises by 29.2 points before 1995 and falls by 6.1 points afterward. Taken alone, these numbers suggest a reversal in technical efficiency. Splitting the term shows instead that structural change was negative in both periods, while within-sector productivity growth was strong enough to offset that drag before 1995 but not afterward.

Why the long horizon matters. Identifying the drag from structural change requires observing the economy before the structural transformation took place. Most studies of Spain begin in 1995, and the last column of Table 1 shows what such a study finds. Evaluated at the production structure of 1995, the sectoral productivity changes of 1995–2019 subtract 5.1 percentage points from aggregate TFP, while subsequent structural change subtracts only 1.0 points. Thus, viewed from 1995, the TFP decline appears to originate within sectors. But by 1995 much of Spain's structural transformation had already occurred. Services already accounted for 64 percent of hours worked, so a study beginning in 1995 necessarily takes this production structure as given. Evaluated at the 1970 structure, the same post-1995 sectoral productivity changes make essentially no contribution to aggregate TFP. The long horizon therefore changes the interpretation of Spain's productivity decline. A sample beginning in 1995 correctly identifies weak productivity performance within sectors, but misses the earlier transformation that greatly amplified its aggregate consequences. Structural transformation did not cause productivity to fall within those sectors; it shifted economic activity toward the sectors in which productivity subsequently performed worst.

Structural transformation and sectoral dynamics. The aggregate patterns of within-sector productivity growth and structural change have a clear sectoral counterpart. Figure 5 shows which industries account for these forces over 1995–2019. The largest productivity losses within sectors (Panel A) come from construction, from hotels and restaurants, and from non-market services, while agriculture, metals, and textiles continue to add to aggregate productivity. The negative contribution of structural change (Panel B) comes largely from the declining weight of these three industries, whose productivity grew, and from the expansion of hotels and restaurants; the remaining in-

dustries partly offset these losses. Appendix C.5 reports the evolution of each industry's share of hours worked, cost-based Domar weights, and productivity.

Figure 5: Industry Contributions in Spain, 1995–2019.

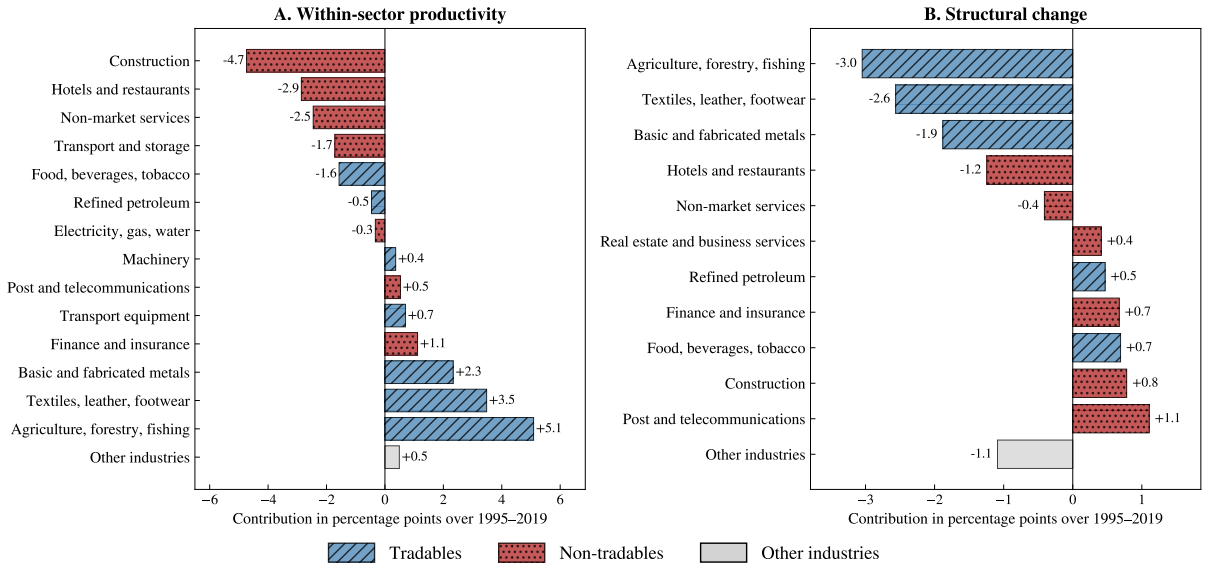


Figure Notes. Exact contribution of each industry with 1970 as the reference period, in percentage points. Industries are ranked from the most negative contribution to the most positive; those contributing less than 0.3 points in absolute value are grouped as other industries and reported at the bottom. Tradables are shown in blue with diagonal lines and non-tradables in red with dots.

Trade integration. Spain's structural transformation unfolded alongside its integration into European and global markets: accession to the European Community in 1986, the completion of the Single Market in 1993, and convergence to the euro in 1999. Between 1970 and 2019, the foreign share of household consumption rose from 2.5 to 13 percent, and the foreign share of intermediate inputs used by Spanish producers from 10 to 23 percent (see Appendix C.6). Trade affects aggregate productivity through two channels. First, it contributes directly through term (V): imported inputs expanded among producers whose inputs earn a social return above their cost, adding 1.1 percentage points to aggregate TFP, almost entirely after 1995. Second, trade can affect aggregate TFP indirectly, by altering the production structure. By shifting domestic demand toward foreign suppliers, and by changing relative prices and incomes, it moves the weights of domestic industries and thus the structural change term. Separating this channel from other sources of structural change would require a general-equilibrium model of trade and structural change and is beyond the scope of this paper.

Robustness. Appendix E shows that the main findings are robust to using alternative measures of distortions and midpoint (Törnqvist) rather than base-period weights. These choices affect the level and composition of measured productivity growth, but not the main conclusions. Across specifications, aggregate TFP is flat between 1988 and

1995 and falls thereafter by less than does the traditional residual. Structural change remains the largest negative term, and factor reallocation remains small.

6.3 The Italian Experience

Italy also experienced a prolonged decline in aggregate productivity, but it began later and had different sources. Figure 6 plots the three measures of aggregate TFP for Italy. All three stop growing in 1995 and remain flat until 2001. By 2019, aggregate TFP is 9 percentage points below its 1995 level, compared with 9.8 under the traditional Solow residual and 7.6 under the series that ignores the production network. The three measures are closer than in Spain because Italy's labor and capital inputs grew much more slowly after 1995.

Figure 6: Aggregate TFP in Italy, 1970–2019.

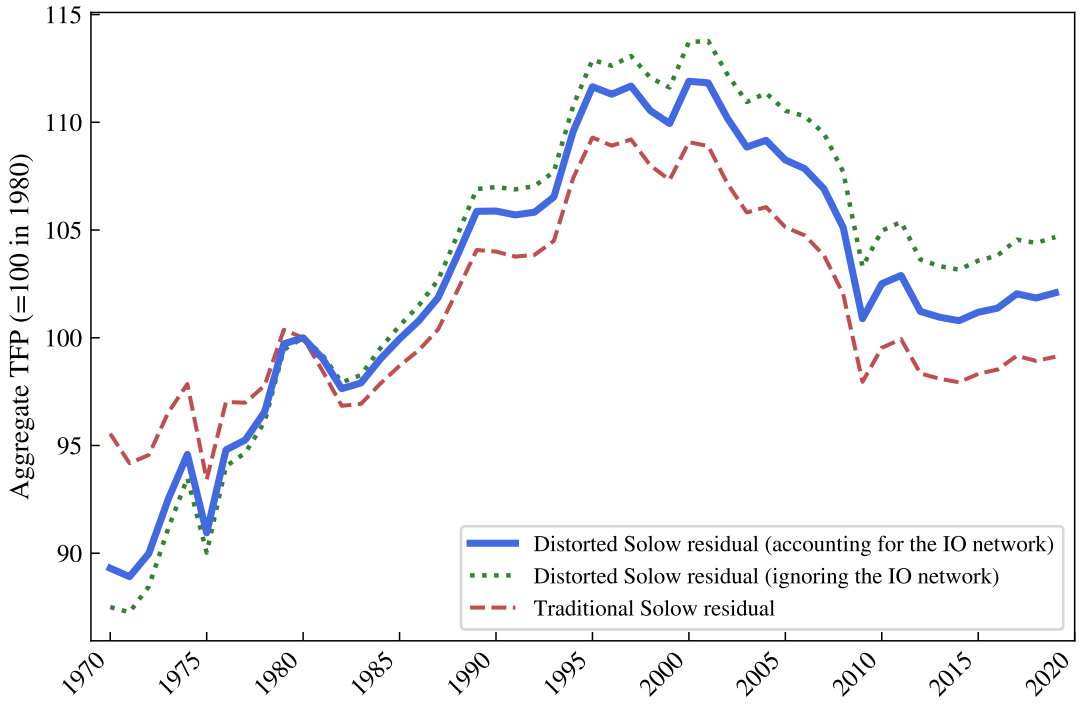


Figure 7 and Table 2 decompose aggregate TFP. Over 1970–2019, the pattern resembles Spain's: within-sector productivity accounts for more than the cumulative gain (+163%), structural change subtracts from it throughout (−72%), and allocative efficiency plays a relatively minor role (+9%). Within-sector productivity grew until 1995, stalled until 2000, and fell over the following decade, so Italy's subsequent productivity decline originated primarily within sectors. With 1970 as the reference period, within-sector productivity subtracts 6.2 points over 1995–2019, almost all of it in non-tradables (6.0 points), compared with essentially no contribution in Spain. Structural change

subtracts 6.7 points before 1995 and 3.0 points after. Imported inputs, which added 1.1 points in Spain, subtracted 0.5 in Italy. Unlike in Spain, the post-1995 decline is therefore already largely visible when sectoral productivity growth is evaluated at the 1970 production structure.

Figure 7: Cumulative Decomposition of Aggregate TFP in Italy, 1970–2019.

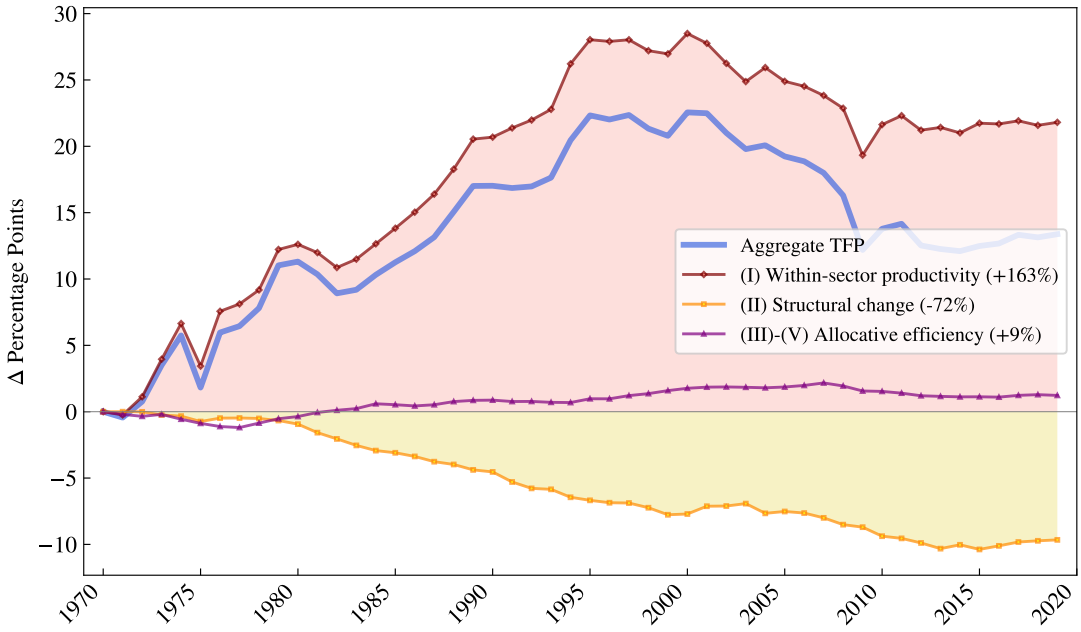


Table 2: Decomposition of Aggregate TFP Growth in Italy, 1970–2019.

Reference period	**1970–1995** $\tau = 1970$	**1995–2019** $\tau = 1970$	**1995–2019** $\tau = 1995$
Δ TFP	**+22.3**	**−9.0**	**−9.0**
Technical efficiency	**+21.4** (+96%)	**−9.2** (+103%)	**−9.2** (+103%)
(I) Within-sector productivity	+28.0 (+126%)	−6.2 (+70%)	−7.3 (+81%)
Tradables	+28.0 (+125%)	−0.2 (+3%)	−0.7 (+8%)
Non-tradables	0.0 (0%)	−6.0 (+67%)	−6.5 (+73%)
(II) Structural change	−6.7 (-30%)	−3.0 (+33%)	−1.9 (+21%)
Tradables	−5.6 (-25%)	−1.0 (+12%)	−0.5 (+6%)
Non-tradables	−1.1 (-5%)	−1.9 (+22%)	−1.4 (+15%)
Allocative efficiency	**+1.0** (+4%)	**+0.3** (-3%)	**+0.3** (-3%)
(III) Factor reallocation	+0.2 (+1%)	−0.3 (+3%)	−0.3 (+3%)
Labor	+0.4 (+2%)	−0.3 (+3%)	−0.3 (+3%)
Capital	−0.3 (-1%)	0.0 (0%)	0.0 (0%)
(IV) Domestic-intermediates reallocation	+0.5 (+2%)	+1.0 (-12%)	+1.0 (-12%)
(V) Foreign-intermediates reallocation	+0.3 (+1%)	−0.5 (+6%)	−0.5 (+6%)

Table Notes. Cumulative changes in percentage points, from Theorem 1 with base-period weights; in parentheses, the share of the change in aggregate TFP in the column, so that when aggregate TFP falls a positive share is a contribution to the decline. The first two columns evaluate terms (I) and (II) with 1970 as reference period; the last column evaluates the 1995–2019 window with 1995 as reference period, as a study that starts in 1995 would. The reference period changes only the split between terms (I) and (II): technical efficiency, their sum, and allocative efficiency, terms (III)–(V), do not depend on it. Tradables are agriculture, mining, and manufacturing; non-tradables are utilities, construction, and services. Each component of term (I) is the productivity growth of that group of industries, and each component of term (II) the change in their cost-based Domar weights since the reference period times their productivity growth; in every column, on average over the window, tradables lost weight and non-tradables gained it.

Figure 8 shows which industries account for these forces. Within sectors (Panel A), the largest productivity losses come from construction, non-market services, and real estate and business services. Manufacturing, which added 5.1 points in Spain, subtracts 1.9 in Italy. The only sizable gains come from wholesale and retail trade and agriculture. Structural change (Panel B) comes mostly from real estate and business services, whose weight rose as their productivity fell. These patterns are consistent with the rise in misallocation across firms within Italian industries documented by Calligaris et al. (2018), and with the slow adoption of information technology and of the management practices it requires (Pellegrino and Zingales, 2017; Schivardi and Schmitz, 2020).

Figure 8: Contributions of Each Industry to Within-Sector Productivity and Structural Change, Italy, 1995–2019.

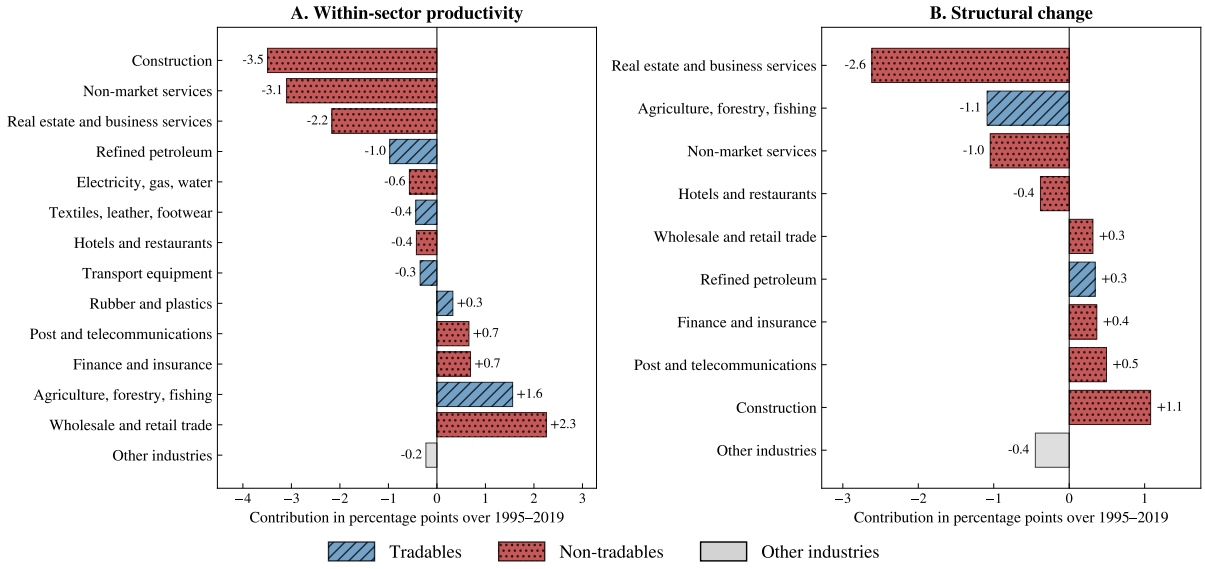


Figure Notes. As in Figure 5, for Italy; the contributions in each panel sum to the term in the second column of Table 2.

6.4 The Role of the Inherited Production Structure

The comparison between Spain and Italy highlights a broader difference in how sectoral productivity growth translated into aggregate TFP after 1995. To make this difference transparent, Figure 9 calculates the contribution to aggregate TFP of the sectoral productivity changes observed over 1995–2019 using the production structure of each year from 1970 to 2018. The figure hence asks: how costly would the observed sectoral productivity changes have been under the production structure observed at different points in time?

The answer differs sharply across the two countries. In Spain, the 1995–2019 sectoral productivity changes contribute essentially nothing to aggregate TFP when evaluated

Figure 9: Contribution of Within-Sector Productivity Growth, 1995–2019.

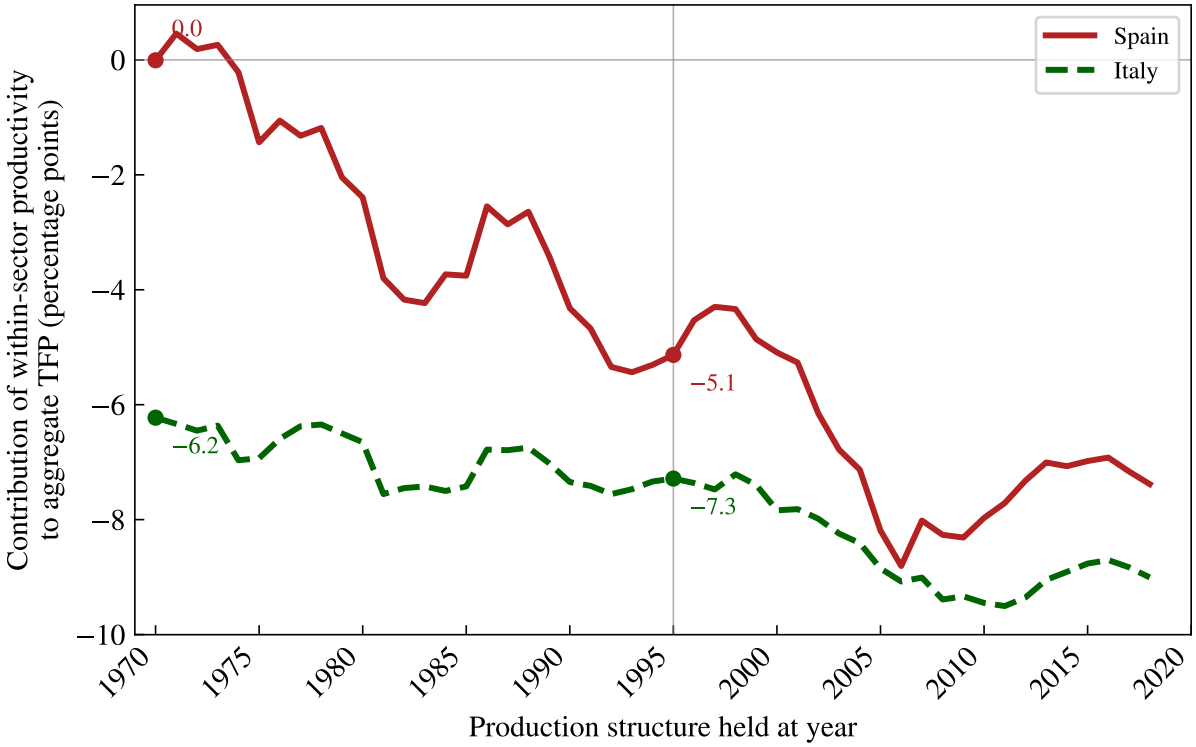


Figure Notes. Term (I) of Theorem 1 over 1995–2019, in percentage points, with the year on the horizontal axis as reference period. Technical efficiency over the window, the sum of terms (I) and (II), does not depend on the reference period, so structural change relative to any year is that sum minus the value plotted. The dots mark the values at 1970 and 1995, those of the second and third columns of Tables 1 and 2. The series ends in 2018, the last year whose weights enter the decomposition. The vertical line marks 1995.

at the 1970 production structure, but subtract 5.1 percentage points at the 1995 structure. The figure therefore illustrates why the starting date of the analysis matters. A study beginning in 1995 necessarily takes the production structure inherited in that year as given and cannot observe how earlier structural transformation changed the contribution of subsequent sectoral productivity growth to aggregate TFP. In Spain, much of that change in exposure had already taken place before the productivity decline began. By the 1988 peak, roughly half of the increase relative to the 1970 structure had already occurred, and the cost kept rising after 1995, to 8.0 points at the 2007 production structure. The post-1995 slowdown therefore looks predominantly like a within-sector productivity decline when viewed from 1995, even though the aggregate cost of that slowdown had been substantially shaped by structural transformation that occurred earlier.

Italy looks very different. The same exercise gives a contribution of −6.2 percentage points at the 1970 structure and −7.3 points at the 1995 structure. Sectoral productivity losses would therefore have generated a large aggregate decline under either production structure. Despite a substantial shift toward services, Italy's production structure changed its exposure to the post-1995 productivity losses much less than Spain's did. A study beginning in 1995 therefore misses much less of the relevant change in exposure

in Italy. The contrast reinforces the diagnosis from Tables 1 and 2: Italy's decline arose primarily within sectors, whereas in Spain much of the aggregate cost of the subsequent slowdown reflected changes in the production structure that had already occurred before 1995.

Figure 9 thus illustrates the delayed effect of structural change emphasized by the decomposition. Changes in the production structure need not lower aggregate productivity when they occur. Instead, they can change how strongly subsequent sectoral productivity growth contributes to aggregate TFP. A decomposition that begins after those structural changes have occurred takes their consequences as part of the initial production structure and cannot separately recover them.

7 Conclusion

In this paper, I reexamine the evolution of aggregate productivity in Spain and Italy between 1970 and 2019 using a growth-accounting framework for open and distorted economies with input-output linkages and aggregate series built from industry data and benchmarked to the national accounts. Measured with cost-based factor shares that account for the propagation of distortions through the entire production network, Spain's productivity stops growing in 1988 but does not begin to decline until 1995, and its decline is about half as large as under the traditional Solow residual. In Italy, correcting for network-adjusted distortions raises measured productivity but leaves the timing of the decline essentially unchanged.

To understand the sources of these productivity declines, I develop a cumulative decomposition of aggregate TFP growth that separates within-sector productivity growth from structural change and from the reallocation of factors and of domestic and foreign intermediate inputs. The comparison between Spain and Italy is revealing. In Spain, within-sector productivity growth had long offset a persistent drag from structural change. After the late 1980s it stalled and turned negative in services and construction, sectors whose weight had risen since 1970, while it continued in industries whose weight had declined. Because much of the transformation occurred before 1995, studies beginning then take the inherited production structure as given and miss how earlier structural transformation altered the aggregate consequences of subsequent sectoral productivity changes. In Italy, by contrast, productivity fell broadly within sectors after 1995, and structural change accounts for a third of the decline, so the diagnosis is much less sensitive to the starting production structure. In neither country did the reallocation of inputs across industries contribute much. These differences caution against single-cause explanations of Southern Europe's productivity declines, even when aggregate productivity trends appear similar across countries.

More generally, the costs of structural change need not appear when resources move. In Spain, the shift toward services before 1995 raised the weight of sectors in which productivity later fell, so the post-1995 slowdown lowered aggregate TFP by more than it would have under the production structure of 1970. The decomposition measures these consequences of observed structural change, not the forces that caused it.

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Appendices

"Rethinking the Southern European Productivity Declines"

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October 31, 2026

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A Omitted Proofs

A.1 Proposition 1 (Distortion Bias in the Traditional Solow Residual)

Proposition 1 (Bias in Traditional Solow Residual)**.** *Assume that there is a representative producer, producing output Y using technology $Y = AF(L_1, \ldots, L_N)$, where $A > 0$ is a productivity parameter and F is a constant-returns-to-scale production function which satisfies standard regularity conditions (i.e., continuity, differentiability, and Inada conditions). Moreover, assume that the representative producer is monopolistic and prices its output charging a markup $\mu \geq 1$ over marginal cost MC; that is, $P = \mu \times MC$. Then,*

(i) The traditional Solow residual, defined as output growth minus revenue-based input growth, can be written as

$$\mathrm{d}\log Y - \sum_{f \in \mathcal{F}} \Lambda_n \mathrm{d}\log L_n = \mathrm{d}\log A + \left(\frac{\mu - 1}{\mu}\right)\Big\{\mathrm{d}\log Y - \mathrm{d}\log A\Big\}, \tag{2}$$

where $\Lambda_n \equiv w_n L_n/(PY)$ is the revenue-based share of input n.

(ii) With labor and capital as the only inputs, if the capital share is computed residually as $1 - \Lambda_L$, so that revenue shares sum to one, the traditional Solow residual can be written as

$$\mathrm{d}\log Y - \Lambda_L \mathrm{d}\log L - (1 - \Lambda_L)\mathrm{d}\log K = \mathrm{d}\log A + (\mu - 1)\,\Lambda_L(\mathrm{d}\log L - \mathrm{d}\log K), \tag{3}$$

Proof. Taking logs and differentiating both sides of the production function with respect to time yields

$$\mathrm{d}\log Y = \mathrm{d}\log A + \sum_{n \in N}\left(\frac{\partial F}{\partial L_n}\frac{L_n}{F}\right)\mathrm{d}\log L_n, \tag{A.1}$$

where $\mathrm{d}\log X \equiv \frac{\mathrm{d}\log X}{\mathrm{d}t}$.

The cost-minimization problem of the representative producer may be written as

$$\begin{aligned} \min_{\boldsymbol{L} \geq \mathbf{0}} \quad & C(\boldsymbol{w}) \equiv \sum_{n \in N} w_n L_n \\ \text{s.t.} \quad & Y = AF(L_1, \ldots, L_N) \geq \overline{Y}, \end{aligned}$$

where w_n is the price of input n, L_n is the quantity demanded of input n, and $\overline{Y}$ is a minimal-output constraint.

Letting λ denote the Lagrange multiplier associated with the constraint, a generic first-order condition for an interior demand reads as:

$$w_n = \lambda A \frac{\partial F}{\partial L_n} \qquad \Longleftrightarrow \qquad w_n L_n = \lambda A \frac{\partial F}{\partial L_n} L_n.$$

By the Envelope Theorem, it follows that $\lambda = MC$. Hence, we have

$$\begin{aligned} w_n L_n &= MC \times A \frac{\partial F}{\partial L_n} L_n \\ &= \frac{\mu}{P} \times A \frac{\partial F}{\partial L_n} L_n \qquad \text{(using the price-setting condition)} \end{aligned}$$

Rearranging and doing simple algebraic manipulations, it follows that

$$\frac{\partial F}{\partial L_n} \frac{L_n}{F} = \mu \times \frac{w_n L_n}{PY} \equiv \tilde{\Lambda}_n,$$

where $\tilde{\Lambda}_n \equiv \mu \times w_n L_n/(PY)$ and $\Lambda_n \equiv w_n L_n/(PY)$ are the cost- and revenue-based shares of input n, respectively.

The fact that $\tilde{\Lambda}_n$ is the cost-based share of input n follows from noting that with a CRS production function, the profit rate s_π is given by

$$s_\pi = 1 - \frac{1}{\mu},$$

which implies that the markup can be written as

$$\mu = (1 - s_\pi)^{-1} = \left(1 - \frac{\sum_{k \in N} w_k L_k}{PY}\right)^{-1} = \frac{PY}{\sum_{k \in N} w_k L_k}.$$

Hence,

$$\tilde{\Lambda}_n = \mu \times \frac{w_n L_n}{PY} = \frac{w_n L_n}{\sum_{k \in N} w_k L_k},$$

which is the cost share of input n.

Plugging cost-based shares into expression (A.1) and rearranging, we obtain that the distorted Solow residual correctly measures TFP growth:

$$\mathrm{d} \log Y - \sum_{n \in N} \tilde{\Lambda}_n \mathrm{d} \log L_n = \mathrm{d} \log A. \tag{A.2}$$

To obtain the bias associated with the traditional Solow residual, defined as output growth minus revenue-weighted input growth, we can algebraically manipulate ex-

pression (A.2). That is, we can write

$$\begin{aligned} \mathrm{d}\log Y - \sum_{n\in N}\Lambda_n \mathrm{d}\log L_n &= \mathrm{d}\log A + \sum_{n\in N}\left(\tilde{\Lambda}_n - \Lambda_n\right)\mathrm{d}\log L_n \\ &= \mathrm{d}\log A + (\mu-1)\sum_{n\in N}\Lambda_n \mathrm{d}\log L_n. \end{aligned}$$

The equation in the main text is the discrete-time analog of this last expression.

For part (ii), CRS imply $\tilde{\Lambda}_L + \tilde{\Lambda}_K = 1$, so that $\tilde{\Lambda}_K - (1-\Lambda_L) = -(\tilde{\Lambda}_L - \Lambda_L)$ and

$$\begin{aligned} \mathrm{d}\log Y - \Lambda_L \mathrm{d}\log L - (1-\Lambda_L)\mathrm{d}\log K &= \mathrm{d}\log A + (\tilde{\Lambda}_L - \Lambda_L)\mathrm{d}\log L + (\tilde{\Lambda}_K - 1 + \Lambda_L)\mathrm{d}\log K \\ &= \mathrm{d}\log A + (\mu-1)\Lambda_L(\mathrm{d}\log L - \mathrm{d}\log K). \end{aligned}$$

Equation (3) in the main text is the discrete-time analog. □

A.2 Propositions 2 and 3 (Network Bias in Distorted Solow Residual)

Proposition 2 (TFP Measurement in a Vertical Chain). *Assume that sector 2 produces an intermediate input from labor with technology $y_2 = A_2 L$, and sector 1 produces the final good from capital and the intermediate input with technology $y_1 = A_1 K^\alpha x^{1-\alpha}$, $0 < \alpha < 1$, where all of sector 2's output is sold to sector 1, $x = y_2$. Each sector i prices its output at a markup $\mu_i \geq 1$ over marginal cost, and factor markets are competitive, with prices w_K and w_L. Then:*

(i) Real GDP is $Y = y_1 = AK^\alpha L^{1-\alpha}$, with $A := A_1 A_2^{1-\alpha}$.

(ii) The shares of capital and labor in nominal GDP are $\Lambda_K := w_K K/(PY) = \alpha/\mu_1$ and $\Lambda_L := w_L L/(PY) = (1-\alpha)/(\mu_1\mu_2)$.

(iii) With the factor shares that correct each factor's share only for the markup of the sector that employs it, $\hat{\Lambda}_K := \mu_1\Lambda_K$ and $\hat{\Lambda}_L := \mu_2\Lambda_L$,

$$\underbrace{\Delta\log Y - \hat{\Lambda}_K\Delta\log K - \hat{\Lambda}_L\Delta\log L}_{\substack{\text{Distorted Solow residual}\\ \text{(ignoring the IO network)}}} = \underbrace{\Delta\log A}_{\text{TFP growth}} + \underbrace{(1-\alpha)\left(1-\frac{1}{\mu_1}\right)\Delta\log L}_{\text{Network bias}}. \tag{6}$$

(iv) With the network-adjusted cost-based shares, $\tilde{\Lambda}_K := \mu_1\Lambda_K$ and $\tilde{\Lambda}_L := \mu_1\mu_2\Lambda_L$,

$$\underbrace{\Delta\log Y - \tilde{\Lambda}_K\Delta\log K - \tilde{\Lambda}_L\Delta\log L}_{\substack{\text{Distorted Solow residual}\\ \text{(accounting for the IO network)}}} = \underbrace{\Delta\log A}_{\text{TFP growth}}. \tag{7}$$

Proof. (i) follows from substituting $x = y_2 = A_2 L$ into sector 1's technology.

To establish (ii), note that with constant returns to scale, each sector's cost is its revenue divided by its markup, and with Cobb–Douglas technologies each input receives a constant fraction of that cost. Sector 1's revenue is nominal GDP, PY, and its cost is PY/μ_1, of which capital receives a fraction α and the intermediate input a fraction $1-\alpha$. Hence $w_K K = \alpha PY/\mu_1$, and sector 2's revenue is $(1-\alpha)PY/\mu_1$. Sector 2's cost is its revenue divided by μ_2, all of it paid to labor, so $w_L L = (1-\alpha)PY/(\mu_1\mu_2)$.

(iii) By (ii), $\hat{\Lambda}_K = \alpha$ and $\hat{\Lambda}_L = (1-\alpha)/\mu_1$. Taking log changes of (i), $\Delta \log Y = \Delta \log A + \alpha \Delta \log K + (1-\alpha)\Delta \log L$, so that

$$\Delta \log Y - \hat{\Lambda}_K \Delta \log K - \hat{\Lambda}_L \Delta \log L = \Delta \log A + \left(1 - \alpha - \frac{1-\alpha}{\mu_1}\right)\Delta \log L.$$

(iv) By (ii), $\tilde{\Lambda}_K = \alpha$ and $\tilde{\Lambda}_L = 1-\alpha$, the elasticities of real GDP with respect to capital and labor in (i), and the result follows from the log change of (i). □

The sign of the network bias does not depend on the vertical chain. Consider a closed economy with N producers. Producer i has a constant-returns-to-scale technology $y_i = A_i F_i(x_{i1}, \ldots, x_{iN}, L_{i1}, \ldots, L_{iF})$, where x_{ik} is its use of good k and L_{if} its use of primary factor f, prices its output at a markup $\mu_i \geq 1$ over marginal cost, and pays competitive factor prices w_f. Let $q_i := y_i - \sum_k x_{ki}$ be its output sold to final uses, real GDP Y the Divisia index of final uses and $PY = \sum_i p_i q_i$ nominal GDP. Let $\lambda_i := p_i y_i/(PY)$ be the revenue-based Domar weight of producer i, $\Omega_{ik} := p_k x_{ik}/(p_i y_i)$ and $\Omega_{if} := w_f L_{if}/(p_i y_i)$ its revenue-based input shares, $\tilde{\lambda}' = b'(I - \mathrm{diag}(\mu)\Omega)^{-1}$ the cost-based Domar weights, with $b_i := p_i q_i/(PY)$, and $\tilde{\Lambda}_f := \sum_i \tilde{\lambda}_i \mu_i \Omega_{if}$ the cost-based factor shares of Definition 1, so that $\Delta \log \mathrm{TFP} := \Delta \log Y - \sum_f \tilde{\Lambda}_f \Delta \log L_f$. Finally, let $\phi_{ik} := x_{ki}/y_i$ be the share of producer i's output sold to producer k, $\phi_i^Y := q_i/y_i$ the share sold to final uses, and $\kappa_i := \mu_i \tilde{\lambda}_i/\lambda_i$ the network-adjusted wedge of Definition 2.

Proposition 3 (Network Bias in a General Input-Output Network). *(i) The cost-based factor shares scale each producer's factor payments by its network-adjusted wedge,*

$$\tilde{\Lambda}_f = \sum_{i=1}^{N} \kappa_i \frac{w_f L_{if}}{PY}, \qquad \kappa_i = \mu_i \Big[\phi_i^Y + \sum_{k=1}^{N} \phi_{ik} \kappa_k \Big].$$

(ii) With the factor shares that correct each producer's payments only for its own markup, $\hat{\Lambda}_f := \sum_i \mu_i w_f L_{if}/(PY)$,

$$\underbrace{\Delta \log Y - \sum_{f=1}^{F} \hat{\Lambda}_f \Delta \log L_f}_{\substack{\text{Distorted Solow residual} \\ \text{(ignoring the IO network)}}} = \underbrace{\Delta \log \text{TFP}}_{\text{TFP growth}} + \underbrace{\sum_{f=1}^{F} \sum_{i=1}^{N} (\kappa_i - \mu_i) \frac{w_f L_{if}}{PY} \Delta \log L_f}_{\text{Network bias}},$$

where $\kappa_i - \mu_i = \mu_i \sum_k \phi_{ik}(\kappa_k - 1) \geq 0$. The network bias is therefore non-negative whenever factors grow, and it is zero when every producer sells only to final uses or to producers with $\kappa_k = 1$.

(iii) With the factor shares that measure each factor's cost over total factor cost, $s_f := w_f L_f / \sum_g w_g L_g$, the Solow residual equals TFP growth for all factor growth rates only if the average network-adjusted wedges of the factors, $\bar{\kappa}_f := \tilde{\Lambda}_f / \Lambda_f = \sum_i s_{fi} \kappa_i$, are equal across factors. In Proposition 2, $\kappa_1 = \mu_1$ and $\kappa_2 = \mu_1 \mu_2$, and thus the bias is

$$-\frac{\alpha(1-\alpha)(\mu_2 - 1)}{1 - \alpha + \alpha \mu_2} \Big(\Delta \log K - \Delta \log L \Big),$$

negative when capital grows faster than labor.

Proof. (i) Since $\lambda_i \Omega_{if} = w_f L_{if}/(PY)$, the definition of κ_i gives $\tilde{\Lambda}_f = \sum_i (\mu_i \tilde{\lambda}_i / \lambda_i) \lambda_i \Omega_{if} = \sum_i \kappa_i w_f L_{if}/(PY)$. The cost-based Domar weights satisfy $\tilde{\lambda}' = b' + \tilde{\lambda}' \text{diag}(\mu) \Omega$, that is, $\tilde{\lambda}_i = b_i + \sum_k \tilde{\lambda}_k \mu_k \Omega_{ki}$. Multiplying by μ_i/λ_i,

$$\kappa_i = \mu_i \Big[\frac{b_i}{\lambda_i} + \sum_k \frac{\mu_k \tilde{\lambda}_k}{\lambda_k} \frac{\lambda_k \Omega_{ki}}{\lambda_i} \Big] = \mu_i \Big[\phi_i^Y + \sum_k \kappa_k \phi_{ik} \Big],$$

because $b_i/\lambda_i = q_i/y_i$ and $\lambda_k \Omega_{ki}/\lambda_i = p_i x_{ki}/(p_i y_i)$. This is Lemma 1 for a closed economy.

(ii) Adding and subtracting $\sum_f \tilde{\Lambda}_f \Delta \log L_f$,

$$\Delta \log Y - \sum_f \hat{\Lambda}_f \Delta \log L_f = \Delta \log \text{TFP} + \sum_f \Big(\tilde{\Lambda}_f - \hat{\Lambda}_f \Big) \Delta \log L_f,$$

and $\tilde{\Lambda}_f - \hat{\Lambda}_f = \sum_i (\kappa_i - \mu_i) w_f L_{if}/(PY)$ by (i). Since $\phi_i^Y + \sum_k \phi_{ik} = 1$, (i) gives $\kappa_i - \mu_i = \mu_i \sum_k \phi_{ik}(\kappa_k - 1)$, and $\kappa_k \geq 1$ when all markups are at least one.

(iii) Since $\sum_g w_g L_g/(PY) = \sum_g \Lambda_g$, $s_f = \Lambda_f / \sum_g \Lambda_g$, while $\tilde{\Lambda}_f = \bar{\kappa}_f \Lambda_f$ by (i). In a closed economy the cost-based factor shares sum to one (Baqaee and Farhi, 2020). If $\bar{\kappa}_f = \bar{\kappa}$ for all f, then $\bar{\kappa} = 1/\sum_g \Lambda_g$ and $\tilde{\Lambda}_f = s_f$. Otherwise $\tilde{\Lambda}_f / s_f = \bar{\kappa}_f \sum_g \Lambda_g$ differs across factors, and the residual computed with s_f differs from $\Delta \log \text{TFP}$ by $\sum_f (\tilde{\Lambda}_f - s_f) \Delta \log L_f$, which is not zero for all factor growth rates. In the vertical chain, sector 1 sells only to final uses and sector 2 only to sector 1, so $\kappa_1 = \mu_1$ and $\kappa_2 = \mu_2 \kappa_1$. By Proposition 2(ii), $s_L = \Lambda_L/(\Lambda_K + \Lambda_L) = (1-\alpha)/(1-\alpha+\alpha\mu_2)$ and $s_K = 1 - s_L$, and the bias $(\tilde{\Lambda}_K - s_K)\Delta \log K + (\tilde{\Lambda}_L - s_L)\Delta \log L$ equals the expression in (iii). □

In the vertical chain, Proposition 3(ii) reduces to Proposition 2(iii): the bias is $(\kappa_2 - \mu_2)\Lambda_L \Delta \log L = (1-\alpha)(1 - 1/\mu_1)\Delta \log L$. In the general economy, $\Delta \log \text{TFP}$ includes allocative efficiency, which Section 4 separates from technical efficiency. In the vertical chain nothing can be reallocated, so it equals the growth of technology. In an open economy the sums over producers run over domestic producers, ϕ_i^Y includes exports, and the cost-based factor shares sum to less than one, so that the cost shares of (iii) are biased even when the average network-adjusted wedges of the factors are equal.

A.3 Proposition 4 (Trade-Induced Reallocation and TFP)

Proof. Part (i) follows from equation (B.6). For part (ii), with two sectors the weights sum to one in every period, so $\lambda_{1,t-1} - \lambda_{1,\tau} = -(\lambda_{2,t-1} - \lambda_{2,\tau})$ and the structural-change term in equation (B.7) equals $(\lambda_{2,t-1} - \lambda_{2,\tau})(\Delta \log A_{2,t} - \Delta \log A_{1,t})$, which is strictly negative whenever the sector whose weight rose has the lower productivity growth. For part (iii), let the original equilibrium (at $t-1$) be parametrized by

$$(p_1^w, p_2^w, \alpha, \phi, L_\ell, A_1, A_2) = (1, 1, 0.5, 0.7, 1, 1, 1).$$

Then $L_{\ell,1,t-1} = L_{\ell,2,t-1} = \lambda_{1,t-1} = \lambda_{2,t-1} = 0.5$. Consider two alternative transitions:

(a) No terms-of-trade change. Maintain all parameters the same, except technological parameters. Let $(A_{1,t}, A_{2,t}) = (0.99, 1.02)$. Using the equilibrium characterization, $\Delta \log \text{TFP}_t = +0.0049 > 0$ and $\Delta \log U_t = +0.0052 > 0$.

(b) Terms-of-trade change. Maintain all parameters the same, except technological parameters and the terms of trade. Let $(A_{1,t}, A_{2,t}) = (0.99, 1.02)$ and $p_2^w = 0.8$. Using the equilibrium characterization, labor reallocates toward sector 1, the sector with negative productivity growth, $(L_{\ell,1,t}, L_{\ell,2,t}) = (0.656, 0.344)$, aggregate TFP growth on impact is $\Delta \log \text{TFP}_t = +0.0049 > 0$, as in (a), by part (i), and

welfare rises by $\Delta \log U_t = +0.0202 > 0.0052$. If productivity keeps changing at the same rates in $t+1$ with world prices unchanged, aggregate TFP growth is $+0.0002$, against $+0.0049$ under the pre-shock production structure, and the structural-change term is $-0.0046 < 0$, as part (ii) states.

This verifies (iii) and illustrates (i) and (ii). □

A.4 Lemma 1 (Cumulative Wedges)

Proof. The cost-based Domar weights of domestic producers are computed with the local cost-based Leontief inverse, which written out reads

$$\tilde{\lambda}_{i,t}^{Y_c} = \Omega_{Y_c,i,t} + \sum_{k \in \mathcal{I}_c} \tilde{\lambda}_{k,t}^{Y_c} \mu_{k,t} \Omega_{ki,t}, \qquad i \in \mathcal{I}_c, \tag{A.3}$$

where $\Omega_{Y_c,i,t} = p_{i,t} q_{ci,t} / (P_{Y_c,t} Y_{c,t})$ and $q_{ci,t} = y_{i,t} - \sum_{k \in \mathcal{I}_c} x_{ki,t}$ is producer i's output net of sales to domestic producers. The revenue-based weights satisfy the same identity without the wedges, $\lambda_{i,t}^{Y_c} = \Omega_{Y_c,i,t} + \sum_{k \in \mathcal{I}_c} \lambda_{k,t}^{Y_c} \Omega_{ki,t} = p_{i,t} y_{i,t} / (P_{Y_c,t} Y_{c,t})$. Dividing (A.3) by $\lambda_{i,t}^{Y_c}$ and using $\Omega_{Y_c,i} / \lambda_i^{Y_c} = q_{ci} / y_i = \phi_i^Y$, $\lambda_k^{Y_c} \Omega_{ki} / \lambda_i^{Y_c} = x_{ki} / y_i = \phi_{ik}$, and $\tilde{\lambda}_k^{Y_c} \mu_k / \lambda_k^{Y_c} = \kappa_k$,

$$\frac{\tilde{\lambda}_{i,t}^{Y_c}}{\lambda_{i,t}^{Y_c}} = \phi_{i,t}^Y + \sum_{k \in \mathcal{I}_c} \phi_{ik,t} \kappa_{k,t}.$$

Multiplying by $\mu_{i,t}$ gives the recursion. If $\mu_{i,t} = 1$ for all i, (A.3) coincides with the identity for the revenue-based weights, so $\tilde{\lambda}^{Y_c} = \lambda^{Y_c}$ and $\kappa_i = 1$. If $\mu_{i,t} \geq 1$ for all i, iterating (A.3) writes $\tilde{\lambda}_i^{Y_c}$ as a sum of products of non-negative terms, each weakly larger than its counterpart in the iteration for $\lambda_i^{Y_c}$, so $\tilde{\lambda}_i^{Y_c} \geq \lambda_i^{Y_c}$ and $\kappa_i \geq \mu_i$. If $\phi_i^Y = 1$, the recursion gives $\kappa_i = \mu_i$. □

A.5 Theorem 1 (Decomposition of Aggregate TFP Growth)

Proof. Nominal GDP of country c equals the value added by domestic producers:

$$P_{Y_c} Y_c = \sum_{i \in \mathcal{I}_c} p_i y_i - \sum_{i \in \mathcal{I}_c} \sum_{j \in \mathcal{I}} p_j x_{ij}.$$

Real GDP growth is measured by the Divisia quantity index with base-period weights: the change in nominal GDP at period-$(t-1)$ prices, divided by nominal GDP, with each

quantity change expressed as a log change,

$$\Delta \log Y_{c,t} = \sum_{i \in \mathcal{I}_c} \lambda_{i,t-1}^{Y_c} \Delta \log y_{i,t} - \sum_{i \in \mathcal{I}_c} \sum_{j \in \mathcal{I}} \lambda_{i,t-1}^{Y_c} \Omega_{ij,t-1} \Delta \log x_{ij,t}, \tag{A.4}$$

where

$$\lambda_{i,t-1}^{Y_c} \equiv \frac{p_{i,t-1} y_{i,t-1}}{P_{Y_c,t-1} Y_{c,t-1}}, \qquad \Omega_{ij,t-1} \equiv \frac{p_{j,t-1} x_{ij,t-1}}{p_{i,t-1} y_{i,t-1}}.$$

Expression (A.4) is the measurement convention for real GDP adopted throughout, so it holds as an identity. Each sector $i \in \mathcal{I}_c$ produces gross output according to a constant-returns-to-scale production function

$$y_{i,t} = A_{i,t} F_i(\{L_{fi,t}\}_{f \in \mathcal{F}_c}, \{x_{ij,t}\}_{j \in \mathcal{I}}).$$

A first-order log-linear approximation around $t-1$ gives

$$\Delta \log y_{i,t} = \Delta \log A_{i,t} + \sum_{f \in \mathcal{F}_c} \varepsilon_{if,t-1} \Delta \log L_{fi,t} + \sum_{j \in \mathcal{I}} \varepsilon_{ij,t-1} \Delta \log x_{ij,t} + r_{i,t} \tag{A.5}$$

where

$$\varepsilon_{if} \equiv \frac{\partial \log y_i}{\partial \log L_{fi}}, \qquad \varepsilon_{ij} \equiv \frac{\partial \log y_i}{\partial \log x_{ij}},$$

and $r_{i,t} = \mathcal{O}(\|\Delta\|^2)$ collects second-order terms. More explicitly,

$$r_{i,t} = \frac{1}{2} \sum_{f \in \mathcal{F}_c} \Delta \varepsilon_{if,t} \Delta \log L_{fi,t} + \frac{1}{2} \sum_{j \in \mathcal{I}} \Delta \varepsilon_{ij,t} \Delta \log x_{ij,t} + O(\|\Delta\|^3),$$

where $\Delta \varepsilon_{if,t}$ and $\Delta \varepsilon_{ij,t}$ denote the changes in output elasticities between $t-1$ and t.

Sector i's price is $p_i = \mu_i \cdot \mathrm{mc}_i$. Cost minimization implies

$$p_i \frac{\partial y_i}{\partial L_{fi}} = \mu_i w_f, \qquad p_i \frac{\partial y_i}{\partial x_{ij}} = \mu_i p_j.$$

Multiplying by L_{fi}/y_i and x_{ij}/y_i, respectively, yields

$$\varepsilon_{if} = \frac{L_{fi}}{y_i} \frac{\partial y_i}{\partial L_{fi}} = \frac{\mu_i w_f L_{fi}}{p_i y_i} = \mu_i \Omega_{if} \equiv \tilde{\Omega}_{if},$$

and

$$\varepsilon_{ij} = \frac{x_{ij}}{y_i}\frac{\partial y_i}{\partial x_{ij}} = \frac{\mu_i p_j x_{ij}}{p_i y_i} = \mu_i \Omega_{ij} \equiv \tilde{\Omega}_{ij}.$$

Substituting $\varepsilon_{if} = \tilde{\Omega}_{if}$ and $\varepsilon_{ij} = \tilde{\Omega}_{ij}$ into (A.5), we obtain

$$\Delta \log y_{i,t} = \Delta \log A_{i,t} + \sum_{f \in \mathcal{F}_c} \tilde{\Omega}_{if,t-1} \Delta \log L_{fi,t} + \sum_{j \in \mathcal{I}} \tilde{\Omega}_{ij,t-1} \Delta \log x_{ij,t} + r_{i,t}. \tag{A.6}$$

The remainder is now

$$r_{i,t} = \frac{1}{2}\sum_f \Delta\tilde{\Omega}_{if,t} \Delta \log L_{fi,t} + \frac{1}{2}\sum_j \Delta\tilde{\Omega}_{ij,t} \Delta \log x_{ij,t} + O(\|\Delta\|^3),$$

which equals zero for Cobb–Douglas production functions.

Substituting (A.6) into (A.4) and collecting terms gives

$$\begin{aligned}\Delta \log Y_{c,t} = &\sum_{i \in \mathcal{I}_c} \lambda_{i,t-1}^{Y_c} \Delta \log A_{i,t} + \sum_{f \in \mathcal{F}_c}\sum_{i \in \mathcal{I}_c} \lambda_{i,t-1}^{Y_c} \tilde{\Omega}_{if,t-1} \Delta \log L_{fi,t} \\ &+ \sum_{j \in \mathcal{I}}\sum_{i \in \mathcal{I}_c} \lambda_{i,t-1}^{Y_c} (\tilde{\Omega}_{ij,t-1} - \Omega_{ij,t-1}) \Delta \log x_{ij,t} + \sum_{i \in \mathcal{I}_c} \lambda_{i,t-1}^{Y_c} r_{i,t}.\end{aligned} \tag{A.7}$$

Equation (A.7) weights sectoral productivity growth by revenue-based Domar weights. Its domestic intermediate term, $j \in \mathcal{I}_c$, still contains the growth of goods that are themselves produced by domestic sectors, and (A.6) ties that growth to the productivity and the inputs of their producers. Making this propagation explicit is what converts the revenue-based weights into cost-based ones.

Write the growth of each domestic input as the growth of the supplier's output plus the change in the buyer's share,

$$\Delta \log x_{ij,t} = \Delta \log y_{j,t} + \hat{x}_{ij,t}, \qquad \hat{x}_{ij,t} := \Delta \log x_{ij,t} - \Delta \log y_{j,t}, \qquad j \in \mathcal{I}_c. \tag{A.8}$$

Substituting (A.8) into (A.6), the growth of each domestic producer's output is the cost-share-weighted growth of domestic goods it uses plus everything else,

$$\Delta \log y_{i,t} = \sum_{j \in \mathcal{I}_c} \tilde{\Omega}_{ij,t-1} \Delta \log y_{j,t} + g_{i,t}, \tag{A.9}$$

$$g_{i,t} \equiv \Delta \log A_{i,t} + \sum_{f \in \mathcal{F}_c} \tilde{\Omega}_{if,t-1} \Delta \log L_{fi,t} + \sum_{j \in \mathcal{I}_c} \tilde{\Omega}_{ij,t-1} \hat{x}_{ij,t} + \sum_{j \notin \mathcal{I}_c} \tilde{\Omega}_{ij,t-1} \Delta \log x_{ij,t} + r_{i,t}.$$

In matrix form, with $\tilde{\boldsymbol{\Omega}}^c$ the domestic block of the cost-based input-output matrix and $\tilde{\boldsymbol{\Psi}}^c = (\boldsymbol{I} - \tilde{\boldsymbol{\Omega}}^c)^{-1}$ the local cost-based Leontief inverse, which exists because every row of

$\tilde{\mathbf{\Omega}}^c$ sums to less than one, each producer spending part of its cost on primary factors or imports, (A.9) reads $\Delta \log \boldsymbol{y}_t = \tilde{\mathbf{\Omega}}^c_{t-1} \Delta \log \boldsymbol{y}_t + \boldsymbol{g}_t$, that is,

$$\Delta \log \boldsymbol{y}_t = \tilde{\mathbf{\Psi}}^c_{t-1} \boldsymbol{g}_t. \tag{A.10}$$

Substituting (A.8) into (A.4), splitting the intermediate-input term into domestic inputs ($j \in \mathcal{I}_c$) and imported inputs ($j \notin \mathcal{I}_c$), and collecting the terms in $\Delta \log y_{j,t}$,

$$\begin{aligned}
\Delta \log Y_{c,t} = &\sum_{j \in \mathcal{I}_c} \lambda^{Y_c}_{j,t-1} \Delta \log y_{j,t} - \sum_{i \in \mathcal{I}_c} \sum_{j \in \mathcal{I}_c} \lambda^{Y_c}_{i,t-1} \Omega_{ij,t-1} (\Delta \log y_{j,t} + \hat{x}_{ij,t}) \\
&- \sum_{i \in \mathcal{I}_c} \sum_{j \notin \mathcal{I}_c} \lambda^{Y_c}_{i,t-1} \Omega_{ij,t-1} \Delta \log x_{ij,t} \\
= &\sum_{j \in \mathcal{I}_c} \left(\lambda^{Y_c}_{j,t-1} - \sum_{i \in \mathcal{I}_c} \lambda^{Y_c}_{i,t-1} \Omega_{ij,t-1} \right) \Delta \log y_{j,t} \\
&- \sum_{i \in \mathcal{I}_c} \sum_{j \in \mathcal{I}_c} \lambda^{Y_c}_{i,t-1} \Omega_{ij,t-1} \hat{x}_{ij,t} - \sum_{i \in \mathcal{I}_c} \sum_{j \notin \mathcal{I}_c} \lambda^{Y_c}_{i,t-1} \Omega_{ij,t-1} \Delta \log x_{ij,t}.
\end{aligned} \tag{A.11}$$

Since $p_j y_j = p_j q_{cj} + \sum_{i \in \mathcal{I}_c} p_j x_{ij}$, the coefficient on $\Delta \log y_{j,t}$, $\lambda^{Y_c}_{j,t-1} - \sum_{j \in \mathcal{I}_c} \lambda^{Y_c}_{i,t-1} \Omega_{ij,t-1}$, is the final-use share $\Omega_{Y_c,j,t-1} = p_{j,t-1} q_{cj,t-1} / (P_{Y_c,t-1} Y_{c,t-1})$, so the first sum of (A.11) is the inner product of the final-use shares with the vector of output growth. Substituting (A.10) for that vector,

$$\sum_{j \in \mathcal{I}_c} \Omega_{Y_c,j,t-1} \Delta \log y_{j,t} = \mathbf{\Omega}_{Y_c,t-1} \Delta \log \boldsymbol{y}_t = \mathbf{\Omega}_{Y_c,t-1} \tilde{\mathbf{\Psi}}^c_{t-1} \boldsymbol{g}_t = \sum_{i \in \mathcal{I}_c} \tilde{\lambda}^{Y_c}_{i,t-1} g_{i,t}.$$

Substituting into (A.11), collecting terms using $\tilde{\lambda}^{Y_c}_i \tilde{\Omega}_{ik} = \tilde{\lambda}^{Y_c}_i \mu_i \Omega_{ik} = \kappa_i \lambda^{Y_c}_i \Omega_{ik}$, and subtracting $\sum_f \tilde{\Lambda}^{Y_c}_{f,t-1} \Delta \log L_{f,t}$ from both sides yields

$$\begin{aligned}
\Delta \log \mathrm{TFP}_{c,t} = &\sum_{i \in \mathcal{I}_c} \tilde{\lambda}^{Y_c}_{i,t-1} \Delta \log A_{i,t} + \sum_{f \in \mathcal{F}_c} \left[\sum_{i \in \mathcal{I}_c} \tilde{\lambda}^{Y_c}_{i,t-1} \mu_{i,t-1} \Omega_{if,t-1} \Delta \log L_{fi,t} - \tilde{\Lambda}^{Y_c}_{f,t-1} \Delta \log L_{f,t} \right] \\
&+ \sum_{i \in \mathcal{I}_c} \sum_{j \in \mathcal{I}_c} \lambda^{Y_c}_{i,t-1} (\kappa_{i,t-1} - 1) \Omega_{ij,t-1} (\Delta \log x_{ij,t} - \Delta \log y_{j,t}) \\
&+ \sum_{i \in \mathcal{I}_c} \sum_{j \notin \mathcal{I}_c} \lambda^{Y_c}_{i,t-1} (\kappa_{i,t-1} - 1) \Omega_{ij,t-1} \Delta \log x_{ij,t} + \mathcal{R}_{c,t},
\end{aligned} \tag{A.12}$$

with $\mathcal{R}_{c,t} = \sum_{i \in \mathcal{I}_c} \tilde{\lambda}^{Y_c}_{i,t-1} r_{i,t} = \mathcal{O}(\|\Delta\|^2)$, and $\hat{x}_{ij,t}$ written out using (A.8). The third and fourth sums are terms (IV) and (V) of (10) for period t. When all wedges equal one, $\tilde{\lambda}^{Y_c} = \lambda^{Y_c}$ and $\kappa_i = 1$, and (A.12) reduces to (A.7) net of the factor term.

Because aggregate factor growth is the Divisia index of the factor's sectoral uses, $\Delta \log L_{f,t} = \sum_{i \in \mathcal{I}_c} s_{fi,t-1} \Delta \log L_{fi,t}$ with $s_{fi,t-1} = \lambda^{Y_c}_{i,t-1} \Omega_{if,t-1} / \Lambda^{Y_c}_{f,t-1}$, and $\tilde{\Lambda}^{Y_c}_{f,t-1} = \overline{\kappa}_{f,t-1} \Lambda^{Y_c}_{f,t-1}$, so that $\tilde{\Lambda}^{Y_c}_{f,t-1} s_{fi,t-1} = \overline{\kappa}_{f,t-1} \lambda^{Y_c}_{i,t-1} \Omega_{if,t-1}$. Hence, the bracket in (A.12) for factor f is

$$\sum_{i \in \mathcal{I}_c} \kappa_{i,t-1} \lambda^{Y_c}_{i,t-1} \Omega_{if,t-1} \Delta \log L_{fi,t} - \tilde{\Lambda}^{Y_c}_{f,t-1} \sum_{i \in \mathcal{I}_c} s_{fi,t-1} \Delta \log L_{fi,t}$$
$$= \sum_{i \in \mathcal{I}_c} \lambda^{Y_c}_{i,t-1} \Omega_{if,t-1} (\kappa_{i,t-1} - \overline{\kappa}_{f,t-1}) \Delta \log L_{fi,t}.$$

Since

$$\sum_{i \in \mathcal{I}_c} \lambda^{Y_c}_{i,t-1} \Omega_{if,t-1} (\kappa_{i,t-1} - \overline{\kappa}_{f,t-1}) = \tilde{\Lambda}^{Y_c}_{f,t-1} - \overline{\kappa}_{f,t-1} \Lambda^{Y_c}_{f,t-1} = 0,$$

$\Delta \log L_{f,t}$ can be subtracted inside the sum without changing its value, which gives the factor reallocation term in the decomposition.

To get terms (I) and (II), sum (A.12) over $t \in (\tau, T]$ and note that $\tilde{\lambda}^{Y_c}_{i,t-1} = \tilde{\lambda}^{Y_c}_{i,\tau} + (\tilde{\lambda}^{Y_c}_{i,t-1} - \tilde{\lambda}^{Y_c}_{i,\tau})$, so that the cumulative technical-efficiency term splits as stated in the decomposition, which gives (10). The sum of terms (I) and (II) does not involve τ and is therefore invariant to the reference period.

Finally, the residual is

$$\mathcal{R}_{c,t} = \frac{1}{2} \sum_{i \in \mathcal{I}_c} \tilde{\lambda}^{Y_c}_{i,t-1} \left(\sum_{f \in \mathcal{F}_c} \Delta \tilde{\Omega}_{if,t} \Delta \log L_{fi,t} + \sum_{j \in \mathcal{I}} \Delta \tilde{\Omega}_{ij,t} \Delta \log x_{ij,t} \right) + O(\|\Delta\|^3),$$

which is zero when all sectors have Cobb–Douglas production, and which is absorbed into $\Delta \log A_{i,t}$ when sectoral productivity is measured as in Definition 3. □

B Structural Change and TFP: Mechanisms and Examples

This appendix formalizes the example of Section 2.3 and provides simple theoretical examples to clarify the different components of the TFP decomposition of Theorem 1.

B.1 Trade-Induced Structural Change

This section formalizes the example of Section 2.3 using a small open-economy model. The goal is not to decompose aggregate TFP into channels, but to show—using exact growth accounting—that changes in international relative prices can lower the path of aggregate TFP through equilibrium reallocation, even in the absence of distortions or intermediate inputs.

Environment. Consider a small open economy with two production sectors, indexed by $i \in \{1, 2\}$, three factors of production (labor and two sector-specific factors), and a representative household.

Each sector produces a tradable good using a constant-returns-to-scale technology:

$$y_i = A_i L_{\ell,i}^{\phi} F_i^{1-\phi}, \quad \phi \in (0, 1), \tag{B.1}$$

where A_i denotes sector-specific productivity, $L_{\ell,i}$ is labor employed in sector i, F_i is a sector-specific fixed factor, and ϕ is a factor-intensity parameter that is common across sectors. Without loss of generality, fixed factors are normalized to one, yielding

$$y_i = A_i L_{\ell,i}^{\phi}. \tag{B.2}$$

Foreign producers supply perfect substitutes for each good at exogenous world prices (p_1^w, p_2^w). The economy is perfectly competitive and undistorted.

The representative household has Cobb–Douglas preferences

$$U(c_1, c_2) = c_1^{\alpha} c_2^{1-\alpha}, \qquad \alpha \in (0, 1), \tag{B.3}$$

and faces the budget constraint

$$\sum_i p_i^w c_i \leq w L_\ell + \sum_i r_i, \tag{B.4}$$

where w is the wage, L_ℓ is aggregate labor supply, which is supplied inelastically, and r_i denotes the rents accruing to fixed factors.

Equilibrium. An equilibrium consists of a set of prices $(\boldsymbol{p}^w, \boldsymbol{r}, w) \in \mathbb{R}^5_{++}$ and allocations $(\boldsymbol{c}, \boldsymbol{y}, \boldsymbol{L}_\ell, \mathbf{nx}) \in \mathbb{R}^8_+$ such that:

1. World prices (p_1^w, p_2^w) are exogenous.
2. Producers maximize profits taking prices as given.
3. The household maximizes utility subject to its budget constraint.
4. Labor and goods markets clear:

$$\sum_i L_{\ell,i} = L_\ell, \qquad c_i + \mathrm{nx}_i = y_i.$$

5. The balance-of-payments condition holds:

$$\sum_i p_i^w \mathrm{nx}_i = 0.$$

Characterization. Any equilibrium of the small open economy is characterized by:

$$\begin{aligned}
&(p_1^w, p_2^w) \gg \mathbf{0} \text{ given},\\
&w = \phi p_i^w A_i L_{\ell,i}^{\phi-1}, \qquad i = 1, 2,\\
&r_i = p_i^w y_i - w L_{\ell,i},\\
&L_{\ell,i} = \frac{(A_i \xi_i)^{\frac{1}{1-\phi}}}{\sum_{j\in\{1,2\}} (A_j \xi_j)^{\frac{1}{1-\phi}}} L_\ell, \qquad \xi_i \equiv \frac{p_i^w}{p_1^w}, \quad \xi_1 = 1,\\
&y_i = A_i L_{\ell,i}^{\phi},\\
&c_i = \frac{\alpha_i}{\xi_i}\Big(w L_\ell + r_1 + r_2\Big), \qquad \alpha_i = \begin{cases} \alpha, & i = 1 \\ 1-\alpha, & i = 2 \end{cases},\\
&\mathrm{nx}_i = y_i - c_i.
\end{aligned}$$

World relative prices are summarized by $\xi_2 = p_2^w / p_1^w$, which is the terms of trade.

Special Cases. When world relative prices coincide with autarky prices, the no-trade allocation is an equilibrium. Labor is allocated according to expenditure shares, and production and consumption coincide. As $\phi \to 1$, the model converges to a Ricardian environment with full specialization. Sector-specific fixed factors prevent complete specialization for $\phi < 1$, ensuring that changes in relative prices induce *partial* reallocation of labor across sectors. These special cases clarify how the terms of trade discipline factor allocation in equilibrium.

Exact Aggregate TFP Accounting. Nominal output (GDP) is

$$PY = \sum_i p_i^w y_i.$$

Real output growth is defined using a Divisia quantity index with base-period weights:

$$\Delta \log Y_t = \sum_i \lambda_{i,t-1} \Delta \log y_{i,t}, \qquad \lambda_{i,t-1} \equiv \frac{p_{i,t-1}^w y_{i,t-1}}{P_{t-1} Y_{t-1}}.$$

Since the economy is undistorted, revenue- and cost-based factor shares coincide and aggregate TFP growth is given by the traditional Solow residual:

$$\Delta \log \text{TFP}_t = \Delta \log Y_t - \sum_f \Lambda_{f,t-1} \Delta \log L_{f,t},$$

where $\Lambda_{f,t}$ denotes revenue-based factor shares and the growth of each aggregate factor is the Divisia index of its sectoral uses. The sector-specific factors do not move, so $\Delta \log L_{F_i,t} = 0$. For labor,

$$\Delta \log L_{\ell,t} = \sum_i s_{i,t-1} \Delta \log L_{\ell,i,t}, \qquad s_{i,t-1} \equiv \frac{w L_{\ell,i,t-1}}{w L_{\ell,t-1}} = \frac{L_{\ell,i,t-1}}{L_\ell},$$

where the payment share equals the employment share because the wage is common to both sectors. Using the sectoral production function,

$$\Delta \log y_{i,t} = \Delta \log A_{i,t} + \phi \Delta \log L_{\ell,i,t},$$

aggregate TFP growth can be written exactly as

$$\Delta \log \text{TFP}_t = \sum_i \lambda_{i,t-1} \Delta \log A_{i,t} + \phi \sum_i \lambda_{i,t-1} \Delta \log L_{\ell,i,t} - \Lambda_{\ell,t-1} \sum_i s_{i,t-1} \Delta \log L_{\ell,i,t}.$$

Since labor is paid the value of its marginal product, $w L_{\ell,i} = \phi p_i^w y_i$, the labor share is $\Lambda_\ell = \phi$ and the output shares satisfy $\phi \lambda_i = \Lambda_i s_i$. The last two terms therefore cancel, and

$$\Delta \log \text{TFP}_t = \sum_i \lambda_{i,t-1} \Delta \log A_{i,t}. \tag{B.5}$$

This is an exact accounting identity *given the index choice* and involves no approximation: the reallocation of labor between the two sectors has no effect on aggregate TFP on impact, because the value of output lost in the contracting sector equals the value of output gained in the expanding one. This is Hulten's theorem for this economy.

Remark (physical factor totals). If aggregate labor were measured instead by its physical total, $L_\ell = \sum_i L_{\ell,i}$, which is constant in this economy, the labor term of the Solow residual would be zero and (B.5) would acquire the additional term

$$\phi \sum_i \lambda_{i,t-1} \Delta \log L_{\ell,i,t} = \phi \left[\sum_i s_{i,t-1} \Delta \log L_{\ell,i,t} - \Delta \log L_{\ell,t} \right], \tag{B.6}$$

the difference between the Divisia index of sectoral labor use and the log change of the total. This composition effect has three properties. First, it is never positive: because total labor is fixed, $\sum_i s_{i,t-1} L_{\ell,i,t}/L_{\ell,i,t-1} = 1$ so by Jensen's inequality

$$\sum_i s_{i,t-1} \Delta \log L_{\ell,i,t} \leq \log \sum_i s_{i,t-1} \frac{L_{\ell,i,t}}{L_{\ell,i,t-1}} = 0,$$

with equality if and only if no labor moves. Second, it is of second order: a second-order expansion of $\log \sum_i L_{\ell,i,t}$, which is constant, around $t-1$ gives

$$(\text{B.6}) = -\frac{\phi}{2} \text{Var}_{s_{t-1}} (\Delta \log L_{\ell,i,t}) + \mathcal{O}(\|\Delta\|^3),$$

where $\text{Var}_{s_{t-1}} (\Delta \log L_{\ell,i,t}) \equiv \sum_i s_{i,t-1} (\Delta \log L_{\ell,i,t} - \sum_j s_{j,t-1} \Delta \log L_{\ell,j,t})^2$ is the employment-share-weighted variance of sectoral labor growth. Third, it therefore depends only on how unevenly labor moves across sectors, and takes the same value whether labor moves toward sector 1 or toward sector 2. It vanishes in continuous time, since $\sum_i s_i \mathrm{d} \log L_{\ell,i} = \mathrm{d} L_\ell / L_\ell = 0$. It is the difference between two index numbers for the same aggregate, not an effect of reallocation on productive efficiency, and it is absent when the definition of TFP uses the growth of each factor as a Divisia index of sectoral uses.

The reallocation affects the path of aggregate TFP through the weights in (B.5). Adding and subtracting the weights of an initial period τ, cumulative technical efficiency over $(\tau, T]$ satisfies the exact identity

$$\sum_{t=\tau+1}^{T} \sum_i \lambda_{i,t-1} \Delta \log A_{i,t} = \sum_{t=\tau+1}^{T} \sum_i \lambda_{i,\tau} \Delta \log A_{i,t} + \sum_{t=\tau+1}^{T} \sum_i (\lambda_{i,t-1} - \lambda_{i,\tau}) \Delta \log A_{i,t}. \tag{B.7}$$

The first term is the path of aggregate TFP with the production structure held at its level in τ; the second is structural change. From the equilibrium characterization, an increase in ξ_2 raises $L_{\ell,2}$ and hence λ_2, so that after a deterioration in the terms of trade of good 1 the weight of sector 2 in the second term rises. If sector 2's productivity growth is lower than sector 1's, the structural-change term is negative in every subsequent period.

Proposition 4 (Trade-Induced Reallocation and TFP)**.** *In the small open economy:*

(i) *On impact, a change in the terms of trade leaves aggregate TFP growth unchanged: it equals* $\sum_i \lambda_{i,t-1} \Delta \log A_{i,t}$*, as under constant world prices, despite the reallocation of labor along the production possibility frontier.*

(ii) *If sectoral productivity grows at constant rates after the change and the sector whose weight rose has the lower rate, the structural-change term in* (B.7) *is strictly negative in every subsequent period, and aggregate TFP growth is lower than it would have been with the pre-shock production structure.*

Proof. See Appendix A.3. □

Interpretation. The proposition separates two effects of a terms-of-trade shock. On impact, the reallocation of labor along the production possibility frontier does not change aggregate TFP: labor is paid its marginal product in both sectors, so the value of output lost in the contracting sector equals the value gained in the expanding one. Over time, however, the shock shifts the weight of aggregate TFP toward the sector with relatively lower productivity growth, so that the fixed-weight component of aggregate TFP growth remains positive while the structural-change term is negative and accumulates, leading to a lower path for the level of aggregate TFP. In the parametrization of Appendix A.3, a terms-of-trade change that moves 16 percent of the labor force leaves aggregate TFP growth on impact at 0.5 percent, its value without the change, and lowers aggregate TFP growth in every later period by 0.46 percentage points relative to the pre-shock structure. Measured with physical factor totals, aggregate TFP would in addition fall on impact by the composition effect (B.6), a discrete-time index-number effect that carries no information about productive efficiency.

B.2 Decomposing Aggregate TFP: Illustrative Examples

This appendix presents five stylized examples, one for each term of Theorem 1. The goal is to illustrate how the aggregate TFP decomposition maps into simple economic environments: each economy is the simplest in which exactly one term is nonzero, and together the examples show what each term captures. In each example, aggregate TFP is computed from the Divisia definition

$$\Delta \log \mathrm{TFP}_t = \Delta \log Y_t - \sum_f \tilde{\Lambda}_{f,t-1} \Delta \log L_{f,t},$$

$$\Delta \log Y_t = \sum_i \lambda_{i,t-1} \Delta \log y_{i,t} - \sum_{i \in \mathcal{I}_c} \sum_j \lambda_{i,t-1} \Omega_{ij,t-1} \Delta \log x_{ij,t},$$

where the second sum subtracts all intermediate inputs j used by domestic sectors i, both domestic ($j \in \mathcal{I}_c$) and imported ($j \notin \mathcal{I}_c$), and the growth of each aggregate factor is the Divisia index of its sectoral uses. Each table below reports the terms of Theorem 1, grouped into technical efficiency, terms (I)–(II), and allocative efficiency, terms (III)–(V).

In the examples without domestic intermediate inputs (Examples 2, 3, and 5), the cost-based Domar weights of domestic producers equal the revenue-based ones, $\tilde{\lambda}_i = \lambda_i$, whatever the markups; in the two with a production network (Examples 1 and 4), they are computed with the local cost-based Leontief inverse, and each example reports the resulting network-adjusted wedges κ_i. Cost-based aggregate factor shares are $\tilde{\Lambda}_f = \sum_i \tilde{\lambda}_i \mu_i \Omega_{if}$, and revenue-based factor shares are $\Lambda_f = \sum_i \lambda_i \Omega_{if}$.

Example 1: Within-sector productivity (closed vertical economy with no distortions). Households consume good 1, produced from good 2 with $y_1 = A_1 x_{12}$. Good 2 is produced from labor with $y_2 = A_2 L$. The wage is the numeraire. Labor is fixed and employed only in sector 2, and all of good 2 is sold to sector 1, so nothing is reallocated and terms (III)–(V) are zero. Both Domar weights equal one, $\tilde{\boldsymbol{\lambda}} = \boldsymbol{\lambda} = (1, 1)$ because each sector's sales equal GDP. The decomposition reduces to Hulten's theorem,

$$\Delta \log \mathrm{TFP}_t = \underbrace{\sum_{i \in \{1,2\}} \tilde{\lambda}_{i,\tau} \Delta \log A_{i,t}}_{\text{within-sector technical efficiency}} = \Delta \log A_{1,t} + \Delta \log A_{2,t},$$

with $\tau = t - 1$. Term (II) is zero here not only because the example spans one period but because the weights never move. Under the following parametrization,

Parameter	Description	Value
L_ℓ	labor endowment, all employed in sector 2	1
(μ_1, μ_2)	sectoral distortions	$(1, 1)$
(A_1, A_2)	sectoral productivities	$(1, 1) \to (1, 1.1)$

the equilibria are:

	Equilibria under selected parametrizations						
	w	p_1	p_2	y_1	$y_2 = x_{12}$	nominal GDP	real GDP
$t-1$	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
t	1.0000	0.9091	0.9091	1.1000	1.1000	1.0000	1.1000

The following table reports the decomposition.

Term	Contribution
Aggregate TFP growth	+0.0953
Technical efficiency	+0.0953
(I) within-sector productivity	+0.0953
(II) structural change	0
Allocative efficiency	0
(III) factor reallocation	0
(IV) domestic-intermediates reallocation	0
(V) foreign-intermediates reallocation	0

In this example, productivity in sector 2 rises by 10% and aggregate TFP rises by 0.0953, all of it term (I): the upstream gain raises real GDP one for one because the intermediate is fully used downstream.[11] Followed for more periods, term (II) would stay zero as well: the cost-based Domar weights are pinned at one by the structure of the chain, so productivity growth never changes the weights that later productivity growth receives.

Example 2: Structural change (open economy, no distortions, no intermediates). Take the small open economy of Appendix B.1: two sectors producing $y_i = A_i L_{\ell i}^{\varphi} L_{F_i}^{1-\varphi}$ with mobile labor and a sector-specific fixed factor, world prices, and no distortions. The terms of trade move against good 2 while productivity changes. The example is followed for a second period in which world prices stay at their new level and productivity keeps changing at the same rates, with the window starting at $\tau = t - 1$. Since $\boldsymbol{\kappa} = \mathbf{1}$, terms (III)–(V) vanish, and since there are no intermediates, $\tilde{\boldsymbol{\lambda}} = \boldsymbol{\lambda}$. The decomposition reduces to

$$\sum_{t=\tau-1}^{T} \Delta \log \mathrm{TFP}_t = \underbrace{\sum_{t} \sum_{i \in \{1,2\}} \lambda_{i,\tau} \Delta \log A_{i,t}}_{\text{(I) within-sector productivity}} + \underbrace{\sum_{t} \sum_{i \in \{1,2\}} (\lambda_{i,t-1} - \lambda_{i,\tau}) \Delta \log A_{i,t}}_{\text{(II) structural change}}.$$

Under the following parametrization, the equilibria are:

[11] Nominal GDP is the wage bill, one in both periods, because the wage is the numeraire and labor is the only primary factor. Both prices fall and both outputs rise by 10%.

Parameter	Description	Value
α	weight on good 1 in Cobb–Douglas utility	0.5
ϕ	labor share in production (all sectors)	0.7
L_ℓ	total labor endowment	1
(L_{F_1}, L_{F_2})	endowments of fixed factors	$(1,1)$
(μ_1, μ_2)	sectoral distortions	$(1,1)$
(A_1, A_2)	sectoral productivities	$(1,1) \to (0.99, 1.02) \to (0.99^2, 1.02^2)$
(p_1, p_2)	world prices	$(1,1) \to (1, 0.8) \to (1, 0.8)$

	Equilibria under selected parametrizations						
	$L_{\ell 1}$	$L_{\ell 2}$	w	y_1	y_2	nominal GDP	real GDP
$t-1$	0.5000	0.5000	0.8618	0.6156	0.6156	1.2311	1.0000
t	0.6557	0.3443	0.7865	0.7368	0.4835	1.1236	0.9696
$t+1$	0.6329	0.3671	0.7870	0.7116	0.5159	1.1243	0.9691

The following table reports the decomposition.

Term	**Contribution** ($\tau = t-1$)	
	t	$t+1$
Aggregate TFP growth	+0.0049	+0.0002
Technical efficiency	+0.0049	+0.0002
(I) within-sector productivity	+0.0049	+0.0049
(II) structural change	0	−0.0047
Allocative efficiency	0	0
(III) factor reallocation	0	0
(IV) domestic-intermediates reallocation	0	0
(V) foreign-intermediates reallocation	0	0

In this example, the terms-of-trade change moves 16 percent of the labor force into sector 1. On impact aggregate TFP rises by 0.0049, exactly as it would have without the change: labor is paid its marginal product in both sectors, so the reallocation has no effect on TFP.[12] In the following period the reallocation shows up: the Domar weights have moved from (0.50, 0.50) to (0.66, 0.34), toward the sector whose productivity is

[12]Real GDP at base-period weights falls by 3.1 log points in the same period because the Divisia index of labor falls by 5.1; that fall is the composition effect of Appendix B.1, and the Solow residual nets it out. Measured with the physical labor total instead, aggregate TFP would fall by 0.0308.

falling, and structural change is $0.156 \times \log 0.99 - 0.156 \times \log 1.02 = -0.0047$, so that aggregate TFP grows by only 0.0002. This effect is identified here as term (II). Absent the split into terms (I) and (II), the technical efficiency term would suggest that productivity growth stalled (+0.0002), although each sector's productivity grows exactly as before; what changed is that the earlier reallocation of economic activity across sectors moved the weights with which sectoral productivity contributes to aggregate TFP.

Example 3: Factor reallocation (distorted, closed economy with no intermediates). Consider the two sectors of Example 2 in a closed economy, $y_i = A_i L_{\ell i}^{\varphi} L_{F_i}^{1-\varphi}$, where sector 2 sells at a markup and households have Cobb–Douglas preferences. The markup rises from 1.5 to 2.5 with productivity constant. Without intermediates, $\tilde{\lambda} = \lambda$ and $\kappa_i = \mu_i$, and $\overline{\kappa}_\ell = \sum_i s_{\ell i} \mu_i$ is the payment-share-weighted average markup. The fixed factors do not move, so the decomposition reduces to

$$\Delta \log \mathrm{TFP}_t = \underbrace{\sum_{i \in \{1,2\}} \lambda_{i,t-1} \Omega_{i\ell,t-1} (\mu_{i,t-1} - \overline{\kappa}_{\ell,t-1})(\Delta \log L_{\ell i,t} - \Delta \log L_{\ell,t})}_{\text{(III) factor reallocation}}.$$

Under the following parametrization, the equilibria are:

Parameter	Description	Value
α	weight on good 1 in Cobb–Douglas utility	0.5
ϕ	labor share in production (all sectors)	0.7
L_ℓ	total labor endowment	1
(L_{F_1}, L_{F_2})	endowments of fixed factors	$(1, 1)$
(μ_1, μ_2)	sectoral distortions	$(1, 1.5) \to (1, 2.5)$
(A_1, A_2)	sectoral productivities	$(1, 1)$

	Equilibria under selected parametrizations								
	$L_{\ell 1}$	$L_{\ell 2}$	p_1	p_2	w	y_1	y_2	nominal GDP	real GDP
$t-1$	0.6000	0.4000	1.0000	1.3282	0.8159	0.6994	0.5266	1.3987	1.0000
t	0.7143	0.2857	1.0000	1.8991	0.7743	0.7902	0.4161	1.5803	0.9448

The following table reports the decomposition.

In this example, the higher markup drives labor out of sector 2, where the social return to labor exceeds its wage, and aggregate TFP falls by 0.0358.

Example 4: Reallocation in domestic intermediates (closed economy with distortions). Consider an economy in which sector 1 produces an intermediate from all of the

Term	Contribution
Aggregate TFP growth	−0.0358
Technical efficiency	0
(I) within-sector productivity	0
(II) structural change	0
Allocative efficiency	−0.0358
(III) factor reallocation	−0.0358
(IV) domestic-intermediates reallocation	0
(V) foreign-intermediates reallocation	0

economy's labor, $y_1 = A_1 L$, and sells it to sectors 2 and 3, which produce the two final goods from a sector-specific fixed factor and the intermediate, $y_i = A_i L_{F_i}^{1/2} x_{i1}^{1/2}$. Sector 3 sells at a markup, whereas sectors 1 and 2 price at cost. The wage is the numeraire. No factor moves and sector 1's output is unchanged, so terms (I), (III), and (V) are zero. Note that $\kappa_2 = 1$, $\kappa_3 = \mu_3$, and $\kappa_1 = \phi_{12} + \phi_{13}\mu_3$, where $\phi_{1j} = x_{j1}/y_1$ is the share of sector 1's output sold to sector j. The decomposition reduces to

$$\Delta \log \mathrm{TFP}_t = \underbrace{\sum_{i \in \{1,2\}} \lambda_{3,t-1}(\mu_{3,t-1} - 1)\Omega_{31,t-1}\Delta \log x_{31,t}}_{\text{(IV) reallocation in domestic intermediates}}.$$

Under the following parametrization, the equilibria are:

Parameter	Description	Value
α	weight on good 2 in Cobb–Douglas utility	0.5→0.4
$L, (L_{F_1}, L_{F_2})$	factor endowments	$1, (1,1)$
(μ_1, μ_2, μ_3)	sectoral distortions	$(1,1,1.5)$
(A_1, A_2, A_3)	sectoral productivities	$(1,1,1)$

	Equilibria under selected parametrizations										
	x_{21}	x_{31}	y_1	y_2	y_3	p_1	p_2	p_3	w	nominal GDP	real GDP
$t-1$	0.6000	0.4000	1.0000	0.7746	0.6325	1.0000	1.5492	1.8974	1.0000	2.4000	1.0000
t	0.5000	0.5000	1.0000	0.7071	0.7071	1.0000	1.4142	2.1213	1.0000	2.5000	1.0188

The following table reports the decomposition.

In this example, the demand shift moves the intermediate from sector 2 to sector 3, and aggregate TFP rises by 0.0186. The intermediate now serves the buyer where its social return exceeds its cost, $\lambda_3(\kappa_3 - 1)\Omega_{31}\Delta \log x_{31} = 0.5 \times 0.5 \times \frac{1}{3} \times 0.223$.

Example 5: Imported inputs (open economy with foreign intermediates). Consider now a small open economy with a single sector producing a tradable final good accord-

Term	Contribution
Aggregate TFP growth	+0.0186
Technical efficiency	0
(I) within-sector productivity	0
(II) structural change	0
Allocative efficiency	+0.0186
(III) factor reallocation	0
(IV) domestic-intermediates reallocation	+0.0186
(V) foreign-intermediates reallocation	0

ing to $y = AL^{\phi}m^{1-\phi}$, where m denotes foreign intermediates. Imported intermediates are purchased at exogenous world price p_m^w, and the tradable good sells at exogenous world price p^w, since there is a perfect substitute available abroad. With price-taking firms—both p^w and p_m^w are exogenous—the wedge $\mu > 1$ cannot represent market power in the output market. Instead, μ should be thought of as an output tax or an input tax that drives a wedge between the world price and marginal cost, leading the representative firm to under-employ inputs relative to an undistorted benchmark. The revenue from the wedge is rebated lump-sum to the household. The first-order condition gives

$$m = \left(\frac{(1-\phi)Ap^w}{\mu p_m^w}\right)^{1/\phi}, \qquad w = \frac{\phi p^w y}{\mu L}.$$

The representative household supplies labor inelastically ($L = 1$), owns the firm, and has utility $U = c$. Household income equals $wL + \pi + T$. In equilibrium, profits are zero ($\pi = 0$) and the lump-sum rebate is $T = p^w(\mu - 1)y/\mu$. The trade balance requires the country to export part of y to finance imports:

$$p^w(y - c) = p_m^w m \qquad \Longrightarrow \qquad c = y - \frac{p_m^w}{p^w}m,$$

so GDP $= p^w c = p^w y - p_m^w m$.

For Cobb–Douglas, $\Omega_m = p^w m/(p^w y) = (1-\phi)/\mu$ and $\lambda_Y = p^w y/\text{GDP}$, which are constant across periods. Productivity is constant and labor is fixed, so the TFP decomposition reduces to

$$\Delta \log \text{TFP}_t = \underbrace{\lambda_{m,t-1}(\mu_{t-1} - 1)\Omega_{m,t-1}\Delta \log m_t}_{\text{(V) reallocation in foreign intermediates}}.$$

Under the following parametrizations, we have the equilibria:

Parameter	Description	Value
ϕ	labor share in gross output	0.5
L	total labor endowment	1
A	productivity	1
μ	cost wedge	1.5
p^w	world output price	1
$(p^w_{m,t-1}, p^w_{m,t})$	world input price	$(0.333, 0.300)$

	Equilibria under selected parametrizations					
	m	y	c	w	nominal GDP	real GDP
$t-1$	1.0000	1.0000	0.6667	0.3333	0.6667	1.0000
t	1.2346	1.1111	0.7407	0.3704	0.7407	1.0541

The following table reports the decomposition.

Term	Contribution
Aggregate TFP growth	+0.0527
Technical efficiency	0
(I) within-sector productivity	0
(II) structural change	0
Allocative efficiency	+0.0527
(III) factor reallocation	0
(IV) domestic-intermediates reallocation	0
(V) foreign-intermediates reallocation	+0.0527

In this example, aggregate TFP increases by roughly 5 percentage points, all of it term (V): imported-input use rises by 21 log points and, because the producer sells above marginal cost, each additional unit raises real GDP by more than its cost.

C Data Appendix

This appendix describes the data sources, how they are harmonized into a consistent panel, and how the aggregate series are constructed, together with the sectoral and trade evidence used in the main text.

C.1 Sources and Sample

Table 3 lists the data sources. The analysis combines world input-output tables, which record production linkages, domestic and international trade flows, and final uses, with industry accounts, which record each industry's output and input volumes, factor payments, factor services, and capital stocks. The input-output tables come from three releases: the Long-Run WIOD for 1970–1994, the 2013 WIOD release for 1995–2010, and Eurostat's FIGARO tables for 2011–2019. For Spain and Italy, the industry accounts come from the 2024 EU KLEMS release from 1995 onward and, before 1995, from its 2008 release, the only EU KLEMS vintage with industry data back to 1970. Output, intermediate-input, and value-added volumes from 1995 onward come from the national accounts. The other countries enter the analysis only through the world production network, and their industry accounts come from WIOD. The remaining sources provide the depreciation rates and inputs to the user cost of capital, described in Appendix D.

The sample consists of 23 countries and a rest-of-world region, which together account for over 85 percent of world GDP (Table 4), and the 23 industries common to all sources (Table 5).

Table 3: Data Sources.

Source	Used for	Years
INPUT-OUTPUT TABLES		
Long-Run WIOD (Woltjer et al., 2021)	World input-output tables	1970–1994
WIOD, 2013 release (Timmer et al., 2015)	World input-output tables (reference levels)	1995–2010
FIGARO (Rémond-Tiedrez and Rueda-Cantuche, 2019)	World input-output tables	2011–2019
INDUSTRY ACCOUNTS: SPAIN AND ITALY		
EU KLEMS & INTANProd, 2024 release (Bontadini et al., 2023)	Labor compensation, hours, factor services, capital stocks	1995–2019
EU KLEMS, 2008 release (Timmer et al., 2007; O'Mahony and Timmer, 2009)	The same, and output volumes	1970–1994
Fundación BBVA–Ivie (Fundación BBVA and Ivie, 2026)	Spain's capital stocks	1970–1994
ISTAT (ISTAT, 2026); Eurostat (Eurostat, 2026b)	Output, intermediate-input, and value-added volumes	1995–2019
INDUSTRY ACCOUNTS: OTHER COUNTRIES		
WIOD Socio-Economic Accounts (Timmer et al., 2016)	Volumes and factor accounts	1970–2014
DEPRECIATION AND COST OF CAPITAL		
BEA detailed fixed assets (U.S. Bureau of Economic Analysis, 2026)	Depreciation rates by industry and asset	1969–2019
Macrohistory Database (Jordà et al., 2017, 2019); FRED (Moody's, 2026; Board of Governors of the Federal Reserve System, 2026)	Bond yields, equity premium, corporate spread	1970–2019
Eurostat financial accounts (Eurostat, 2026a); Kuvshinov and Zimmermann (2022)	Debt share of non-financial corporations	1970–2019
Oxford CBT Tax Database (Oxford University Centre for Business Taxation, 2017)	Corporate tax rates and depreciation allowances	1981–2017
World Bank, World Development Indicators (World Bank, 2026)	Euro–dollar rate; price indices for alternative user costs	1970–2019

Table 4: Country and Region Sample.

Australia (AUS)	Denmark (DNK)	Ireland (IRL)	Sweden (SWE)
Austria (AUT)	Spain (ESP)	Italy (ITA)	United States (USA)
Belgium (BEL)	Finland (FIN)	Japan (JPN)	Rest of world (ROW)
Brazil (BRA)	France (FRA)	South Korea (KOR)	
Canada (CAN)	United Kingdom (GBR)	Mexico (MEX)	
China (CHN)	Greece (GRC)	Netherlands (NLD)	
Germany (DEU)	India (IND)	Portugal (PRT)	

Table Notes. ISO-3 country codes are listed in parentheses. The countries listed in this table appear in the Long-Run WIOD, in the 2013 WIOD release and in FIGARO. Countries that appear in only some of these sources (e.g., Hong Kong, Taiwan) are included in the "Rest of world" region.

Table 5: Industry Sample: WIOD and FIGARO Industries.

WIOD industry (ISIC Rev. 3)		**Corresponding FIGARO industries (NACE Rev. 2)**
AtB	Agriculture, Hunting, Forestry and Fishing	Crop and animal production, forestry, fishing (A01–A03)
C	Mining and Quarrying	Mining and quarrying (B)
D15t16	Food, Beverages and Tobacco	Food, beverages and tobacco (C10–C12)
D17t19	Textiles, Leather, and Footwear	Textiles, apparel and leather (C13–C15)
D21t22	Pulp, Paper, Printing and Publishing	Paper (C17), printing (C18), publishing (J58)
D23	Coke, Refined Petroleum and Nuclear Fuel	Coke and refined petroleum (C19)
D24	Chemicals and Chemical Products	Chemicals (C20); pharmaceuticals (C21)
D25	Rubber and Plastics	Rubber and plastics (C22)
D26	Other Non-Metallic Mineral	Other non-metallic minerals (C23)
D27t28	Basic Metals and Fabricated Metal	Basic metals (C24), fabricated metal products (C25)
D29	Machinery, Nec	Machinery n.e.c. (C28), repair and installation of machinery (C33)
D30t33	Electrical and Optical Equipment	Computer, electronic and optical products (C26), electrical equipment (C27)
D34t35	Transport Equipment	Motor vehicles (C29), other transport equipment (C30)
Dnec	Wood, Manufacturing, Nec, Recycling	Wood (C16), furniture and other manufacturing (C31–C32)
E	Electricity, Gas and Water Supply	Electricity and gas (D35), water supply (E36)
F	Construction	Construction (F)
G	Wholesale and Retail Trade	Wholesale and retail trade, including motor vehicles (G45–G47), repair of household goods (S95)
H	Hotels and Restaurants	Accommodation and food services (I)
I60t63	Transport and Storage	Land, water and air transport, warehousing (H49–H52), travel agencies (N79)
I64	Post and Telecommunications	Postal and courier activities (H53), telecommunications (J61)
J	Financial Intermediation	Financial and insurance services (K64–K66)
K	Real Estate, Renting and Business Activities	Real estate (L), professional, scientific and technical activities (M69–M75), rental, employment, security and office services (N77, N78, N80–N82), computer services (J62–J63)
LtQ	Public Administration, Education, Health and Other Services	Public administration (O84), education (P85), health and social work (Q86–Q88), arts, recreation and other services (R90–R93, S94, S96), sewerage and waste (E37–E39), film and broadcasting (J59–J60), households and extraterritorial organizations (T, U)

Table Notes. The first two columns give the 23 industries of the panel, those of the Long-Run WIOD and the 2013 WIOD release, in the ISIC Rev. 3 classification. The last column gives the industries of FIGARO, in NACE Rev. 2, assigned to each of them to harmonize the two classifications, following the content of each industry in ISIC Rev. 3. The same correspondence is applied to the other sources classified in NACE Rev. 2 (the 2024 EU KLEMS release and the Eurostat and ISTAT national accounts). Where such a source combines industries that belong to different panel industries, they are divided in proportion to their aggregate value added. The text refers to LtQ as non-market services and to K as real estate and business services.

C.2 Harmonization

The panel follows one principle: a reference source determines the level and cross-sectional structure of each series, while the other sources extend it using their time-series variation. This preserves the levels of the preferred source without discarding the longer histories available in earlier vintages. The reference is the 2013 WIOD release for the input-output tables and for the accounts of the other countries, and, for Spain and Italy, the 2024 EU KLEMS release for factor accounts and their national accounts for output and input volumes.

Input-output tables. The three vintages differ both in levels and in the allocation of purchases between domestic and foreign suppliers. Over their overlapping years, the Long-Run WIOD assigns Spain and Italy an imported share of intermediate inputs 2 to 3 percentage points below that of the 2013 release. Simply appending the vintages would therefore create breaks in trade shares and production linkages in 1995 and 2011. Rescaling each seller's row to the reference level, as in Lewis et al. (2022), removes the level discrepancy but leaves the earlier vintage's domestic-import split unchanged, so the difference in import shares persists.

I therefore harmonize both levels and sourcing patterns. First, each seller's sales are scaled to the reference level of gross output. Second, each buyer's domestic and imported purchases are adjusted separately using factors estimated over the overlapping years so that their relative shares align with the reference source. These factors are held fixed backward in time, preserving the earlier vintage's within-source dynamics. Third, the table is rebalanced so that each industry's sales equal its uses. FIGARO is linked forward to the 2013 release in the same way. The resulting import shares evolve smoothly through both splices: their changes in 1995 and 2011 are comparable to those in adjacent years. Taxes less subsidies on products paid on intermediate inputs are treated as part of the wedge between the buyer's price and the seller's cost.

Industry accounts. Labor compensation and capital stocks enter as ratios to value added and are applied to the value added of the input-output panel, placing factor payments and the production network on a common nominal basis. Industries in the 2024 EU KLEMS release are aggregated to the 23 industries of the panel; the eight categories spanning two panel industries are divided according to their 1995 value-added shares. For the remaining countries, series unavailable before 1995 or after 2014 are held at their nearest available values, while the rest-of-world region takes cross-country averages of factor shares and industry output prices. These countries affect the productivity measures for Spain and Italy only through the prices used to deflate

imported inputs. Their wedges do not enter because the measures use the domestic cost-based Leontief inverse of Lemma 1, which stops at imported inputs. Using the world inverse instead, and therefore allowing round trips through foreign producers, changes no producer's weight by more than 0.0002.

Foreign output prices determine only how intermediate-input reallocation is divided between domestic and imported inputs. As a check, if every input purchased by a buyer is instead assumed to grow at the buyer's aggregate intermediate-input volume—so that no foreign output prices are used—aggregate TFP, terms (I)–(III), and the sum of terms (IV)–(V) are unchanged. For Spain over 1995–2019, only the split of the latter two terms changes, from +1.3 and +1.1 percentage points to +0.2 and +2.1. The extrapolations for other countries and the rest-of-world averages therefore affect only the allocation of intermediate-input reallocation between its domestic and foreign components, not the aggregate productivity results.

Volumes. From 1995 onward, the volumes of gross output, intermediate inputs, and value added for Spain and Italy come from the national accounts. Where the Spanish accounts do not report gross output and intermediate inputs at previous-year prices, I construct them by double deflation. Each industry's input price is built from the output prices of its domestic suppliers and, for imported inputs, those of euro-area producers. As a validation exercise, applying the same procedure to countries for which official volume measures are available reproduces annual growth rates with mean absolute errors of 1 to 3 percentage points. Before 1995, volumes are extended using the industry indices from the 2008 EU KLEMS release.

C.3 Growth-Accounting Variables

Table 6 maps the theoretical objects to the data. The three aggregate productivity measures use the same real GDP and aggregate factor-growth series, which are Divisia aggregates of the industry data, and differ only in the factor shares used to weight labor and capital. The traditional residual uses income shares, with the capital share defined as one minus the labor share. The network-adjusted residual uses cost-based factor shares that account for each producer's wedge and its propagation through the input-output network. The intermediate measure corrects each producer's factor payments only by its own wedge and therefore ignores network propagation. The cost-based shares use the cost-based Leontief inverse of the domestic network because Theorem 1 substitutes the technologies of domestic producers only.

Table 6: Growth-Accounting Variables.

Model Variable	Data Counterpart	Source
COMMON VARIABLES		
$\Delta \log Y$	Divisia of GO_QI net of II_QI	ISTAT, Eurostat, EU KLEMS
$\Delta \log L_K$	Divisia of CAP_QI	
$\Delta \log L_L$	Divisia of LAB_QI	
TRADITIONAL SOLOW RESIDUAL		
Λ_L	Labor compensation over value added at basic prices	WIOD, KLEMS
Λ_K	$1 - \Lambda_L$	
DISTORTED SOLOW RESIDUAL, ACCOUNTING FOR THE IO NETWORK		
Λ_L	Revenue-based labor share, $\sum_i \lambda_i \Omega_{iL}$	WIOD, KLEMS, BEA
Λ_K	Revenue-based capital share, $\sum_i \lambda_i \Omega_{iK}$	
$\tilde{\Lambda}_L$	Cost-based labor share, $\sum_i \tilde{\lambda}_i \mu_i \Omega_{iL}$ (local inverse)	
$\tilde{\Lambda}_K$	Cost-based capital share, $\sum_i \tilde{\lambda}_i \mu_i \Omega_{iK}$ (local inverse)	
DISTORTED SOLOW RESIDUAL, IGNORING THE IO NETWORK		
$\hat{\Lambda}_L$	Own-wedge labor share, $\sum_i \lambda_i \mu_i \Omega_{iL}$	WIOD, KLEMS, BEA
$\hat{\Lambda}_K$	Own-wedge capital share, $\sum_i \lambda_i \mu_i \Omega_{iK}$	

Table Notes. $\Delta \log X_t = \log X_t - \log X_{t-1}$. Common variables are Divisia aggregates of the 23 industries' volume indices (ISTAT and Eurostat for output, intermediates, and value added, EU KLEMS 2024 industry accounts for factor services, chained with the 2008 release before 1995); real GDP is benchmarked to the national accounts' growth of real value added.

C.4 Construction of the Aggregate Series

Intermediate inputs. The growth of each input flow is measured from the input-output table after deflating by the seller's output price. All input flows purchased by a given industry are then adjusted by a common factor so that their aggregate grows at the same rate as that industry's intermediate-input volume in the industry accounts. The input-output tables therefore determine the relative growth of sourcing across suppliers, which enters terms (IV) and (V) in the TFP decomposition, while the industry accounts determine the growth of total intermediate-input use, which enters the sectoral productivity residuals and real GDP.

Real GDP. The industry data determine the composition of real output growth, while the national accounts determine its aggregate growth. Each year, the output growth of all industries is therefore adjusted by a common factor so that their Divisia aggregate equals the growth of real value added in the national accounts. Because the adjustment is common across industries, it leaves the reallocation terms unchanged. The adjustment is small: 1 percentage point over 1970–2019 in Spain and 1.2 points in Italy. Without it, the post-1995 decline is 4.4 percentage points rather than 4.5 in Spain and 8.9 rather than 9 in Italy.

Labor and capital. Aggregate labor and capital are the Divisia aggregates of the corresponding industry series, as in the theory and the EU KLEMS accounting methodology (O'Mahony and Timmer, 2009). I do not benchmark them to the published EU KLEMS

total-economy indices because, across releases, those indices are not always constructed as aggregates of the industry series used here. The choice matters little for Spain: benchmarking labor to the total-economy index from 1996 changes the post-1995 decline by 0.3 percentage points. It matters more for Italy, where the corresponding decline falls to 6.5 points, although within-sector productivity remains the largest component of the decomposition.

C.5 Sectoral Employment Shares and Productivity

Figure 10 shows the share of each of the 23 industries in total hours worked in Spain from 1970 to 2019. Labor shifted steadily away from agriculture and manufacturing toward services, especially real estate and business services, trade, hotels and restaurants, and non-market services. Construction followed a different pattern, rising sharply during the housing boom and falling after.

Figure 10: Labor Reallocation in Spain, 1970–2019.

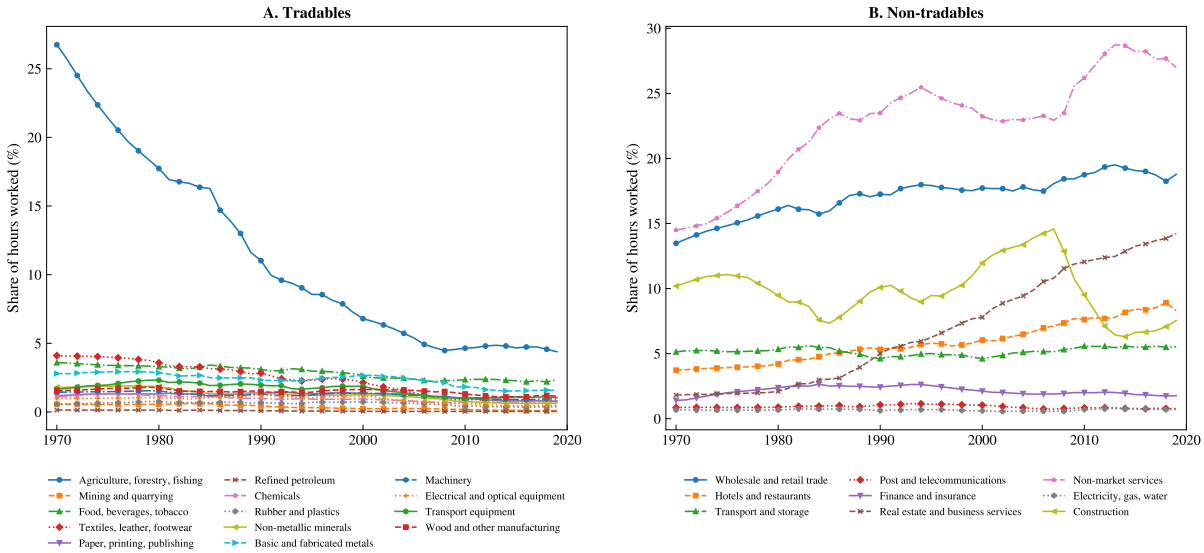
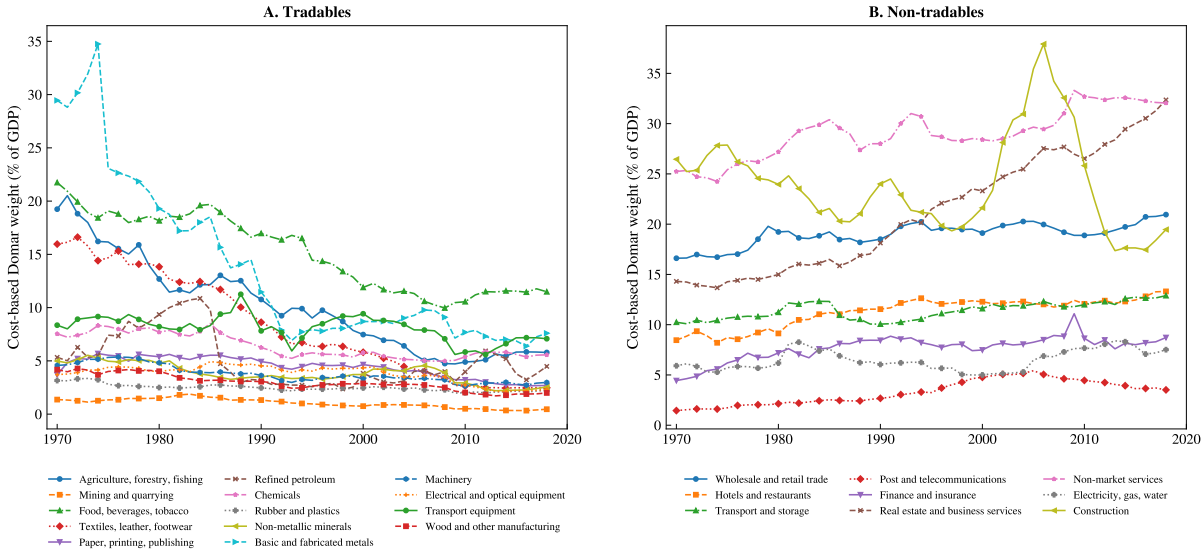


Figure 11 shows the corresponding cost-based Domar weights, which determine the contribution of each industry's productivity growth to aggregate TFP. Because these weights are nominal, they reflect relative prices and production linkages as well as the allocation of primary factors. Agriculture, metals, and textiles fell from 19, 29, and 16 percent of GDP in 1970 to 6, 8, and 2 percent in 2018, while real estate and business services rose from 14 to 32 percent, non-market services from 25 to 32, and hotels and restaurants from 8 to 13. Construction rose from 21 percent in 1995 to 38 in 2006 before falling back to 19 in 2018.

Figure 12 reports cumulative productivity growth by industry, measured as the gross-output Solow residual with the cost-based shares of Definition 3. Over 1970–2019,

Figure 11: Cost-Based Domar Weights by Industry, Spain, 1970–2019.



productivity rose most in agriculture, telecommunications, and much of manufacturing, and fell in non-market services, hotels and restaurants, real estate and business services, and construction. The industries that gained weight were therefore generally those in which productivity grew more slowly or declined. After 1995, productivity continued to rise in agriculture, telecommunications, and finance, while falling in hotels and restaurants, construction, transport, and non-market services. Real estate and business services had already lost about 15 log points before 1995 and were roughly flat after.

Figure 12: Sectoral Productivity Growth in Spain, 1970–2019.

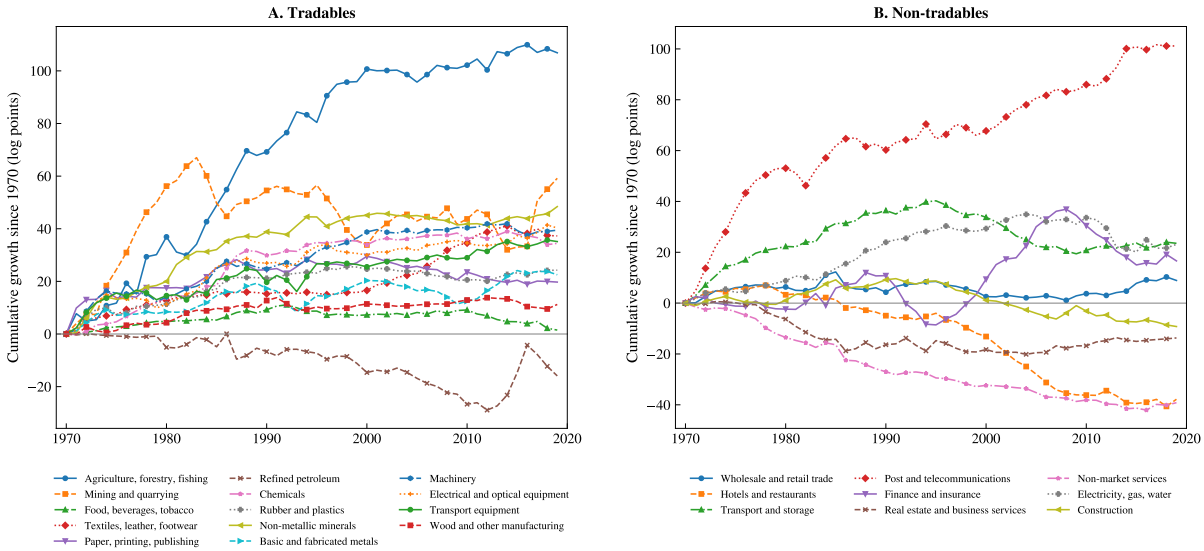


Figure Notes. Cumulative productivity growth since 1970 of each of the 23 industries, in log points: the gross-output Solow residual with cost-based input shares of Definition 3, with base-period weights, which enters terms (I) and (II) of Theorem 1. The left panel shows tradables (agriculture, mining, and manufacturing) and the right panel non-tradables (services, utilities, and construction), as in Table 1.

C.6 Trade Integration

Figure 13 shows the foreign share of household consumption and intermediate inputs in Spain and Italy from 1970 to 2019. In Spain, the foreign share of household consumption rose from 2.5 percent in 1970 to 13 percent by 2007 and remained at that level until 2019; in Italy, it rose from around 5 percent to 12 percent in 2019. The foreign share of intermediate inputs purchased by domestic producers more than doubled in Spain, from 10 percent to 23 percent, and rose from 13 percent to 19 percent in Italy.

Figure 13: Trade Statistics in Spain and Italy, 1970–2019.

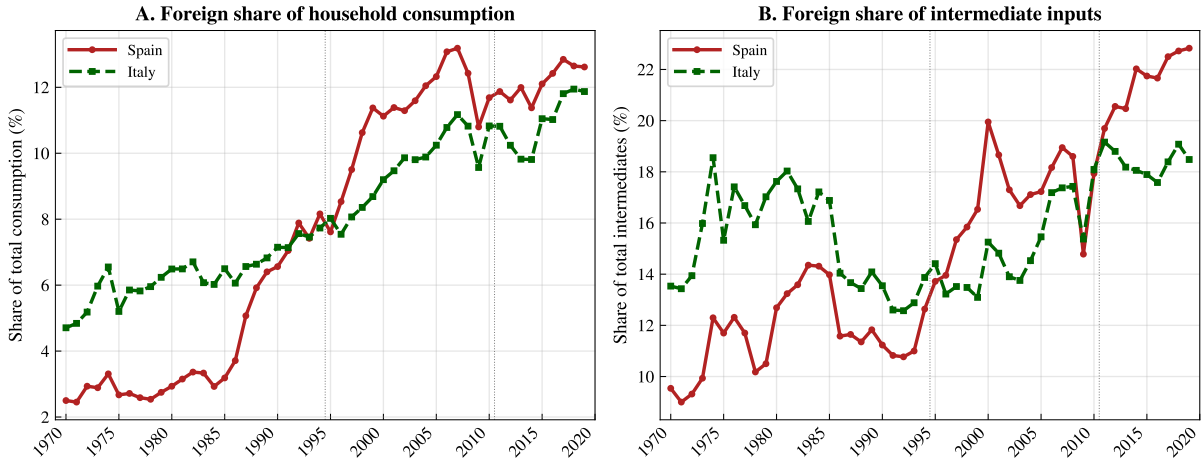


Figure Notes. Shares computed from the harmonized input-output panel (Long-Run WIOD, WIOD 2013, FIGARO).

Figure 14 shows the foreign share of Spain's final demand and intermediate use of each tradable product every ten years from 1970 to 2010. Import penetration increased broadly across manufacturing. The foreign share of intermediate use rose from 14 to 68 percent in transport equipment, from 16 to 59 in electrical and optical equipment, from 29 to 55 in chemicals and from 4 to 34 in textiles, and that of final use from 2 to 79 percent in textiles, from 38 to 83 in electrical equipment and from 6 to 56 in chemicals. The figure stops in 2010 because FIGARO, which covers the later years, allocates imports across products differently from the WIOD tables, making product-level shares not comparable even though the aggregate trade shares and the weights entering the decomposition remain smooth.

Figure 14: Import Penetration by Tradable Product, Spain, 1970–2010.

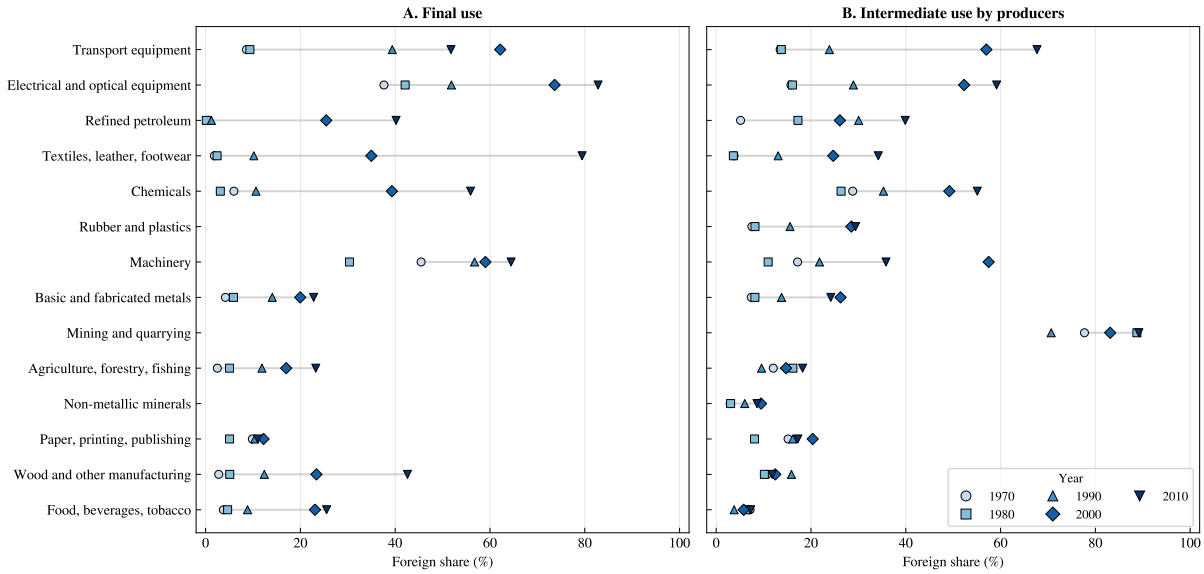


Figure Notes. Foreign share of Spain's final use—household and government consumption and fixed investment—and of intermediate use by domestic producers, for each tradable product, every ten years from 1970 to 2010.

D Estimation of the Wedges

Distortions μ_i are wedges between price and marginal cost, capturing market imperfections such as market power and distortionary taxes. I recover them from wedge margins, defined as the share of revenues remaining after expenditures on intermediate inputs, labor, and the user cost of capital:

$$\text{Wedge margin}_i := \frac{p_i y_i - \sum_{j \in \mathcal{I}} p_j x_{ij} - w_i \ell_i - r_i k_i}{p_i y_i},$$

where $p_i y_i$ denotes sales, $\sum_j p_j x_{ij}$ expenditures on intermediate inputs, $w_i \ell_i$ labor costs, and $r_i k_i$ the user cost of capital.

Under constant returns to scale, wedge margins map directly into distortions:

$$\text{Wedge margin} = 1 - \frac{1}{\mu} \qquad \Longleftrightarrow \qquad \mu = \frac{1}{1 - \text{Wedge margin}}.$$

More generally,

$$\text{Wedge margin} = 1 - \frac{\text{RS}}{\mu} \qquad \Longleftrightarrow \qquad \mu = \frac{\text{RS}}{1 - \text{Wedge margin}},$$

where RS denotes the local returns to scale.[13] Away from constant returns, the wedge margin therefore does not identify the price–marginal cost ratio without additional information on RS. I accordingly refer to these measures as wedges rather than markups.

An alternative is to estimate markups from an output elasticity and the corresponding input revenue share (Hall, 1988; De Loecker and Warzynski, 2012). With only 23 industries, however, production-function elasticities are too imprecisely estimated for this approach to be informative. I therefore estimate wedge margins. Their measurement requires capital stocks, depreciation rates, and the required return on capital. Capital stocks come from EU KLEMS for Spain and Italy and from the WIOD Socio-Economic Accounts for the remaining countries. The construction of industry-specific depreciation rates is described below.

D.1 The User Cost of Capital

The market cost of capital. For Spain and Italy, the baseline user cost is the tax-adjusted rental rate of Hall and Jorgenson (1967). The rental rate combines the real after-tax required return with asset-specific depreciation rates and the tax treatment of

[13] This relation follows from cost minimization and standard regularity conditions on the production function, including differentiability and quasi-concavity (see Hasenzagl and Pérez, 2026).

depreciation allowances. The rental rate per unit of capital of producer i in year t is

$$r_{i,t} = \sum_j s_{ij,t}\Big(\max\{\rho_t, 0\} + \delta_j\Big)\frac{1-\tau_t z_{j,t}}{1-\tau_t},$$
$$\rho_t = \frac{D_t}{D_t+E_t}(b_t+s_t)(1-\tau_t) + \frac{E_t}{D_t+E_t}(b_t+e_t) - \pi_t,$$

where j indexes assets, $s_{ij,t}$ is asset j's share in the producer's capital stock, δ_j its depreciation rate, τ_t the country's corporate tax rate, $z_{j,t}$ the present value of the tax depreciation allowances on a unit of asset j, and ρ_t the country's real after-tax cost of capital. The cost of capital weights the cost of debt and of equity by the financing structure of the country's non-financial corporations: b_t is the country's long-term government bond yield, s_t the spread of the Moody's Baa corporate bond yield over the 10-year US Treasury yield, e_t the country's historical equity premium, and $D_t/(D_t+E_t)$ the share of debt securities and loans in the liabilities of non-financial corporations. Interest is deductible from taxable profits, so that debt costs $(b_t+s_t)(1-\tau_t)$ after tax.

The equity premium is the mean log excess return of the country's equities over its long-term government bonds from 1900 to year $t-1$, in the returns of Jordà et al. (2019), so that it is the country's own and uses no future data; it averages 3.6 percent in Spain and 4.1 percent in Italy over 1970–2019. No national corporate bond spread exists for 1970–2019, so the spread is the US one throughout. Expected capital-goods inflation is the trailing three-year mean of the realized change in the price of the market economy's capital stock, so that the expectation uses no future data. The rental rate is common to the producers of a country and year up to their asset composition.

The tax adjustment $(1-\tau z_j)/(1-\tau)$ converts the after-tax cost of asset j into the pre-tax return that the asset must earn, accounting for depreciation allowances. Corporate tax rates and depreciation rules come from the Oxford CBT Tax Database. Depreciation allowances are mapped into structures, machinery and equipment, and intangible assets. Outside the years covered by the database, I hold the nearest available tax parameters fixed. Accounting for taxes raises the capital-weighted rental rate by 6.5 percent on average in Spain and 3 percent in Italy relative to a rental rate without taxes.

Eurostat's financial accounts provide the debt share of non-financial corporations from 1995 onward. Before 1995, I backcast it using the ratio of business loans to business loans plus stock-market capitalization. Over 1995–2016, where both measures are available, the proxy tracks the Eurostat series closely: their correlation is 0.93 in Spain and 0.95 in Italy. I scale its pre-1995 movements using the relationship between the two series estimated over their overlap.

The real after-tax required return averages 2.8 percent in Spain and 3.3 percent in Italy, compared with 4.5 and 5.6 percent before taxes. It is negative in Spain through

1981 and in Italy in 1974–1982, when high inflation drove measured real interest rates below zero in economies with heavily regulated credit markets. Because a negative required return is not a plausible opportunity cost of capital, I impose a lower bound at zero. The floor binds in 12 years in Spain and 9 in Italy. The robustness appendix considers an international cost of capital, which is never negative, and a constant required return; neither materially changes the post-1995 results. Non-market services—public administration, education, health and other services—have no wedge because their output is valued at cost in the national accounts.

The estimated net return. For the other countries, which enter only through the world production network and for which the series above are not available, and also as a robustness check for Spain and Italy, I estimate the net rate of return on capital from the industry data with the method of van Vlokhoven (2022), which exploits cross-sectional variation in input choices. With constant returns, producer i prices at its wedge over average cost, $p_i y_i = \mu_i\left[\text{COGS}_i + (r + \delta_i - g_i)\, p_i^K K_i\right]$, where COGS_i denotes the cost of goods sold—intermediate input expenditure plus labor compensation—$p_i^K K_i$ is the nominal capital stock, δ_i is the producer's depreciation rate, g_i is the expected change in the price of its assets relative to the capital-weighted average of its country, and r is the net rate of return, common to the producers of a country and year. Moving the known part of the user cost, $(\delta_i - g_i)\, p_i^K K_i$, into costs gives the regression

$$\frac{p_i y_i}{\text{COGS}_i + (\delta_i - g_i) p_i^K K_i} = \overline{\psi} + \overline{\psi r}\, \frac{p_i^K K_i}{\text{COGS}_i + (\delta_i - g_i) p_i^K K_i} + \varepsilon_i,$$

whose intercept and slope imply the net rate of return,

$$\bar{r} = \frac{\overline{\psi r}}{\overline{\psi}}.$$

The user cost of producer i is $(\bar{r} + \delta_i - g_i)\, p_i^K K_i$. van Vlokhoven (2022) estimates the gross user cost, depreciation included, from a regression of sales over operating expenses on the capital stock over operating expenses across firms within an industry, and lets it vary with each firm's depreciation rate through a control interacted with the capital ratio; he finds that the estimated user cost moves one for one with the depreciation. I impose that relation and estimate only the net return, since estimating the depreciation coefficient is imprecise with only 21 market-economy industries. This allows a common net return across industries whose capital depreciates at different rates. Estimating a single gross return would instead confound the net return with differences in asset compositions and depreciation rates across industries.

The regression is estimated over centered five-year windows. It excludes real estate and non-market services, whose output-to-capital ratios are difficult to interpret as returns on productive capital, and the estimated return is then applied to all industries. Expected asset-price appreciation is measured as a trailing three-year average and enters only relative to the country's capital-weighted average. The estimated net return is bounded below at zero. It averages 5.7 percent in Spain and 7.2 percent in Italy over 1970–2019. The robustness appendix reports the corresponding wedges and decompositions for Spain and Italy.

D.2 Depreciation and Expected Capital-Goods Inflation

Each industry's depreciation rate is a capital-stock-weighted average across ten asset categories. Asset-specific rates are constructed from BEA fixed-asset data as depreciation divided by the beginning-of-period net capital stock plus one-half of current investment, the formula underlying the BEA's implied depreciation rates. The rates therefore vary across industries according to their asset composition and over time as that composition changes. They range from about 1.5 percent per year for dwellings and 2.5 percent for other structures to 11–14 percent for machinery and transport equipment and 27–40 percent for computers and software.

Asset composition comes from the 2024 EU KLEMS capital accounts for Spain and Italy from 1995 onward, from the Fundación BBVA–Ivie database for Spain before 1995, and from the 2008 EU KLEMS release for Italy before 1995 and for the other countries with asset data. Earlier series are chain-linked to the 2024 release in 1995. Where asset composition is unavailable, I use the cross-country mean for the same industry and year. The resulting sales-weighted depreciation rate in Spain falls from 7.0 percent in 1970 to 5.2 percent in 1995 and remains close to that level thereafter. In Italy it is 7.1 percent in 1970, 6.9 percent in 1995, and 6.7 percent in 2019.

The user cost subtracts expected capital-goods appreciation. Realized capital-goods inflation is the log change in each asset's investment price, weighted by its share in the industry's capital stock at the start of the year. Expected inflation is the trailing three-year mean of the capital-weighted average across market-economy industries, so the measure uses no future information. It falls from about 12 percent per year in the 1970s in both countries to 4 percent in the 1990s and approximately zero in the 2010s. Using each industry's own trailing mean instead changes Spain's post-1995 decline from 4.5 to 4.1 percentage points and Italy's from 9.0 to 8.9.

D.3 The Wedges

Baseline wedge measure. I compute wedge margins using each industry's capital stock and rental rate. Negative margins are set to zero, implying a minimum wedge of one, and the resulting wedge distribution is winsorized at the 1st and 99th percentiles.

The floor binds in 15 percent of Spain's industry-years and 20 percent of Italy's, most frequently in capital-intensive industries for which the measured required return exceeds the return implied by revenues. Allowing negative margins instead changes Spain's post-1995 decline from 4.5 to 5.4 percentage points and factor reallocation from -0.7 to -1.6, while Italy's decline becomes 9.1 points. Replacing each industry's wedge with its average over a centered five-year window changes no component by more than half a percentage point in Spain and by less than one point in Italy. The robustness appendix reports results under alternative measures of the required return.

What the wedges measure. The wedge margin is a residual: revenue net of intermediate inputs, labor compensation, and capital evaluated at the required return. It therefore need not represent pure economic profits. Besides market power and taxes, it can capture omitted productive assets such as land and inventories and any difference between the financial-market required return and the return firms actually require. In this sense it is closely related to the factorless income of Karabarbounis and Neiman (2019). Because the required return varies substantially over time while operating margins move more slowly, the measured wedge is inversely related to the market cost of capital. The robustness appendix shows that the main productivity results survive alternative required returns constructed from industry data, international financial markets, and a constant return.

Figure 15 reports the baseline user cost and wedges. Panel A plots the capital-weighted rental rate and the real after-tax cost of capital before imposing the zero lower bound. Where the bound binds, the rental rate is the capital-weighted depreciation rate, about 3.5 percent in Spain and 4 in Italy, adjusted up for taxes to about 4 and 5 percent. The real after-tax cost of capital is negative during parts of the 1970s, rises substantially during the disinflation of the 1980s and 1990s, declines during the pre-crisis boom, and rises again around the sovereign-debt crisis. The aggregate wedge moves inversely with the required return because higher capital costs absorb a larger share of the residual between revenue and measured operating costs. This time-series relationship motivates the alternative required-return measures considered in the robustness appendix.

Figure 15: The User Cost of Capital and the Wedges, 1970–2019.

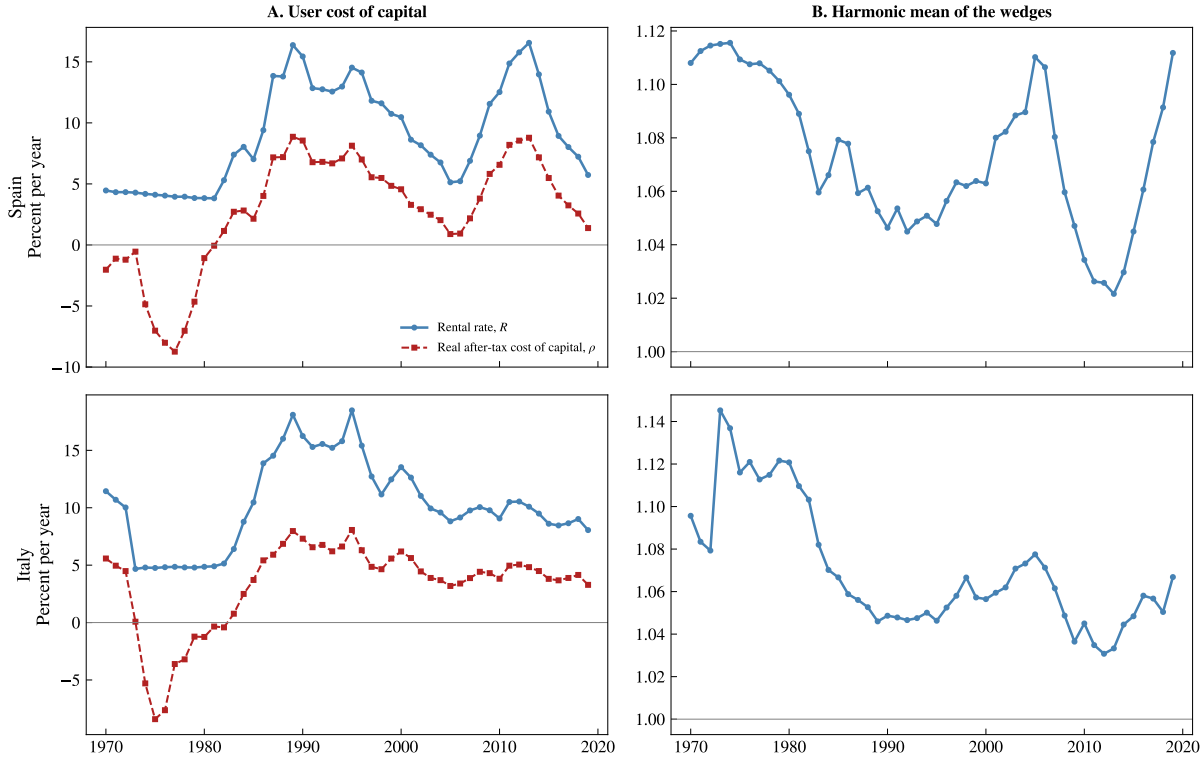


Figure Notes. Baseline estimates, wedge margins with the market cost of capital, for the 23 industries. Panel A: tax-adjusted rental rate per unit of capital, $R_{i,t}$, weighted by the net capital stock at current replacement cost, and the real after-tax cost of capital, ρ_t, before the floor at zero. Panel B: $1/\sum_i (p_i y_i / \sum_j p_j y_j)\,\mu_i^{-1}$, the ratio of aggregate revenue to aggregate cost.

D.4 Income Shares

Figure 16 decomposes value added net of taxes into labor compensation, capital evaluated at the required return, and the residual income associated with the wedges. Overall the evolution of income shares appears reasonable.

Figure 16: Income Shares in Spain and Italy, 1970–2019.

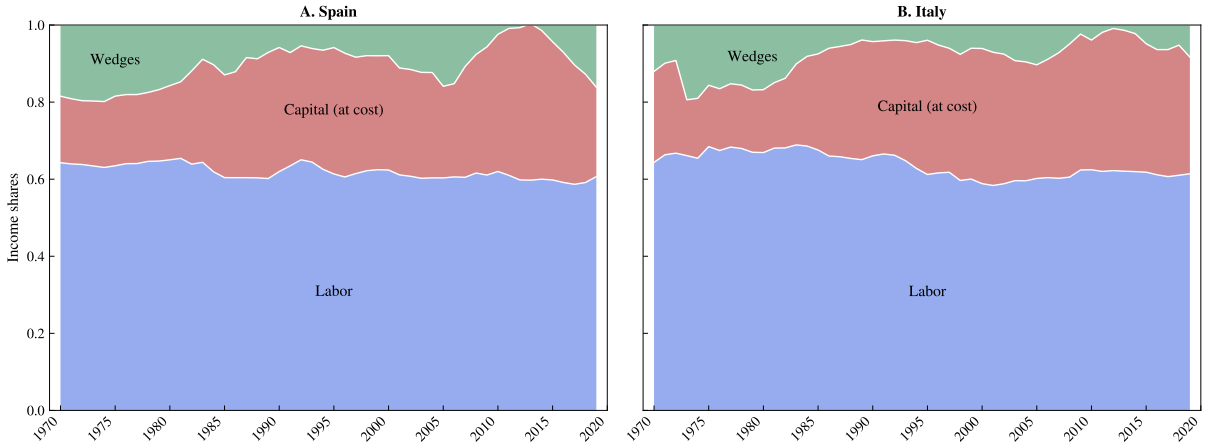


Figure Notes. Income shares are computed after subtracting taxes from total income: labor compensation, capital at cost (the net capital stock at current replacement cost times the gross user cost), and the income of the wedges net of the taxes less subsidies on products paid on intermediate inputs, each summed over industries and divided by value added at basic prices.

E Robustness

This appendix examines the robustness of the main results to alternative measures of distortions and alternative aggregation weights. Other data and estimation choices are discussed where they are introduced: depreciation rates and expected capital-goods inflation in Appendix D.2, the user cost of capital and the treatment of wedge margins in Appendix D, and benchmarking and factor aggregation in Appendix C.4.

Alternative Measures of Distortions

The baseline wedges depend on the required return used in the user cost of capital. I consider three alternatives, each designed to remove a potential source of misspecification in the baseline. The first estimates the net return directly from the industry accounts and therefore does not rely on financial-market measures. The second uses the cost of capital in US markets, removing the effects of domestic financial regulation and country-specific risk premia in Spain and Italy. The third holds the required real return constant at its country-specific sample mean, eliminating all time variation in the required return. Figure 17 shows the corresponding rental rates and aggregate wedges.

Estimated net return. This specification replaces the market cost of capital in Spain and Italy with the net return estimated from the industry data as described in Appendix D. The estimated return is the one that removes the systematic relation between margins and capital intensity across industries. It averages 5.7 percent in Spain and 7.2 percent in Italy, compared with average real market cost of capital before taxes of 4.5 and 5.6 percent. Its time-series behavior differs substantially from the baseline: in Spain it is relatively high during 1995–2010 and low after 2014, whereas the market cost of capital is lowest during the pre-crisis boom and rises during the sovereign-debt crisis. This specification therefore provides a direct check on whether the results are driven by the baseline user cost.

International cost of capital. The second alternative replaces the domestic required return with one constructed from US financial markets: the US long-term government bond yield, the Baa corporate spread, and the historical US equity premium, net of expected US consumer-price inflation, while retaining each country's own financing structure and taxes. Its real after-tax return averages 5 percent in Spain and 4.5 percent in Italy and is never negative. This specification removes the low administered real rates of the 1970s and the country-specific sovereign risk premia that affect the baseline.

Figure 17: User Costs of Capital and Wedges under Four Measures, 1970–2019.

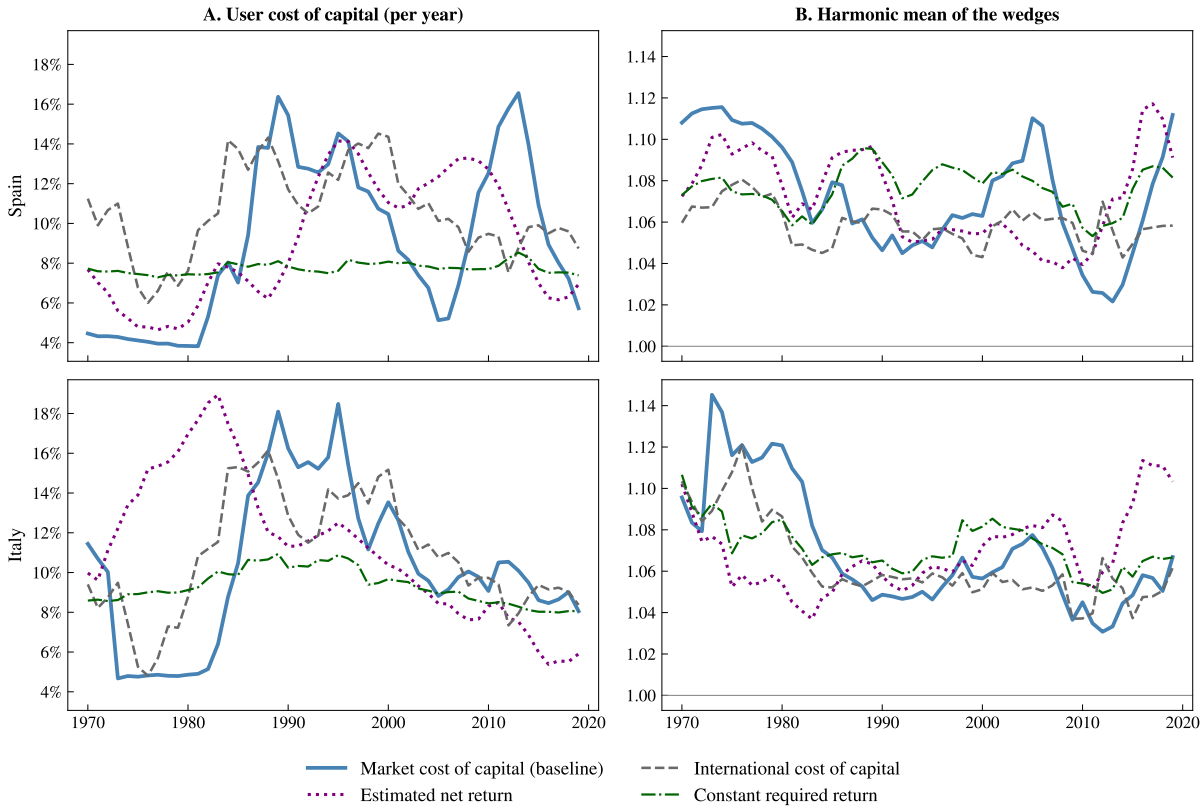


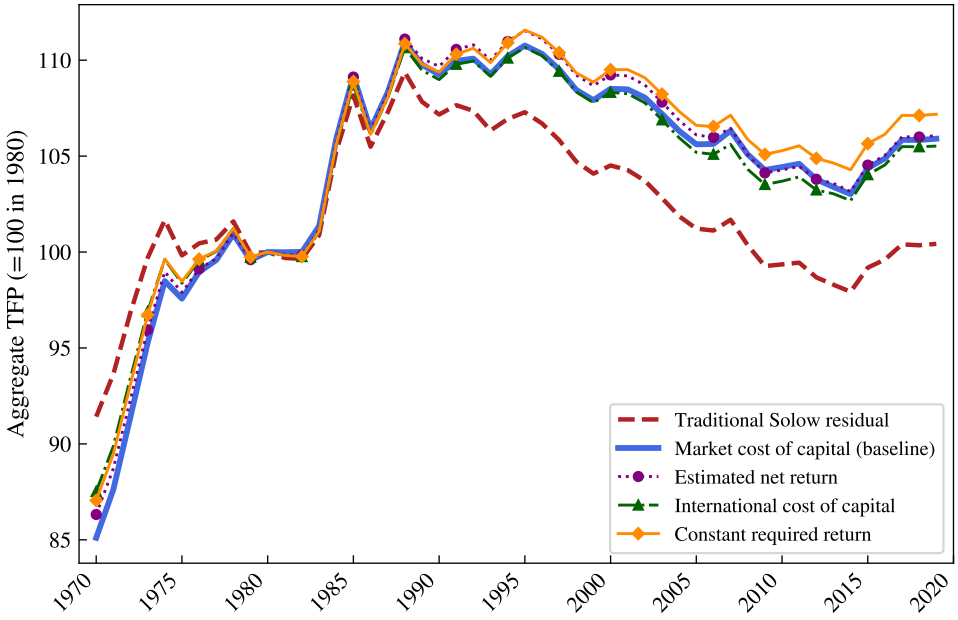
Figure Notes. Panel A: average of the 23 industries' rental rate per unit of capital, weighted by their net capital stocks at current replacement cost; that is, the economy's rental cost of capital per unit of capital. Panel B: sales-weighted harmonic mean of the wedges, the ratio of aggregate revenue to aggregate cost. The baseline is the tax-adjusted rental rate with the market cost of capital.

Constant required return. The third alternative holds the real after-tax required return fixed at its country-specific average over 1970–2019: 2.8 percent in Spain and 3.3 percent in Italy. Taxes and depreciation rates remain unchanged. This specification removes both the negative real rates of the 1970s and all subsequent variation in the financial cost of capital. Two additional specifications are close to the baseline: a market cost of capital without taxes and a user cost equal to the government bond yield plus a fixed three-percentage-point risk premium.

Aggregate TFP. Figure 18 reports aggregate TFP in Spain under the baseline and the three main alternatives. Under every specification, productivity is approximately flat between 1988 and 1995 and declines thereafter by substantially less than the traditional Solow residual implies. From 1995 to 2019, aggregate TFP falls by 4.5 percentage points under the baseline, 5 with the estimated net return, 4.8 with the international cost of capital, and 4 with the constant required return, compared with 6.6 under the traditional residual. The exact peak varies between 1988 and 1995, but under all four measures the two years lie within 0.7 percentage points of one another. The alternatives matter more for measured growth before 1995, when they differ most in the treatment of the 1970s.

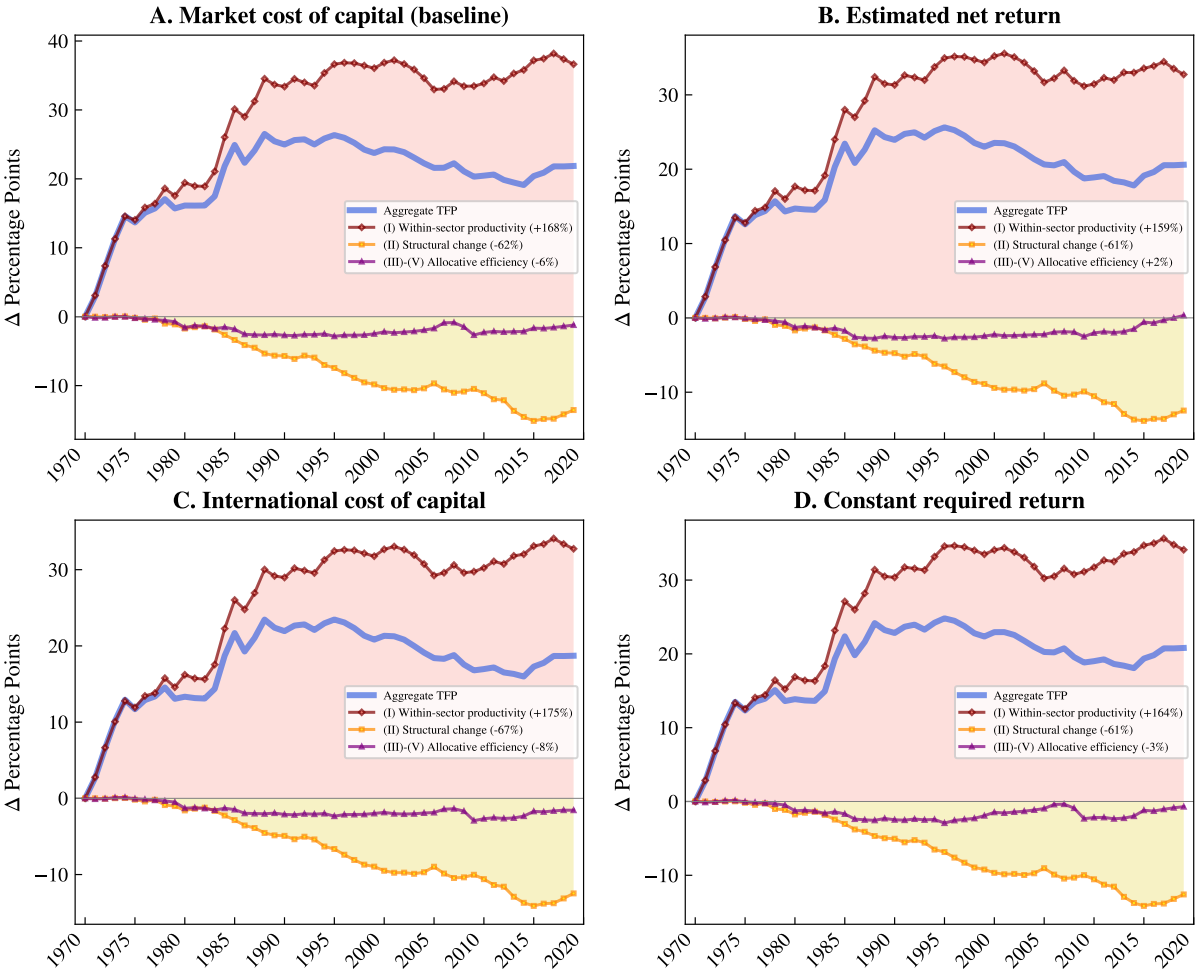
Nevertheless, the long-run conclusion remains the same: conventional factor shares understate productivity growth when wedge income is attributed to capital and capital deepens rapidly. Measures based on gross margins or accounting profits are not useful alternatives for this purpose because they leave all or part of the required return on capital inside the wedge and therefore fail to net capital deepening appropriately.

Figure 18: Aggregate TFP in Spain with Alternative Measures of Distortions.



Aggregate TFP Decomposition. Figure 19 reports the corresponding decompositions for Spain. The main conclusions are stable across all distortion measures. Over 1970–2019, within-sector productivity remains the dominant positive force, structural change remains the largest negative component, and allocative efficiency across industries remains comparatively small. Structural change subtracts between 12.5 and 13.5 percentage points over the full sample across the four specifications. Over 1995–2019 it subtracts between 5.7 and 5.9 points.

Figure 19: Decompositions of Aggregate TFP in Spain with Several Distortion Measures.



Italy. Figure 20 plots aggregate TFP in Italy under the same measures, together with the traditional Solow residual. The alternative measures have even less effect on the timing and magnitude of the post-1995 decline. Under every specification, aggregate TFP stops growing in 1995, remains flat until 2001, and declines by 8.6 to 9.8 percentage points between 1995 and 2019. The alternatives mainly affect measured productivity growth before 1995.

Figure 20: Aggregate TFP in Italy with Alternative Measures of Distortions.

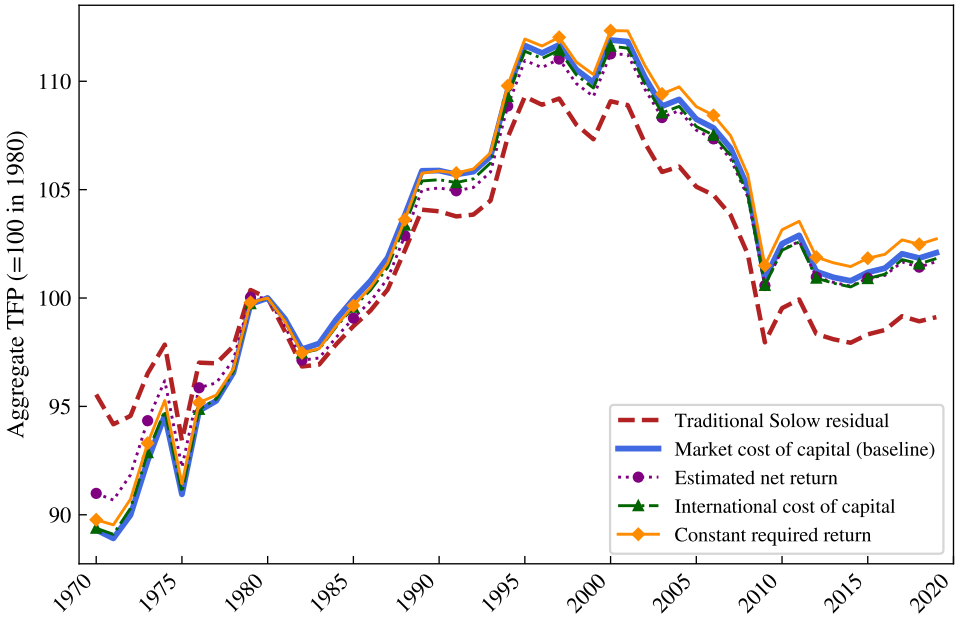


Figure 21 reports the corresponding decompositions. Across all measures, within-sector productivity accounts for roughly 68 to 72 percent of Italy's post-1995 decline and structural change for 33 to 38 percent, while allocative efficiency plays almost no role.

Weighting Conventions

The main text uses base-period weights. As an alternative, I recompute aggregate TFP and the decomposition using midpoint (Törnqvist) weights, which average the network weights of every two consecutive years. Theorem 1 remains exact under either convention provided the sectoral Solow residuals are constructed using the corresponding weights. Figure 22 reports the distorted Solow residual under the baseline measure of distortions for the two countries using both base-period and midpoint weights.

Figure 23 shows the three main channels of the decomposition under midpoint weights. For Spain, the main conclusions are unchanged. Over 1970–2019, within-sector productivity is +32.6 percentage points rather than +36.6 under base-period weights, structural change is −13.0 rather than −13.5, and allocative efficiency is +0.1 rather than −1.2. Over 1995–2019, the decline is −5.6 rather than −4.5, with the difference concentrated in the within-sector term. For Italy, within-sector productivity and structural change move more, to +17.8 and −7.6 from +21.8 and −9.7, but the main conclusions also remain unchanged.

Figure 21: Decompositions of Aggregate TFP in Italy with Several Distortion Measures.

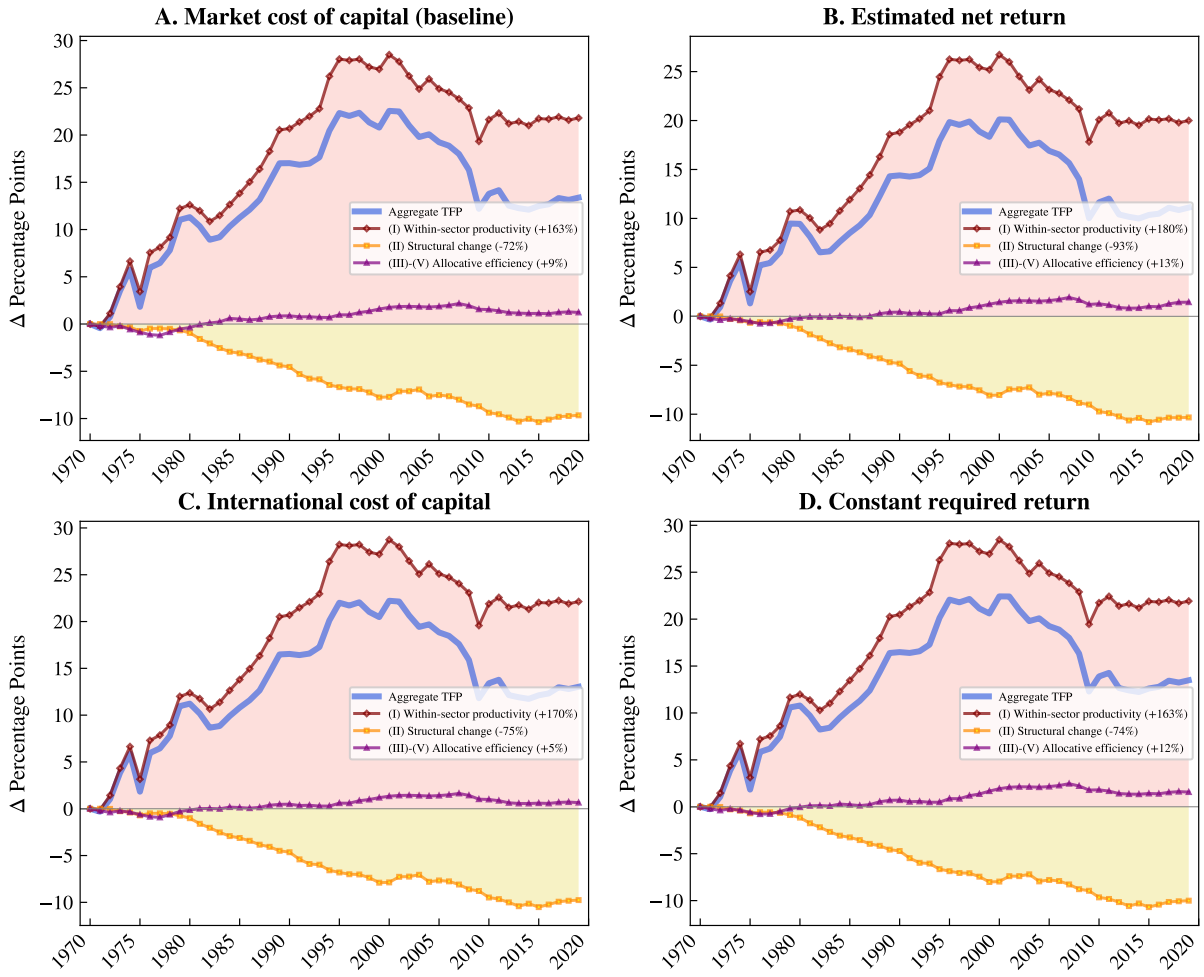


Figure 22: Aggregate TFP Indices: Base-period vs. Midpoint Weights.

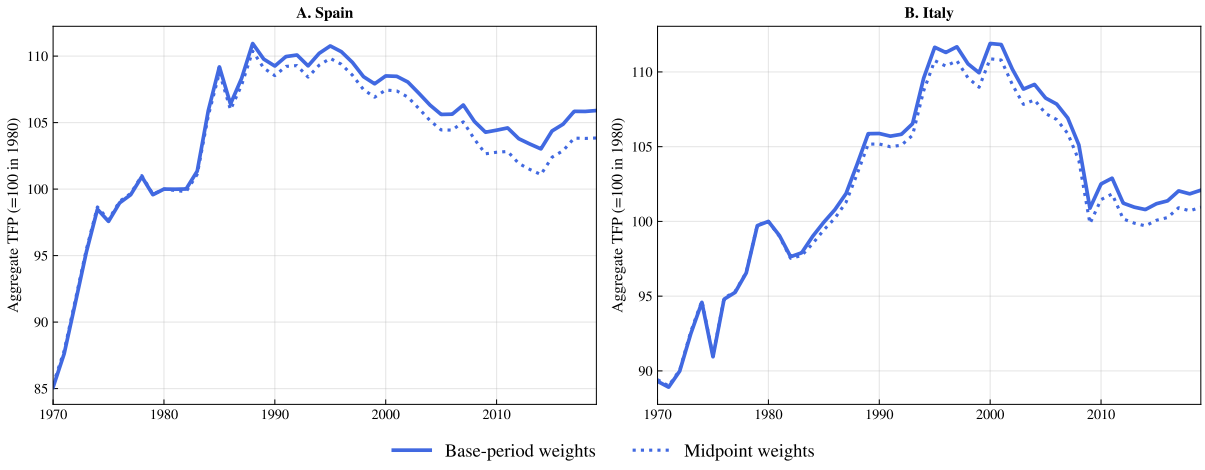


Figure Notes. Aggregate TFP measured with the distorted Solow residual accounting for the IO network, with base-period weights and with midpoint weights, which average the network weights of $t-1$ and t.

Figure 23: Cumulative Decomposition of Aggregate TFP with Midpoint Weights.

